Appendices

There are altogether 18 appendices in the back of this book, covering in more detail various topics ranging from generation and recombination, analytic short-channel threshold model, quantum mechanical solution in weak inversion, emitter and base series resistance, to unity-gain frequencies of MOSFET and bipolar transistors. They usually involve mathematical treatments too tedious and lengthy to be included in the main text. Ten of the 18 appendices are new additions to the second edition.

2 Basic Device Physics

This chapter reviews the basic concepts of semiconductor device physics. Starting with electrons and holes and their transport in silicon, we focus on the most elementary types of devices in VLSI technology: p–n junction, metal–oxide–semiconductor (MOS) capacitor, and metal-semiconductor contacts. The rest of the chapter deals with subjects of importance to VLSI device reliability: high-field effects, the Si–SiO_2 system, and dielectric breakdown.

2.1 Electrons and Holes in Silicon

The first section covers energy bands in silicon, Fermi level, n-type and p-type silicon, electrostatic potential, drift and diffusion current transport, and basic equations governing VLSI device operation. These will serve as the basis for understanding the more advanced device concepts discussed in the rest of the book.

2.1.1 Energy Bands in Silicon

The starting material used in the fabrication of VLSI devices is silicon in the crystalline form. The silicon wafers are cut parallel to either the ⟨111⟩ or ⟨100⟩ planes (Sze, 1981), with ⟨100⟩ material being the most commonly used. This is largely due to the fact that ⟨100⟩ wafers, during processing, produce the lowest charges at the oxide–silicon interface as well as higher mobility (Balk *et al.*, 1965). In a silicon crystal each atom has four valence electrons to share with its four nearest neighboring atoms. The valence electrons are shared in a paired configuration called a *covalent bond*. ***The most important result of the application of quantum mechanics to the description of electrons in a solid is that the allowed energy levels of electrons are grouped into bands*** (Kittel, 1976). ***The bands are separated by regions of energy that the electrons in the solid cannot possess: forbidden gaps.*** The highest energy band that is completely filled by electrons at 0 K is called the *valence band*. The next higher energy band, separated by a forbidden gap from the valence band, is called the *conduction band*, as shown in Fig. 2.1.

2.1.1.1 Bandgap of Silicon

What sets a semiconductor such as silicon apart from a metal or an insulator is that at absolute zero temperature, the valence band is completely filled with electrons, while

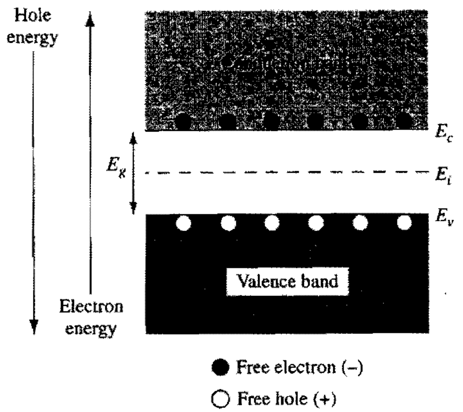

Figure 2.1. Energy-band diagram of silicon.

the conduction band is completely empty, and that the separation between the conduction band and valence band, or the *bandgap*, is on the order of 1 eV. On one hand, no electrical conduction is possible at 0 K, since there are no current carriers in the conduction band, whereas the electrons in the completely filled valence band cannot be accelerated by an electric field and gain energy. On the other hand, the bandgap is small enough that at room temperature a small fraction of the electrons are excited into the conduction band, leaving behind vacancies, or *holes*, in the valence band. This allows limited conduction to take place from the motion of both the electrons in the conduction band and the holes in the valence band. In contrast, an insulator has a much larger forbidden gap of at least several electron volts, making room-temperature conduction virtually impossible. Metals, on the contrary, have partially filled conduction bands even at absolute zero temperature, so that the electrons can gain an infinitesimal amount of energy from the applied electric field. This makes them good conductors at any temperature.

As shown in Fig. 2.1, the energy of the electrons in the conduction band increases upward, while the energy of the holes in the valence band increases downward. The bottom of the conduction band is designated E_c, and the top of the valence band E_v. Their separation, or the bandgap, is $E_g = E_c - E_v$. For silicon, E_g is 1.12 eV at room temperature or 300 K. The bandgap decreases slightly as the temperature increases, with a temperature coefficient of $dE_g/dT \approx -2.73 \times 10^{-4}$ eV/K for silicon near 300 K. Other important physical parameters of silicon and silicon dioxide are listed in Table 2.1 (Green, 1990).

2.1.1.2 Density of States

The density of available electronic states within a certain energy range in the conduction band is determined by the number of different momentum values that can be acquired by electrons in this energy range. Based on quantum mechanics, there is one allowed state in a phase space of volume $(\Delta x\,\Delta p_x)(\Delta y\,\Delta p_y)(\Delta z\,\Delta p_z) = h^3$, where p_x, p_y, p_z are the x-, y-, z-components of the electron momentum and h is Planck's constant. If we

Table 2.1 Physical Properties of Si and SiO_2 at Room Temperature (300 K)

Property	Si	SiO_2
Atomic/molecular weight	28.09	60.08
Atoms or molecules/cm^3	5.0×10^{22}	2.3×10^{22}
Density (g/cm^3)	2.33	2.27
Crystal structure	Diamond	Amorphous
Lattice constant (Å)	5.43	–
Energy gap (eV)	1.12	8–9
Dielectric constant	11.7	3.9
Intrinsic carrier concentration (cm^{-3})	1.0×10^{10}	–
Carrier mobility (cm^2/V-s)	Electron: 1430	–
	Hole: 480	–
Effective density of states (cm^{-3})	Conduction band, N_c: 2.9×10^{19}	–
	Valence band, N_v: 3.1×10^{19}	–
Breakdown field (V/cm)	3×10^5	$>10^7$
Melting point (°C)	1415	1600–1700
Thermal conductivity (W/cm-°C)	1.5	0.014
Specific heat (J/g-°C)	0.7	1.0
Thermal diffusivity (cm^2/s)	0.9	0.006
Thermal expansion coefficient (°C^{-1})	2.5×10^{-6}	0.5×10^{-6}

let $N(E)\,dE$ be the number of electronic states per unit volume with an energy between E and $E + dE$ in the conduction band, then

$$N(E)\,dE = 2g\,\frac{dp_x\,dp_y\,dp_z}{h^3}, \tag{2.1}$$

where $dp_x\,dp_y\,dp_z$ is the volume in the momentum space within which the electron energy lies between E and $E + dE$, g is the number of equivalent minima in the conduction band, and the factor of two arises from the two possible directions of electron spin. The conduction band of silicon has a sixfold degeneracy, so $g = 6$. Note that MKS units are used here (e.g., length must be in meters, not centimeters).

If the electron kinetic energy is not too high, one can consider the energy–momentum relationship near the conduction-band minima as being parabolic and write

$$E - E_c = \frac{p_x^2}{2m_x} + \frac{p_y^2}{2m_y} + \frac{p_z^2}{2m_y}, \tag{2.2}$$

where $E - E_c$ is the electron kinetic energy, and m_x, m_y, m_z are the effective masses. The constant energy surface in momentum space is an ellipsoid with the lengths of the symmetry axes proportional to the square roots of m_x, m_y, and m_z. For the silicon conduction band in the <100> direction, two of the effective masses are the transverse mass $m_t = 0.19m_0$, and the third is the longitudinal mass $m_l = 0.92m_0$, where m_0 is the free electron mass. The volume of the ellipsoid given by Eq. (2.2) in momentum space is $(4\pi/3)(8m_xm_ym_z)^{1/2}(E-E_c)^{3/2}$. Therefore, the volume $dp_xdp_ydp_z$ within which the electron energy lies between E and $E + dE$ is $4\pi(2m_xm_ym_z)^{1/2}(E-E_c)^{1/2}dE$ and Eq. (2.1) becomes

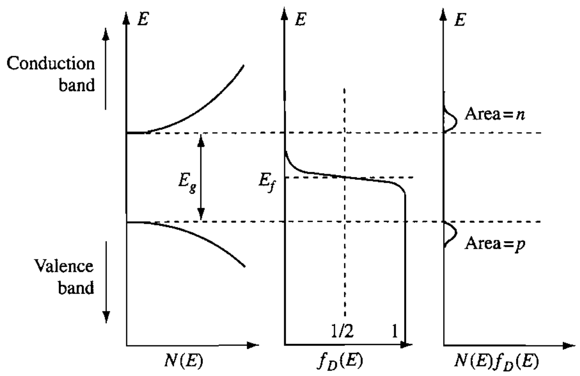

Figure 2.2. Schematic plots of density of states, Fermi–Dirac distribution function, and their products versus electron energy in a band diagram. (After Sze, 1981.)

$$N(E)\,dE = \frac{8\pi g\sqrt{2m_x m_y m_z}}{h^3}\sqrt{E-E_c}\,dE = \frac{8\pi g\sqrt{2m_t^2 m_l}}{h^3}\sqrt{E-E_c}\,dE. \quad (2.3)$$

The 3-D electron density of states in an energy diagram is then a parabolic function with its downward apex at the conduction-band edge, and vice versa for the hole density of states in the valence band. These are shown schematically in Fig. 2.2 (Sze, 1981).

2.1.1.3 Statistical Distribution Function

The energy distribution of electrons in a solid is governed by the laws of Fermi–Dirac statistics. For a system in thermal equilibrium, the principal result of these statistics is the *Fermi–Dirac distribution function*, which gives the probability that an electronic state at energy E is occupied by an electron,

$$f_D(E) = \frac{1}{1 + e^{(E-E_f)/kT}}. \quad (2.4)$$

Here $k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant, and T is the absolute temperature. This function contains a parameter, E_f, called the *Fermi level*. The Fermi level is the energy at which the probability of occupation of an energy state by an electron is exactly one-half. At absolute zero temperature, $T = 0$ K, all the states below the Fermi level are filled ($f_D = 1$ for $E < E_f$), and all the states above the Fermi level are empty ($f_D = 0$ for $E > E_f$). At finite temperatures, some states above the Fermi level are filled as some states below become empty. In other words, the probability distribution $f_D(E)$ makes a smooth transition from unity to zero as the energy increases across the Fermi level. The width of the transition is governed by the thermal energy, kT. This is plotted schematically in Fig. 2.2, with a Fermi level in the middle of the forbidden gap (for reasons that will soon be clear). It is important to keep in mind that the thermal energy at room temperature is 0.026 eV, or roughly $\frac{1}{40}$ of the silicon bandgap. In most cases when the energy is at least several kT above or below the Fermi level, Eq. (2.4) can be approximated by the simple formulas

$$f_D(E) \approx e^{-(E-E_f)/kT} \quad \text{for} \quad E > E_f \quad (2.5)$$

and

$$f_D(E) \approx 1 - e^{-(E_f-E)/kT} \quad \text{for} \quad E < E_f. \quad (2.6)$$

Equation (2.6) should be interpreted as stating that the probability of finding a hole (i.e., an empty state *not* occupied by an electron) at an energy $E < E_f$ is $e^{-(E_f-E)/kT}$. The last two equations follow directly from the Maxwell–Boltzmann statistics for classical particles, which is a good approximation to the Fermi–Dirac statistics when the energy is at least several kT away from E_f.

Fermi level plays an essential role in characterizing the equilibrium state of a system. Consider two electronic systems brought into contact with Fermi levels E_{f1} and E_{f2}, and corresponding distribution functions $f_{D1}(E)$ and $f_{D2}(E)$. If $E_{f1} > E_{f2}$, then $f_{D1}(E) > f_{D2}(E)$, which means that at every energy E where electronic states are available in both systems, a larger fraction of the states in system 1 are occupied by electrons than in system 2. Equivalently, a larger fraction of the states in system 2 are empty than in system 1 at energies where electronic states exist. Since the two systems in contact are free to exchange electrons, there is a higher probability for the electrons in system 1 to re-distribute to system 2 than vice versa. This leads to a net electron transport from system 1 to system 2, i.e., current flows (defined in terms of positive charges) from system 2 to system 1. If there are no power sources connected to the systems to sustain the Fermi level imbalance, eventually the two systems will come to an equilibrium and $E_{f1} = E_{f2}$. No further net electron flow takes place once the same fractions of the electronic states in the two systems are occupied at every energy E. Note that this conclusion is reached regardless of the specific density of states in each of the two systems. For example, the two systems can be two metals, a metal and a semiconductor, two semiconductors of different doping or different composition. ***When two systems are in thermal equilibrium with no current flow between them, their Fermi levels must be equal.*** A direct extension is that, ***for a continuous region of metals and/or semiconductors in contact, the Fermi level at thermal equilibrium is flat, i.e., spatially constant, throughout the region.*** The role of Fermi level at the contacts when there is an applied voltage driving a steady-state current is further discussed in Section 2.1.4.5.

2.1.1.4 Carrier Concentration

Since $f_D(E)$ is the probability that an electronic state at energy E is occupied by an electron, the total number of electrons per unit volume in the conduction band is given by

$$n = \int_{E_c}^{\infty} N(E) f_D(E)\,dE. \quad (2.7)$$

Here the upper limit of integration (the top of the conduction band) is taken as infinity. Both the product $N(E)f_D(E)$ and n, p are shown schematically in Fig. 2.2. In general, Eq. (2.7) is a Fermi integral of the order 1/2 and must be evaluated numerically (Ghandhi, 1968). For nondegenerate silicon with a Fermi level at least $3kT/q$ below the edge of the conduction band, the Fermi–Dirac distribution function can be approximated by the Maxwell–Boltzmann distribution, Eq. (2.5). Equation (2.7) then becomes

$$n = \frac{8\pi g\sqrt{2m_t^2 m_l}}{h^3}\int_{E_c}^{\infty}\sqrt{E-E_c}\,e^{-(E-E_f)/kT}dE. \tag{2.8}$$

With a change of variable, the integral can be expressed in the form of a gamma function, $\Gamma(3/2)$, which equals $\pi^{1/2}/2$. The electron concentration in the conduction band is then

$$n = N_c e^{-(E_c-E_f)/kT}, \tag{2.9}$$

where the pre-exponential factor is defined as the *effective density of states*,

$$N_c = 2g\sqrt{m_t^2 m_l}\left(\frac{2\pi kT}{h^2}\right)^{3/2}. \tag{2.10}$$

A similar expression can be derived for the hole density in the valence band,

$$p = N_v e^{-(E_f-E_v)/kT}, \tag{2.11}$$

where N_v is the effective density of states of the valence band, which depends on the hole effective mass and the valence band degeneracy. Both N_c and N_v are proportional to $T^{3/2}$. Their values at room temperature are listed in Table 2.1 (Green, 1990).

For an intrinsic silicon, $n = p$, since for every electron excited into the conduction band, a vacancy or hole is left behind in the valence band. The Fermi level for intrinsic silicon, or the *intrinsic Fermi level*, E_i, is then obtained by equating Eq. (2.9) and Eq. (2.11) and solving for E_f:

$$E_i = E_f = \frac{E_c+E_v}{2} - \frac{kT}{2}\ln\left(\frac{N_c}{N_v}\right). \tag{2.12}$$

By substituting Eq. (2.12) for E_f in Eq. (2.9) or Eq. (2.11), one obtains the intrinsic carrier concentration, $n_i = n = p$:

$$n_i = \sqrt{N_c N_v}\,e^{-(E_c-E_v)/2kT} = \sqrt{N_c N_v}\,e^{-E_g/2kT}. \tag{2.13}$$

Since the thermal energy, kT, is much smaller than the silicon bandgap E_g, ***the intrinsic Fermi level is very close to the midpoint between the conduction band and the valence band.*** In fact, E_i is sometimes referred to as the midgap energy level, since the error in assuming E_i to be $(E_c + E_v)/2$ is only about 0.3 kT. The intrinsic carrier concentration n_i at room temperature is 1.0×10^{10} cm^{-3}, as given in Table 2.1, which is very small compared with the atomic density of silicon.

Equations (2.9) and (2.11) can be rewritten in terms of n_i and E_i:

$$n = n_i e^{(E_f-E_i)/kT}, \tag{2.14}$$

$$p = n_i e^{(E_i-E_f)/kT}. \tag{2.15}$$

These equations give the equilibrium electron and hole densities for any Fermi level position (not too close to the band edges) relative to the intrinsic Fermi level at the midgap. In the next section, we will show how the Fermi level varies with the type and concentration of impurity atoms in silicon. Since any change in E_f causes reciprocal changes in n and p, ***a useful, general relationship is that the product***

$$pn = n_i^2 \tag{2.16}$$

in equilibrium is a constant, independent of the Fermi level position.

2.1.2 n-Type and p-Type Silicon

Intrinsic silicon at room temperature has an extremely low free-carrier concentration; therefore, its resistivity is very high. In practice, intrinsic silicon hardly exists at room temperature, since it would require materials with an unobtainably high purity. Most impurities in silicon introduce additional energy levels in the forbidden gap and can be easily ionized to add either electrons to the conduction band or holes to the valence band, depending on where the impurity level is (Kittel, 1976). ***The electrical conductivity of silicon is then dominated by the type and concentration of the impurity atoms, or dopants***, and the silicon is called *extrinsic*.

2.1.2.1 Donors and Acceptors

Silicon is a column-IV element with four valence electrons per atom. There are two types of impurities in silicon that are electrically active: those from column V such as arsenic or phosphorus, and those from column III such as boron. As is shown in Fig. 2.3, a column-V atom in a silicon lattice tends to have one extra electron loosely bonded after forming covalent bonds with other silicon atoms. In most cases, the thermal energy at room temperature is sufficient to ionize the impurity atom and free the extra electron to the conduction band. Such types of impurities are called *donors*; they become positively charged when ionized. Silicon material doped with column-V impurities or donors is

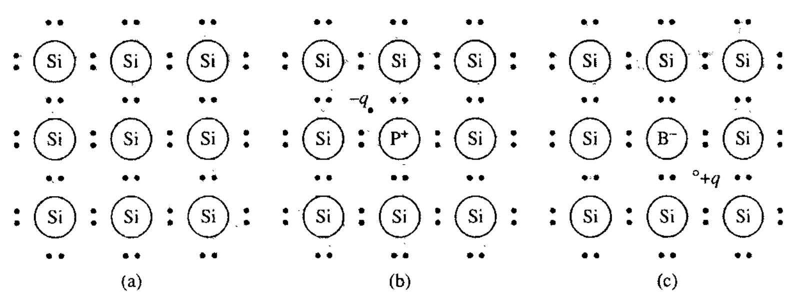

Figure 2.3. Three basic bond pictures of silicon: (a) intrinsic Si with no impurities, (b) n-type silicon with donor (phosphorus), (c) p-type silicon with acceptor (boron). (After Sze, 1981.)

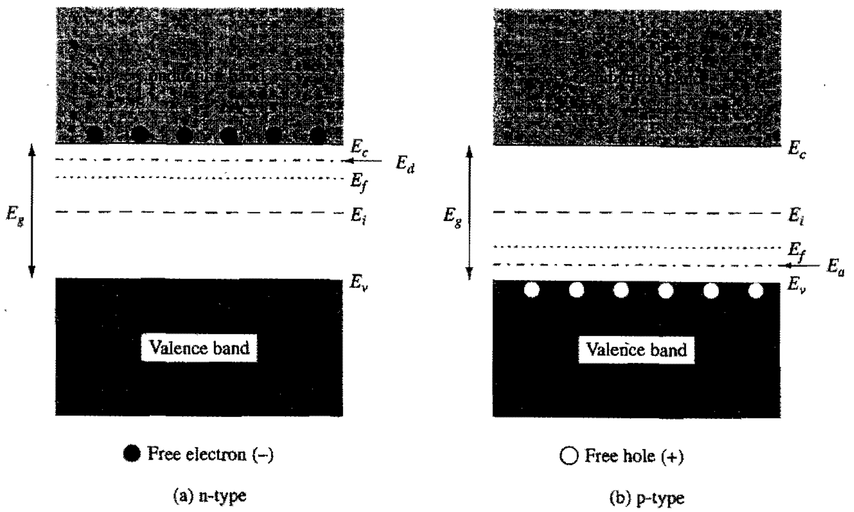

Figure 2.4. Energy-band diagram representation of (a) donor level E_d and Fermi level E_f in n-type silicon, (b) acceptor level E_a and Fermi level E_f in p-type silicon.

called *n-type* silicon, and its electrical conductivity is dominated by electrons in the conduction band. On the other hand, a column-III impurity atom in a silicon lattice tends to be deficient by one electron when forming covalent bonds with other silicon atoms (Fig. 2.3). Such an impurity atom can also be ionized by accepting an electron from the valence band, which leaves a free-moving hole that contributes to electrical conduction. These impurities are called *acceptors*; they become negatively charged when ionized. Silicon material doped with column-III impurities or acceptors is called *p-type* silicon, and its electrical conductivity is dominated by holes in the valence band. It should be noted that impurity atoms must be in a *substitutional* site (as opposed to *interstitial*) in silicon in order to be electrically active.

In terms of the energy-band diagrams in Fig. 2.4, donors add allowed electron states in the bandgap close to the conduction-band edge, while acceptors add allowed states just above the valence-band edge. Donor levels contain positive charge when ionized (emptied). Acceptor levels contain negative charge when ionized (filled). The ionization energies are denoted by $E_c - E_d$ for donors and $E_a - E_v$ for acceptors, respectively. Figure 2.5 shows the donor and acceptor levels of common impurities in silicon and their ionization energies (Sze, 1981). Phosphorus and arsenic are commonly used donors, or n-type dopants, with low ionization energies on the order of $2kT$, while boron is a commonly used acceptor or p-type dopant with a comparable ionization energy. Figure 2.6 shows the *solid solubility* of important impurities in silicon as a function of annealing temperature (Trumbore, 1960). Arsenic, boron, and phosphorus have the highest solid solubility among all the impurities, which makes them the most important doping species in VLSI technology.

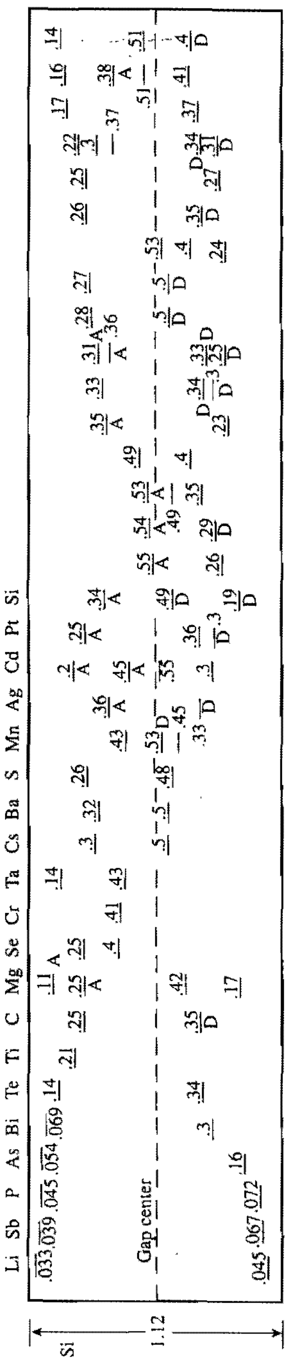

Figure 2.5. Donor and acceptor levels of various impurities in silicon. Numbers next to the level indicate ionization energies $E_c - E_d$ (donors) or $E_a - E_v$ (acceptors) in electron volts. (After Sze, 1981.)

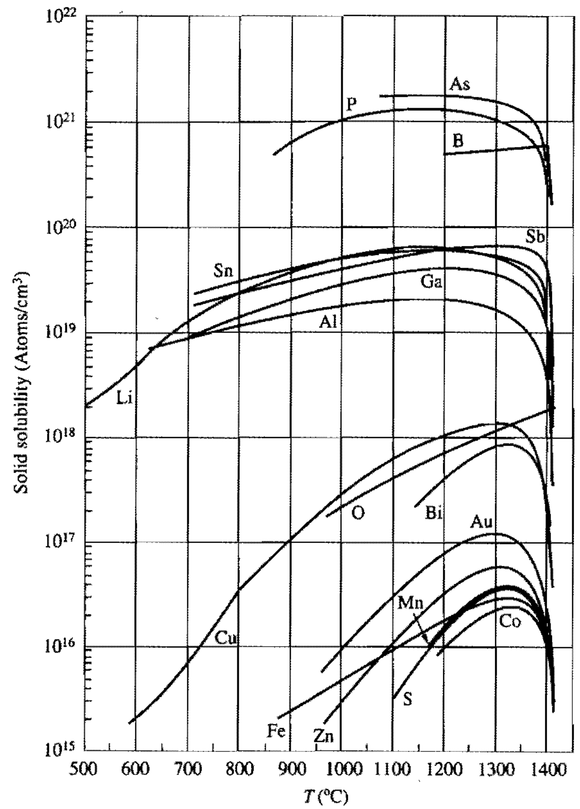

Figure 2.6. Solid solubility of various elements in silicon as a function of temperature. (After Trumbore, 1960.)

2.1.2.2 Fermi Level in Extrinsic Silicon

In contrast to intrinsic silicon, the Fermi level in an extrinsic silicon is not located at the midgap. The Fermi level in n-type silicon moves up towards the conduction band, consistent with the increase in electron density as described by Eq. (2.9). On the other hand, the Fermi level in p-type silicon moves down towards the valence band, consistent with the increase in hole density as described by Eq. (2.11). These cases are depicted in Fig. 2.4. The exact position of the Fermi level depends on both the ionization energy and the concentration of dopants. For example, for an n-type material with a donor impurity concentration N_d, the charge neutrality condition in silicon requires that

$$n = N_d^+ + p, \tag{2.17}$$

where N_d^+ is the density of ionized donors given by

$$N_d^+ = N_d[1 - f_D(E_d)] = N_d\left(1 - \frac{1}{1 + \frac{1}{2}e^{(E_d - E_f)/kT}}\right), \tag{2.18}$$

since the probability that a donor state is occupied by an electron (i.e., in the neutral state) is $f_D(E_d)$. The factor $\frac{1}{2}$ in the denominator of $f_D(E_d)$ arises from the spin degeneracy (up or down) of the available electronic states associated with an ionized donor level[1] (Ghandhi, 1968). Substituting Eq. (2.9) and Eq. (2.11) for n and p in Eq. (2.17), one obtains

$$N_c e^{-(E_c - E_f)/kT} = \frac{N_d}{1 + 2e^{-(E_d - E_f)/kT}} + N_v e^{-(E_f - E_v)/kT}, \tag{2.19}$$

which is an algebraic equation that can be solved for E_f. In n-type silicon, electrons are the majority current carriers, while holes are the minority current carriers, which means that the second term on the right-hand side (RHS) of Eq. (2.19) can be neglected. For shallow donor impurities with low to moderate concentration at room temperature, $(N_d/N_c)\exp[(E_c - E_d)/kT] \ll 1$, a good approximate solution for E_f is

$$E_c - E_f = kT \ln\left(\frac{N_c}{N_d}\right). \tag{2.20}$$

In this case, the Fermi level is at least a few kT below E_d and essentially all the donor levels are empty (ionized), i.e., $n = N_d^+ = N_d$.

It was shown earlier (Eq. (2.16)) that, in equilibrium, the product of majority and minority carrier densities equals n_i^2, independent of the dopant type and Fermi level position. The hole density in n-type silicon is then given by

$$p = n_i^2/N_d. \tag{2.21}$$

Likewise, for p-type silicon with a shallow acceptor concentration N_a, the Fermi level is given by

$$E_f - E_v = kT \ln\left(\frac{N_v}{N_a}\right), \tag{2.22}$$

the hole density is $p = N_a^- = N_a$, and the electron density is

$$n = n_i^2/N_a. \tag{2.23}$$

Figure 2.7 plots the Fermi-level position in the energy gap versus temperature for a wide range of impurity concentration (Grove, 1967). The slight variation of the silicon bandgap with temperature is also incorporated in the figure. It is seen that as the temperature increases, the Fermi level approaches the intrinsic value near midgap. When the intrinsic carrier concentration becomes larger than the doping concentration, the silicon is intrinsic. In an intermediate range of temperature including room temperature, all the donors or acceptors are ionized. The majority carrier concentration is then given by the doping concentration, independent of temperature. For temperatures below this range, *freeze-out* occurs, i.e., the thermal energy is no longer sufficient to ionize all the impurity atoms even with their shallow levels (Sze, 1981). In this case, the

[1] Detailed study showed that there are no other degeneracy with the electronic ground state in a donor except for spin (Ning and Sah, 1971).

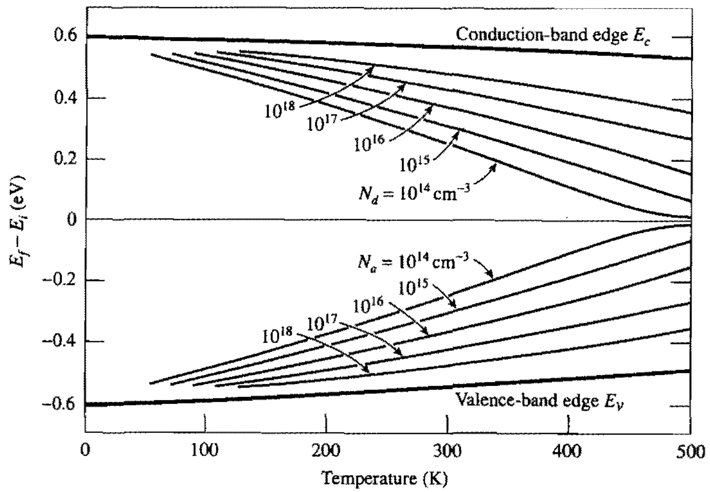

Figure 2.7. The Fermi level in silicon as a function of temperature for various impurity concentrations. (After Grove, 1967.)

majority-carrier concentration is less than the doping concentration, and one would have to solve Eq. (2.19) numerically to find E_f, n, and p (Shockley, 1950).

Instead of using N_c, N_v and referring to E_c and E_v, Eq. (2.20) and Eq. (2.22) can be written in a more useful form in terms of n_i and E_i defined by Eq. (2.12) and Eq. (2.13):

$$E_f - E_i = kT\ln\left(\frac{N_d}{n_i}\right) \tag{2.24}$$

for n-type silicon, and

$$E_i - E_f = kT\ln\left(\frac{N_a}{n_i}\right) \tag{2.25}$$

for p-type silicon. In other words, ***the distance between the Fermi level and the intrinsic Fermi level near the midgap is a logarithmic function of doping concentration***. These expressions will be used extensively throughout the book.

2.1.2.3 Fermi Level in Degenerately Doped Silicon

For heavily doped silicon, the impurity concentration N_d or N_a can exceed the effective density of states N_c or N_v, so that $E_f > E_c$ or $E_f < E_v$ according to Eq. (2.20) and Eq. (2.22). In other words, the Fermi level moves into the conduction band for n^+ silicon, and into the valence band for p^+ silicon. In addition, when the impurity concentration is higher than 10^{18}–10^{19} cm^{-3}, the donor (or acceptor) levels broaden into bands. This results in an effective decrease in the ionization energy until finally the impurity band merges with the conduction (or valence) band and the ionization energy becomes zero. Under these circumstances, the silicon is said to be *degenerate*. Strictly speaking, Fermi statistics

should be used for the electron concentration in calculation of the Fermi level when $E_c - E_f \le kT$ (Ghandhi, 1968). For practical purposes, it is a good approximation [within $(1\text{–}2)kT$] to assume that the Fermi level of the degenerate n^+ silicon is at the conduction-band edge, and that of the degenerate p^+ silicon is at the valence-band edge.

2.1.3 Carrier Transport in Silicon

Carrier transport or current flow in silicon is driven by two different mechanisms: (a) the *drift* of carriers, which is caused by the presence of an electric field, and (b) the *diffusion* of carriers, which is caused by an electron or hole concentration gradient in silicon. The drift current will be discussed first.

2.1.3.1 Drift Current and Mobility

When an electric field is applied to a conducting medium containing free carriers, the carriers are accelerated and acquire a drift velocity superimposed upon their random thermal motion. This is described in more detail in Appendix 3. The drift velocity of holes is in the direction of the applied field, and the drift velocity of electrons is opposite to the field. The velocity of the carriers does not increase indefinitely under field acceleration, since they are scattered frequently and lose their acquired momentum after each collision. At low electric fields, the drift velocity v_d is proportional to the electric field strength $\mathscr{E}$ with a proportionality constant μ, defined as the *mobility*, in units of cm^2/V-s, i.e.,

$$v_d = \mu\mathscr{E}. \tag{2.26}$$

The mobility is proportional to the time interval between collisions and is inversely proportional to the effective mass of the carriers (Appendix 3). Electron and hole mobilities in silicon at low impurity concentrations are listed in Table 2.1. The electron mobility is approximately three times the hole mobility, since the effective mass of electrons in the conduction band is much lighter than that of holes in the valence band.

Figure 2.8 plots the electron and hole mobilities at room temperature versus n-type or p-type doping concentration. At low impurity levels, the mobilities are mainly limited by carrier collisions with the silicon lattice or acoustic phonons (Kittel, 1976). As the doping concentration increases beyond 10^{15}–10^{16}/cm^3, collisions with the charged (ionized) impurity atoms through Coulomb interaction become more and more important and the mobilities decrease. In general, one can use *Matthiessen's rule* to include different contributions to the mobility:

$$\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I} + \cdots, \tag{2.27}$$

where μ_L and μ_I correspond to the lattice- and impurity-scattering-limited components of mobility, respectively. At high temperatures, the mobility tends to be limited by lattice scattering and is proportional to $T^{-3/2}$, relatively insensitive to the doping concentration (Sze, 1981). At low temperatures, the mobility is higher, but is a strong function of

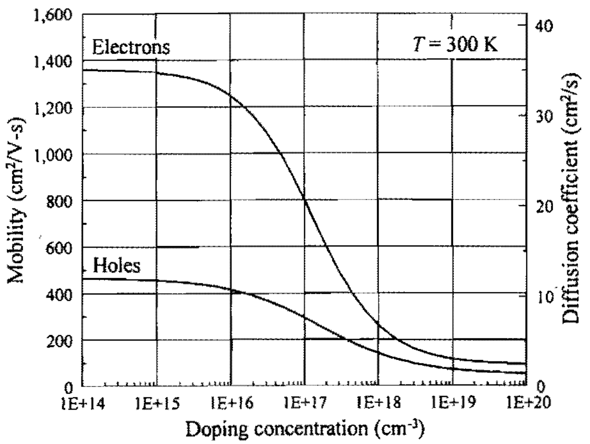

Figure 2.8. Electron and hole mobilities in bulk silicon at 300 K as a function of doping concentration.

doping concentration as it becomes more limited by impurity scattering. What is shown in Fig. 2.8 is the *bulk mobility* applicable to conduction in silicon substrates far from the surface. ***In the inversion layer of a MOSFET device, the current flow is governed by the surface mobility, which is much lower than the bulk mobility.*** This is mainly due to additional scattering mechanisms between the carriers and the $Si–SiO_2$ interface in the presence of high electric fields normal to the surface. Surface scattering adds another term to Eq. (2.27). Carrier mobility in the surface inversion channel of a MOSFET will be discussed in more detail in Section 3.1.5.

2.1.3.2 Resistivity

For a homogeneous n-type silicon with a free-electron density n, the drift current density under an electric field $\mathscr{E}$ is

$$J_{n,drift} = qnv_d = qn\mu_n\mathscr{E}, \tag{2.28}$$

where $q = 1.6 \times 10^{-19}$ C is the electronic charge and μ_n is the electron mobility. The resistivity, ρ_n, of n-type silicon defined by $J_{n,drift} = \mathscr{E}/\rho_n$ (a form of Ohm's law) is then given by

$$\rho_n = \frac{1}{qn\mu_n}. \tag{2.29}$$

Similarly, for p-type silicon,

$$J_{p,drift} = qp\mu_p\mathscr{E} \tag{2.30}$$

and

$$\rho_p = \frac{1}{qp\mu_p}, \tag{2.31}$$

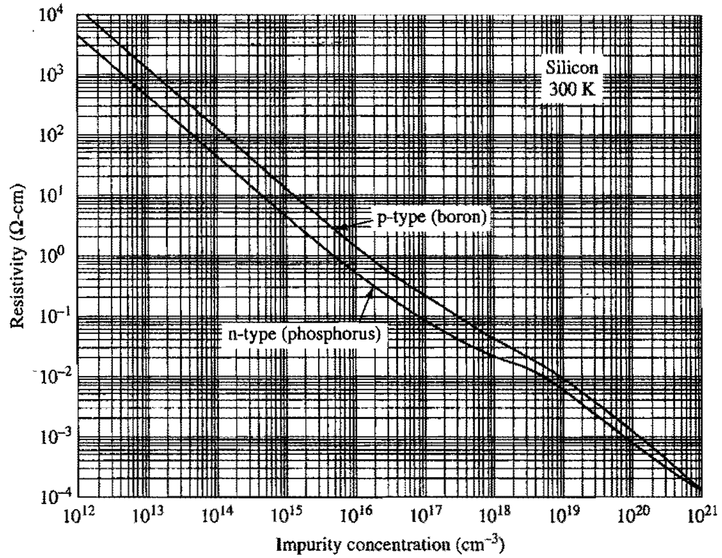

Figure 2.9. Resistivity versus impurity concentration for n-type and p-type silicon at 300 K. (After Sze, 1981.)

where μ_p is the hole mobility. In general, the total resistivity should include both the majority and the minority carrier components:

$$\rho = \frac{1}{qn\mu_n + qp\mu_p}, \tag{2.32}$$

since both electrons and holes contribute to electrical conduction. Figure 2.9 shows the measured resistivity of n-type (phosphorus-doped) and p-type (boron-doped) silicon versus impurity concentration at room temperature.

2.1.3.3 Sheet Resistivity

The resistance of a uniform conductor of length L, width W, and thickness t is given by

$$R = \rho\frac{L}{Wt}, \tag{2.33}$$

where ρ is the resistivity in ohm-centimeters. In a planar IC technology, the thickness of conducting regions is uniform and normally much less than both the length and the width of the regions. It is then useful to define a quantity, called the *sheet resistivity*, as

$$\rho_{sh} = \frac{\rho}{t} \tag{2.34}$$

in units of $\Omega/\square$ (ohms per square). Then

$$R = \frac{L}{W}\rho_{sh}, \tag{2.35}$$

i.e., the total resistance is equal to the number of squares ($L/W = 1$ is one square) of the line times the sheet resistivity. Note that sheet resistivity does not depend on the size of the square. The most common technique of measuring the sheet resistivity of a thin film is the four-point method, in which a small current is passed through the two outer probes and the voltage is measured between the two inner probes (Sze, 1981). If the spacing between the probes is much greater than the film thickness but much smaller than the linear dimension of the conducting film, the resistance measured can be approximated by $V/I = \rho_{sh}(\ln 2)/\pi \approx 0.22\rho_{sh}$, from which ρ_{sh} can be easily determined.

2.1.3.4 Velocity Saturation

The linear velocity–field relationship discussed above is valid only when the electric field is not too high and the carriers are in thermal equilibrium with the lattice. At high fields, the average carrier energy increases and carriers lose their energy by optical-phonon emission nearly as fast as they gain it from the field. This results in a decrease of the mobility as the field increases until finally the drift velocity reaches a limiting value, $v_{sat} \approx 10^7$ cm/s. This phenomenon is called *velocity saturation.*

Figure 2.10 shows the measured velocity–field relationship of electrons and holes in high-purity bulk silicon at room temperature. At low fields, the drift velocity is proportional to the field (45° slope on a log–log scale) with a proportionality constant given by the electron or the hole mobility. When the field becomes higher than 3×10^3 V/cm for electrons, velocity saturation starts to occur. The saturation velocity of holes is similar to or slightly lower than that of electrons, but saturation for holes takes place at a much higher field because of their lower mobility. For more highly doped material, low-field mobilities are lower because of impurity scattering (Fig. 2.8). However, the saturation velocity remains essentially the same, independent of impurity concentration. There is a weak dependence of v_{sat} on temperature. It decreases slightly as the temperature increases (Arora, 1993).

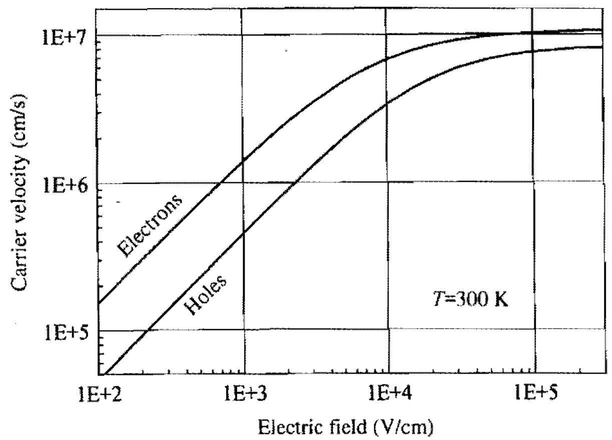

Figure 2.10. Velocity–field relationship of electrons and holes in silicon at 300 K.

2.1.3.5 Diffusion Current

The discussion in this section so far has dealt only with the case when the carrier concentration within the silicon is uniform and the carriers move under the influence of an electric field. ***If the carrier concentration is not uniform, carriers will also diffuse as a result of the concentration gradient.*** This leads to an additional current contribution in proportion to the concentration gradient:

$$J_{n,diff} = q\,D_n \frac{dn}{dx} \tag{2.36}$$

for electrons, and

$$J_{p,diff} = -q\,D_p \frac{dp}{dx} \tag{2.37}$$

for holes. The proportionality constants D_n and D_p are called the electron and hole *diffusion coefficients* and have units of cm^2/s. There is a negative sign in Eq. (2.37), since diffusion current flows in the direction of decreasing hole (positive charge) concentration.

Physically, both drift and diffusion are closely associated with the random thermal motion of carriers and their collisions with the silicon lattice in thermal equilibrium. A simple relationship between the diffusion coefficient and the mobility is derived from the basic principles in Appendix 3 (Muller and Kamins, 1977):

$$D_n = \frac{kT}{q}\mu_n \tag{2.38}$$

for electrons, and

$$D_p = \frac{kT}{q}\mu_p \tag{2.39}$$

for holes. These are known as the *Einstein relations.* The values of diffusion coefficient at room temperature can be read from Fig. 2.8 using the vertical scale on the right-hand side.

2.1.4 Basic Equations for Device Operation

2.1.4.1 Poisson's Equation

One of the key equations governing the operation of VLSI devices is *Poisson's equation.* It comes from Maxwell's first equation, which in turn is based on Coulomb's law for electrostatic force of a charge distribution. Poisson's equation is expressed in terms of the electrostatic potential, which is defined as the potential energy of carriers divided by the electronic charge q. The potential energies of carriers are either at the conduction-band edge or at the valence-band edge, as discussed before in connection with the energy-band diagram, Fig. 2.1. Since one is only interested in the spatial variation of the electrostatic potential, it can be defined with an arbitrary additive constant. It makes no difference whether E_c, E_v, or any other quantity displaced from the band edges by a fixed amount is used to represent the potential. Conventionally, the electrostatic potential is defined in terms of the intrinsic Fermi level,

$$\psi_i = -\frac{E_i}{q}. \tag{2.40}$$

There is a negative sign because E_i is defined as electron energy while ψ_i is defined for a positive charge. The band diagram can thus be considered also as a potential diagram with the potential increasing downward, opposite to the electron energy.

The electric field $\mathscr{E}$, which is defined as the electrostatic force per unit charge, is equal to the negative gradient of ψ_i,

$$\mathscr{E} = -\frac{d\psi_i}{dx}. \tag{2.41}$$

Now we can write Poisson's equation as

$$\frac{d^2\psi_i}{dx^2} = -\frac{d\mathscr{E}}{dx} = -\frac{\rho_{net}(x)}{\varepsilon_{si}}, \tag{2.42}$$

where $\rho_{net}(x)$ is the net charge density per unit volume at x, and ε_{si} is the permittivity of silicon equal to $11.7\varepsilon_0$. Here $\varepsilon_0 = 8.85 \times 10^{-14}$ F/cm is the vacuum permittivity.

Another form of Poisson's equation is *Gauss's law*, which is obtained by integrating Eq. (2.42):

$$\mathscr{E} = \frac{1}{\varepsilon_{si}}\int \rho_{net}(x)\,dx = \frac{Q_s}{\varepsilon_{si}}, \tag{2.43}$$

where Q_s is the integrated charge density per unit area.

There are two sources of charge in silicon: *mobile charge* and *fixed charge*. Mobile charges are electrons and holes, whose densities are represented by n and p. Fixed charges are ionized donor (positively charged) and acceptor (negatively charged) atoms whose densities are represented by N_d^+ and N_a^-, respectively. Equation (2.42) can then be written as

$$\frac{d^2\psi_i}{dx^2} = -\frac{d\mathscr{E}}{dx} = -\frac{q}{\varepsilon_{si}}[p(x) - n(x) + N_d^+(x) - N_a^-(x)]. \tag{2.44}$$

For a homogeneous n-type or p-type silicon with no applied field, the RHS of Eq. (2.44) is zero and the potential is constant throughout the sample.

2.1.4.2 Dielectric Boundary Conditions

Equation (2.42) is one dimensional and is for a homogeneous material, silicon. It is adequate for describing most of the basic device operations. In some cases, e.g., in short-channel MOSFETs, the two-dimensional Poisson's equation is needed. In addition, the device could consist of two different materials with different dielectric constants, as depicted in Fig. 2.11. There are two basic boundary conditions for the components of the electric field across a dielectric interface.

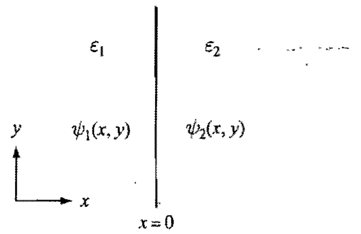

Figure 2.11. Diagram for discussing the boundary conditions of electric field at the interface between two dielectric media.

The two-dimensional Poisson's equation takes the following general form:

$$\frac{\partial(\varepsilon\mathscr{E}_x)}{\partial x} + \frac{\partial(\varepsilon\mathscr{E}_y)}{\partial y} = \rho_{net}, \tag{2.45}$$

where $\mathscr{E}_x = -\partial\psi/\partial x$ and $\mathscr{E}_y = -\partial\psi/\partial y$ are the components of the electric field perpendicular and parallel to the dielectric boundary, respectively. Note that in the geometry defined in Fig. 2.11, ε is a step function of x only, independent of y. If the potential functions in the two dielectric regions are represented by $\psi_1(x, y)$ and $\psi_2(x, y)$, they must be continuous at the interface, i.e., $\psi_1(0, y) = \psi_2(0, y)$. It follows that their y-derivatives or ***the tangential fields are also continuous***,

$$\mathscr{E}_{1y}(0, y) = \mathscr{E}_{2y}(0, y). \tag{2.46}$$

For a finite volume charge density ρ_{net}, the x-derivative in Eq. (2.45) must be finite at $x = 0$. This means that $\varepsilon\mathscr{E}_x$ must be continuous in the x-direction across the dielectric boundary. Therefore,

$$\varepsilon_1\mathscr{E}_{1x}(0, y) = \varepsilon_2\mathscr{E}_{2x}(0, y), \tag{2.47}$$

i.e., ***the perpendicular component of the displacement, $D = \varepsilon\mathscr{E}$, is continuous***. In the presence of a sheet charge at the dielectric interface, ρ_{net} can be considered as a delta function with an area equal to the interface charge density per unit area, Q_{it}. Then the last condition changes to $\varepsilon_2\mathscr{E}_{2x} - \varepsilon_1\mathscr{E}_{1x} = Q_{it}$. Interface trapped charge in an MOS capacitor is discussed in Section 2.3.6.

2.1.4.3 Carrier Concentration as a Function of Electrostatic Potential

Many of the parameters discussed before can be expressed in terms of the electrostatic potential ψ_i. For example, Eq. (2.24) and Eq. (2.25) can be combined into one equation for both n-type and p-type silicon:

$$\psi_B \equiv |\psi_f - \psi_i| = \frac{kT}{q}\ln\left(\frac{N_b}{n_i}\right), \tag{2.48}$$

where $\psi_f = -E_f/q$ is the Fermi potential and N_b is either the donor or the acceptor concentration. Equation (2.48) is a very useful expression relating the separation of the

Fermi potential from the midgap, ψ_B, to the doping concentration (or the ionized dopant concentration in the case of incomplete ionization). It is based on charge neutrality and is valid only if the local mobile carrier concentration equals the ionized dopant concentration, i.e., only if the RHS of Eq. (2.44) is zero and the field is either zero or constant.

In general, for any Fermi level position in the bandgap, carrier densities are given by Eqs. (2.14) and (2.15), which can be expressed in terms of the electrostatic potential:

$$n = n_i e^{q(\psi_i - \psi_f)/kT} \tag{2.49}$$

and

$$p = n_i e^{q(\psi_f - \psi_i)/kT}. \tag{2.50}$$

The last two equations are often referred to as *Boltzmann's relations* and are valid for either n-type or p-type silicon in thermal equilibrium. Note that Eqs. (2.49) and (2.50) are derived from Fermi level and density of states considerations, regardless of charge neutrality. They are generally applicable in the presence of net charge (due to an imbalance between the mobile and the fixed charge densities) and band bending (spatial variation of ψ_i) where $|\psi_f - \psi_i|$ is no longer given by Eq. (2.48).

2.1.4.4 Debye Length

In an inhomogeneous silicon, the doping concentration varies spatially. The bands (E_c, E_v, E_i) are not flat as in the uniformly doped case. Both the intrinsic Fermi level and the bands generally follow the doping variation according to Eq. (2.48). However, if the doping concentration changes abruptly on a very short length scale, the bands do not respond immediately, since both ψ_i and its first-order spatial derivative are continuous owing to the thermal diffusion effects of mobile carriers. The length scale in an n-type silicon, for example, can be estimated by substituting Eq. (2.49) into Eq. (2.44):

$$\frac{d^2\psi_i}{dx^2} = -\frac{q}{\varepsilon_{si}}\left[N_d(x) - n_i e^{q(\psi_i - \psi_f)/kT}\right]. \tag{2.51}$$

Without any applied bias or current, the Fermi level is spatially flat, i.e., ψ_f is independent of x, as discussed in Section 2.1.1.3. For an incremental change of doping concentration $\Delta N_d(x)$ with respect to a uniformly doped background, the corresponding change in the intrinsic potential $\Delta\psi_i(x)$ can be found by expanding the exponential term in Eq. (2.51) and keeping only the first-order term (zeroth-order terms have no spatial dependence):

$$\frac{d^2(\Delta\psi_i)}{dx^2} - \frac{q^2 N_d}{\varepsilon_{si} kT}\Delta\psi_i = -\frac{q}{\varepsilon_{si}}\Delta N_d(x). \tag{2.52}$$

This is a second-order differential equation whose solution $\Delta\psi_i$ takes the form $\exp(-x/L_D)$, where

$$L_D \equiv \sqrt{\frac{\varepsilon_{si} kT}{q^2 N_d}} \tag{2.53}$$

is called the *Debye length*. Physically, this means that ***it takes a distance on the order of L_D for the silicon bands to respond to an abrupt change in N_d***. A small electric field is set up in this region due to the charge imbalance. The Debye length is usually much smaller than the lateral device dimension. For example, L_D = 0.04 μm for $N_d = 10^{16}$ cm^{-3}.

2.1.4.5 Current-Density Equations

The next set of equations are current-density equations. The total current density is the sum of the drift current density given by Eq. (2.28) and Eq. (2.30) and the diffusion current density given by Eq. (2.36) and Eq. (2.37). In other words,

$$J_n = qn\mu_n\mathscr{E} + qD_n\frac{dn}{dx} \tag{2.54}$$

for the electron current density, and

$$J_p = qp\mu_p\mathscr{E} - qD_p\frac{dp}{dx} \tag{2.55}$$

for the hole current density. The total conduction current density is $J = J_n + J_p$.

Using Eq. (2.41) and the Einstein relations, Eq. (2.38) and Eq. (2.39), one can write the current densities as

$$J_n = -qn\mu_n\left(\frac{d\psi_i}{dx} - \frac{kT}{qn}\frac{dn}{dx}\right) \tag{2.56}$$

and

$$J_p = -qp\mu_p\left(\frac{d\psi_i}{dx} + \frac{kT}{qp}\frac{dp}{dx}\right). \tag{2.57}$$

If both n and p take on their equilibrium values, Eqs. (2.49) and (2.50) can be substituted into the above to yield:

$$J_n = -qn\mu_n\frac{d\psi_f}{dx} \tag{2.58}$$

and

$$J_p = -qp\mu_p\frac{d\psi_f}{dx}. \tag{2.59}$$

The total current is then

$$J = J_n + J_p = -\frac{1}{\rho}\frac{d\psi_f}{dx}, \tag{2.60}$$

where ρ is the resistivity of silicon given by Eq. (2.32). Equation (2.60) resembles Ohm's law, $J_{drift} = \mathscr{E}/\rho$, discussed before. With the addition of the diffusion current, ***the total current is proportional to the gradient of the Fermi potential*** instead of proportional to the electric field, $\mathscr{E} = -d\psi_i/dx$. This reinforces and goes farther than the concept on Fermi level and equilibrium discussed in Section 2.1.1.3: for a connected system of metals and/or semiconductors in thermal equilibrium with no current flow, the Fermi level is flat, i.e., spatially constant, throughout the system. It is important to keep in mind that Fermi level difference is the driving force for current flow, much like voltage difference drives currents in a circuit.

Strictly speaking, when current flows, the system is not in equilibrium and the Fermi level is not well defined. The electron distribution function is no longer a function of energy only. It becomes asymmetric in the current flow direction to favor population of the electronic states with a forward momentum. However, if the current is not too large and the net velocity of electron transport is small compared with the thermal velocity, there is only a slight departure from equilibrium. It is then useful to consider a *local* Fermi level, $E_f(x)$ or $\psi_f(x)$, based on the *local* equilibrium state at any given point. In this way, we generalize the Fermi level concept so that the current density equations (2.58) and (2.59) are valid as long as the *local* electron and hole densities are equal to their equilibrium values, Eqs. (2.49) and (2.50).

When an external battery or voltage source is connected to a device, it pumps all the electrons and states at one contact to a higher energy with respect to another contact. By the definition of contacts, both the electron and hole densities equal their equilibrium values at the contacts; therefore, one can define Fermi levels, e.g., E_{f1} and E_{f2}, respectively for the two contacts being considered. Without any externally applied voltage, $E_{f1} = E_{f2}$. ***When a voltage source is connected,*** $E_{f1} - E_{f2} = qV_{app}$, ***where*** V_{app} ***is the applied voltage*** with the lower voltage (higher electron energy) side connected to contact 1. It is this Fermi level imbalance at the contacts, sustained by the external power source, that drives a steady state current in the device. Fermi level will be used to reference terminal voltage and to establish the alignment relationship of electronic energy bands throughout the book.

2.1.4.6 Quasi-Fermi Potentials

The above discussion applies only when both the electron and hole densities take on their local equilibrium values and a local Fermi level can be defined. It is often in VLSI device operation to encounter nonequilibrium situations where the densities of one or both types of carriers depart from their equilibrium values given by Eqs. (2.49) and (2.50). In particular, the minority carrier concentration can be easily overwhelmed by injection from neighboring regions. This happens on a distance scale much larger than VLSI device dimensions. It results from the slow generation–recombination processes (discussed in the next section) that are inefficient to establish equilibrium between electrons and holes. Under these circumstances, while the electrons are in local equilibrium with themselves and so are the holes, electrons and holes are not in equilibrium with each other. In order to extend the kind of relationship between Fermi level and current densities discussed above, one can introduce *separate* Fermi levels for electrons and holes, respectively. They are called *quasi-Fermi levels*, E_{fn} and E_{fp}, defined so as to replace E_f in Eqs. (2.14) and (2.15):

$$n = n_i e^{(E_{fn}-E_i)/kT}, \tag{2.61}$$

$$p = n_i e^{(E_i-E_{fp})/kT}. \tag{2.62}$$

In this regard, quasi-Fermi levels have a similar physical interpretation in terms of the state occupancy as the Fermi level. ***That is, the electron density in the conduction band can be calculated as if the Fermi level is at*** E_{fn}, ***and the hole density in the valence band can be calculated as if the Fermi level is at*** E_{fp}.[2] With these definitions, the current densities, Eqs. (2.56) and (2.57), become

$$J_n = -qn\mu_n \frac{d\phi_n}{dx} \tag{2.63}$$

and

$$J_p = -qp\mu_p \frac{d\phi_p}{dx}, \tag{2.64}$$

where the *quasi-Fermi potentials* ϕ_n and ϕ_p are defined by

$$\phi_n \equiv -\frac{E_{fn}}{q} = \psi_i - \frac{kT}{q}\ln\left(\frac{n}{n_i}\right) \tag{2.65}$$

and

$$\phi_p \equiv -\frac{E_{fp}}{q} = \psi_i + \frac{kT}{q}\ln\left(\frac{p}{n_i}\right). \tag{2.66}$$

Equations (2.63) and (2.64) are more generally applicable than Eqs. (2.58) and (2.59). They show that electron and hole currents are driven by separate entities: ***the gradient of electron quasi-Fermi potential drives the electron current, and the gradient of hole quasi-Fermi potential drives the hole current.*** When current flows, $E_{fn}(x)$ and $E_{fp}(x)$ (or $\phi_n(x)$ and $\phi_p(x)$) should be interpreted in the same sense of *local* equilibrium at point x as with the case of local $E_f(x)$ discussed earlier.

When electrons and holes are not in equilibrium with each other, the *pn* product is given by

$$pn = n_i^2 e^{q(\phi_p-\phi_n)/kT}. \tag{2.67}$$

It equals n_i^2 when $\phi_p = \phi_n = \psi_f$. Quasi-Fermi potentials are used extensively in the rest of the book for current calculations.

2.1.4.7 Continuity Equations

The next set of equations are *continuity equations* based on the conservation of mobile charge:

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - R_n + G_n \tag{2.68}$$

[2] While it appears to be physically inconsistent to have half of the electron states occupied at one energy and half of the electron states empty at a different energy, quasi-Fermi levels are defined mainly for mathematical convenience.

and

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - R_p + G_p, \tag{2.69}$$

where G_n and G_p are the electron and hole generation rates, R_n and R_p are the electron and hole recombination rates, and $\partial J_n/\partial x$ and $\partial J_p/\partial x$ are the net flux of mobile charges in and out of x. At thermal equilibrium, the generation rate is equal to the recombination rate and $np = n_i^2$. When excess minority carriers are injected by light or other means, the recombination rate exceeds the generation rate, which establishes a tendency to return to equilibrium.

In silicon, the probability of direct band-to-band recombination by a *radiative* (transfer of energy to a photon) or *Auger* (transfer of energy to another carrier) process is very low due to its indirect bandgap. Most of the recombination processes take place indirectly via a trap or a deep impurity level near the middle of the forbidden gap. This is often referred to as the *Shockley–Read* recombination (Shockley and Read, 1952). Under low-injection conditions, the recombination rate is inversely proportional to the *minority-carrier lifetime*, τ, which is in the range of 10^{-4} to 10^{-9} s, depending on the quality of the silicon crystal. The *minority-carrier diffusion length*, which is the average distance a minority carrier travels before it recombines with a majority carrier, is given by $L = (D\tau)^{1/2}$, where D is the diffusion coefficient. The diffusion length is typically a few microns to a few millimeters in silicon. (A discussion of the minority-carrier diffusion process can be found in Section 2.2.4.) ***Since L is much larger than the active dimensions of a VLSI device, generation–recombination in general plays very little role in device operation.*** Only in a few special circumstances, such as CMOS latch-up, the SOI floating-body effect, junction leakage current, and radiation-induced soft error, must the generation–recombination mechanism be taken into account.

More detailed discussions on generation and recombination can be found in Appendix 5. In the steady state, $\partial n/\partial t = \partial p/\partial t = 0$. Also, the net electron reduction rate must equal the net hole reduction rate, i.e., $R_n - G_n = R_p - G_p$, so that there is no buildup of net immobile charge with time at any point. Subtracting Eq. (2.69) from Eq. (2.68) then yields $\partial(J_n + J_p)/\partial x = 0$, or continuity of the total current, $J_n + J_p$. In a device region where generation and recombination are negligible, the continuity equations in the steady state are reduced to $dJ_n/dx = dJ_p/dx = 0$, which simply states the conservation of electron current and conservation of hole current, respectively.

2.1.4.8 Dielectric Relaxation Time

In contrast to the minority-carrier lifetime discussed above, the majority-carrier response time is very short in a semiconductor. It can be estimated for a one-dimensional (1-D) homogeneous n-type silicon as follows. Suppose there is a local perturbation in the majority carrier density, Δn. From Poisson's equation, the resulting charge imbalance sets up a field divergence, $\partial\mathscr{E}/\partial x = -q\Delta n/\varepsilon_{si}$, around the point of perturbation. This, in turn, leads to a divergent current according to Ohm's law, $J_n = \mathscr{E}/\rho_n$, which tends to restore the majority carrier concentration back to its equilibrium, charge neutral value. Neglecting R_n and G_n in the continuity equation, Eq. (2.68), one then obtains

$$\frac{\partial\Delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} = \frac{1}{q\rho_n}\frac{\partial\mathscr{E}}{\partial x} = -\frac{\Delta n}{\rho_n\varepsilon_{si}}. \tag{2.70}$$

The solution to this equation takes the form of $\Delta n(t) \propto \exp(-t/\rho_n\varepsilon_{si})$, where $\rho_n\varepsilon_{si}$ is the majority-carrier response time, or *dielectric relaxation time*. ***The majority-carrier response time in silicon is typically on the order of 10^{-12} s, which is shorter than most device switching times.***

Note that $\rho_n\varepsilon_{si}$ is the minimum response time for an ideal 1-D case without any parasitic capacitances. In practice, the majority-carrier response time may be limited by the RC delay of the specific silicon device structure and contacts.

2.2 p–n Junctions

p–n junctions, also called *p–n diodes*, are important devices as well as important components of all MOSFET and bipolar devices. The characteristics of p–n diodes are therefore important in determining the characteristics of VLSI devices and circuits. A p–n diode is formed when one region of a semiconductor substrate is doped n-type and an immediately adjacent region is doped p-type. In practice, a silicon p–n diode is usually formed by counterdoping a local region of a larger region of doped silicon. For instance, a region of a p-type silicon substrate or "well" can be counterdoped with n-type impurities to form the n-type region of a p–n diode. The n-type region thus formed has a donor concentration higher than its acceptor concentration.

A doped semiconductor region is called *compensated* if it contains both donor and acceptor impurities such that neither impurity concentration is negligible compared to the other. For a compensated semiconductor region, it is the *net* doping concentration, i.e., $N_d - N_a$ if it is n-type and $N_a - N_d$ if it is p-type, that determines its Fermi level and its mobile carrier concentration. However, for simplicity, we shall derive the characteristics and behavior of p–n diodes assuming none of the doped regions are compensated, i.e., the n-sides of the diodes have a net donor concentration of N_d and the p-sides have a net acceptor concentration of N_a. The resultant equations can be extended to diodes with compensated doped regions simply by replacing N_d by $N_d - N_a$ for the n-regions and replacing N_a by $N_a - N_d$ for the p-regions.

2.2.1 Energy-Band Diagrams for a p–n Diode

It was shown in Sections 2.1.1.3 and 2.1.4.5 that at thermal equilibrium or when there is no net electron or hole current, the Fermi level is spatially constant. Furthermore, when an external voltage V_{app} is connected to two contacts of a piece of silicon, the Fermi level at the lower voltage contact is shifted relative to the Fermi level at the higher voltage contact by qV_{app}. In this section, we apply these results to establish the energy-band diagrams for a p–n diode under various bias conditions.

Consider a p-silicon region and an n-silicon region physically separate from each other. As discussed in Section 2.1.2.2, the Fermi level for a p-type silicon lies close to its

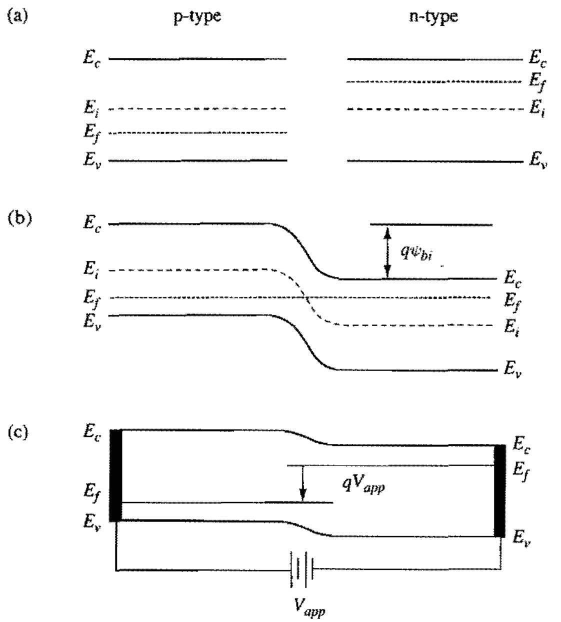

Figure 2.12. Energy-band diagrams for a p–n diode. (a) A p-silicon region and an n-silicon region physically separate from each other. (b) A p–n junction at thermal equilibrium. (c) A p–n diode connected to a battery, with the n-side connected to the negative end and the p-side connected to the positive end of the battery. The solid vertical bars represent the ohmic contacts of the p- and n-regions. For simplicity and clarity of the figure, E_i is not shown.

valence band, and that for an n-type silicon lies close to its conduction band. The energy-band diagrams for the two silicon pieces are illustrated schematically in Fig. 2.12(a).

If the p-silicon and the n-silicon are brought together to form a p–n diode, the resulting energy-band diagram is as shown in Fig. 2.12(b). At thermal equilibrium, the Fermi level must remain flat across the entire p–n diode structure, causing the energy bands of the p-region to lie higher than those of the n-region. Near the physical junction, the energy-bands are bent in order to maintain energy-band continuity between the p-region and the n-region. The band bending implies an electric field, $\mathscr{E} = -d\psi_i/dx$, in this transition region. This electric field causes a drift component of electron and hole currents to flow. At thermal equilibrium, this drift-current component is exactly balanced by a diffusion component of electron and hole currents flowing in the opposite direction caused by the large electron and hole concentration gradients across the junction. The net result is zero electron and hole currents across the p–n junction at thermal equilibrium. On both sides of the band-bending region, the energy bands are flat and there is no electric field. These regions are referred to as the *quasineutral regions*.

If a battery of voltage V_{app} is connected to the diode, with the p-side connected to the positive end of the battery and the n-side connected to the negative end of the battery, the

Fermi level at the n-side contact becomes shifted by qV_{app} relative to the Fermi level at the p-side contact. This is illustrated in Fig. 2.12(c).

2.2.1.1 Built-in Potential

Consider the energy-band diagram in Fig. 2.12(b). The difference between the energy bands on the p-side and the corresponding energy bands on the n-side is E_c(p-side) $- E_c$ (n-side) $= q\psi_{bi}$, where ψ_{bi} is the *built-in potential* of the p–n junction. In this subsection, we want to establish the relationship between ψ_{bi} and the p- and n-side doping concentrations.

To facilitate description of both the n-side and the p-side of a diode simultaneously, when necessary for clarity, we shall distinguish the parameters on the n-side from the corresponding ones on the p-side by adding a subscript n to the symbols associated with the parameters on the n-side, and a subscript p to the symbols associated with the parameters on the p-side (Shockley, 1950). For example, E_{fn} and E_{in} denote the Fermi level and intrinsic Fermi level, respectively, on the n-side, and E_{fp} and E_{ip} denote the Fermi level and intrinsic Fermi level, respectively, on the p-side. Similarly, n_n and p_n denote the electron concentration and hole concentration, respectively, on the n-side, and n_p and p_p denote the electron concentration and hole concentration, respectively, on the p-side. Thus, n_n and p_p signify majority-carrier concentrations, while n_p and p_n signify minority-carrier concentrations.

Consider the n-side of a p–n diode at thermal equilibrium. If the n-side is non-degenerately doped to a concentraon of N_d, then the separation between its Fermi level, which is flat across the diode, and its intrinsic Fermi level is given by Eq. (2.24), namely

$$E_{fn} - E_{in} = kT\ln\left(\frac{n_{n0}}{n_i}\right) = kT\ln\left(\frac{N_d}{n_i}\right), \tag{2.71}$$

where n_{n0} denotes the n-side electron concentration at thermal equilibrium.

Similarly, for the nondegenerately doped p-side of a p–n diode at thermal equilibrium, with a doping concentration of N_a, we have

$$E_{ip} - E_{fp} = kT\ln\left(\frac{p_{p0}}{n_i}\right) = kT\ln\left(\frac{N_a}{n_i}\right), \tag{2.72}$$

where p_{p0} is the p-side hole concentration at thermal equilibrium.

The built-in potential across the p–n diode is

$$q\psi_{bi} = E_{ip} - E_{in} = kT\ln\left(\frac{n_{n0}p_{p0}}{n_i^2}\right). \tag{2.73}$$

Since Eq. (2.16) gives $n_{n0}p_{n0} = n_{p0}p_{p0} = n_i^2$, Eq. (2.73) can also be written as

$$q\psi_{bi} = kT\ln\left(\frac{p_{p0}}{p_{n0}}\right) = kT\ln\left(\frac{n_{n0}}{n_{p0}}\right), \tag{2.74}$$

which relates the built-in potential to the electron and hole densities on the two sides of the p–n diode.

2.2.2 Abrupt Junctions

Analysis of a p–n diode is much simpler if the junction is assumed to be abrupt, i.e., the doping impurities are assumed to change abruptly from p-type on one side to n-type on the other side of the junction. The abrupt-junction approximation is reasonable for modern VLSI devices, where the use of ion implantation for doping the junctions, followed by low-thermal-cycle diffusion and/or annealing, results in junctions that are fairly abrupt. Besides, the abrupt-junction approximation often leads to closed-form solutions which render the device physics much easier to understand.

2.2.2.1 Depletion Approximation

The spatial dependence of the electrostatic potential $\psi_i(x)$ is governed by Poisson's equation, i.e., Eq. (2.44). For a diode at thermal equilibrium, the electron and hole densities, $n(x)$ and $p(x)$, are given by Eqs. (2.49) and (2.50), respectively. As suggested in Fig. 2.12(b), $\psi_i(x)$ is independent of x in the uniformly doped quasineutral regions. Within the band-bending region, $\psi_i(x)$ changes from being $-E_{ip}/q$ at the p-region end of the band-bending region to being $-E_{in}/q$ at the n-region end of the band-bending region. Within the band-bending region, Eq. (2.49) suggests that the electron density drops very rapidly as $\psi_i(x)$ changes, being equal to the ionized donor density at the n-region end and dropping $10\times$ at room temperature for every 60 mV change in $\psi_i(x)$. Thus, the density of electrons within the band-bending region is negligible compared to the density of ionized donors except for a very narrow region adjacent the quasineutral n-region where $q(\psi_i - \psi_f)$ is less than about $3kT$. Similarly, Eq. (2.50) suggests that, within the band-bending region, the density of holes is negligible compared to the density of ionized acceptors except for a very narrow region adjacent to the quasineutral p-region.

A closed-form solution to Poisson's equation can be obtained if the electron and hole densities are assumed to be negligible in the entire band-bending region. This is called the *depletion approximation*. In this case, the abrupt junction is approximated by three regions as illustrated in Fig. 2.13(a). Both the quasineutral p-region, i.e., the region with $x < -x_p$ and the quasineutral n-region, i.e., the region with $x > x_n$, are assumed to be charge-neutral, while the transition region, i.e., the region with $-x_p < x < x_n$, is assumed to be depleted of mobile electrons and holes. As we shall show later, the depletion-layer widths, x_p and x_n, are dependent on the donor concentration N_d on the n-side and the acceptor concentration N_a on the p-side, as well as on the applied voltage V_{app} across the junction. The depletion approximation is quite accurate for all applied voltages except at large forward biases, where the mobile-charge densities are not negligible compared to the ionized impurity concentrations in the transition region. The transition region is often referred to as the *depletion region* or *depletion layer*. Since the transition region is not charge-neutral, it is also referred to as the *space-charge region* or *space-charge layer*.

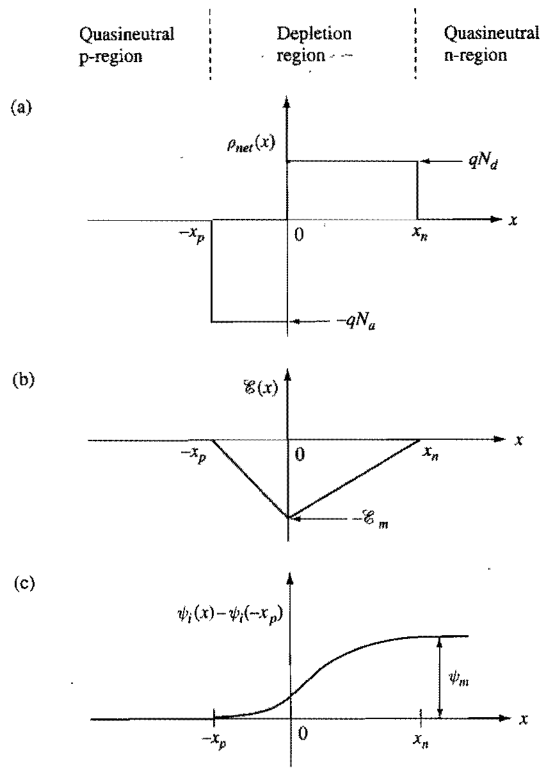

Figure 2.13. Depletion approximation of a p–n junction: (a) charge distribution, (b) electric field, and (c) electrostatic potential.

Poisson's equation, i.e., (2.44), for the depletion region is

$$-\frac{d^2\psi_i}{dx^2} = \frac{d\mathscr{E}}{dx} = \frac{q}{\varepsilon_{si}}\left[p(x) - n(x) + N_d^+(x) - N_a^-(x)\right] = \frac{q}{\varepsilon_{si}}\left[N_d^+(x) - N_a^-(x)\right], \tag{2.75}$$

where N_d^+ is the ionized-donor concentration and N_a^- is the ionized-acceptor concentration, and where the mobile-electron and -hole concentrations have been set to zero, consistent with the depletion approximation. For simplicity, we shall assume that all the donors and acceptors within the depletion region are ionized, and that the junction is abrupt and not compensated, i.e., there are no donor impurities on the p-side and no acceptor impurities on the n-side. With these assumptions, Eq. (2.75) becomes

$$-\frac{d^2\psi_i}{dx^2} = \frac{qN_d}{\varepsilon_{si}} \quad \text{for} \quad 0 \le x \le x_n \tag{2.76}$$

and

$$-\frac{d^2\psi_i}{dx^2} = -\frac{qN_a}{\varepsilon_{si}} \quad \text{for} \quad -x_p \le x \le 0. \tag{2.77}$$

Integrating Eq. (2.76) once from $x = 0$ to $x = x_n$, and Eq. (2.77) once from $x = -x_p$ to $x = 0$, subject to the boundary conditions of $d\psi_i/dx = 0$ at $x = -x_p$ and at $x = x_n$, we obtain the maximum electric field, $\mathscr{E}_m$, which is located at $x = 0$. That is,

$$\mathscr{E}_m \equiv \left|\frac{-d\psi_i}{dx}\right|_{x=0} = \frac{qN_d x_n}{\varepsilon_{si}} = \frac{qN_a x_p}{\varepsilon_{si}}. \tag{2.78}$$

It is clear from Eq. (2.78) that the total space charge inside the n-side of the depletion region is equal (but opposite in sign) to the total space charge inside the p-side of the depletion region. Thus, in Fig. 2.13(a), the two charge distribution plots have the same area. Equation (2.78) could have been obtained directly from Gauss's law, i.e., Eq. (2.43).

Let ψ_m be the total potential drop across the p–n junction, i.e., $\psi_m = [\psi_i(x_n) - \psi_i(-x_p)]$. The total potential drop can be obtained by integrating Eqs. (2.76) and (2.77) twice, the second time from $x = -x_p$ to $x = x_n$. That is,

$$\begin{aligned}\psi_m &= \int_{-x_p}^{x_n} d\psi_i(x) = -\int_{-x_p}^{x_n} \mathscr{E}(x)dx \\ &= \frac{\mathscr{E}_m(x_n + x_p)}{2} = \frac{\mathscr{E}_m W_d}{2},\end{aligned} \tag{2.79}$$

where $W_d = x_n + x_p$ is the total width of the depletion layer. It can be see from Eq. (2.79) that ψ_m is equal to the area in the $\mathscr{E}(x) - x$ plot, i.e., Fig. 2.13(b). Eliminating $\mathscr{E}_m$ from Eqs. (2.78) and (2.79) gives

$$W_d = \sqrt{\frac{2\varepsilon_{si}(N_a + N_d)\psi_m}{qN_aN_d}}. \tag{2.80}$$

This equation relates the total width of the depletion layer to the total potential drop across the junction and to the doping concentrations of the two sides of the diode.

2.2.2.2 Externally Biased Junctions

In the absence of any externally applied voltage, the total electrostatic potential drop ψ_m across a p–n diode is equal to the built-in potential ψ_{bi}, as indicated in Fig. 2.12(b). This built-in potential represents an energy barrier limiting the flow of electrons from the n-side to the p-side and the flow of holes from the p-side to the n-side. An externally applied voltage across a p–n diode shifts the Fermi level at the n-region contact relative to the Fermi level at the p-region contact. If the applied voltage causes ψ_m to be reduced, the diode is said to be *forward biased*. If the applied voltage causes ψ_m to be increased, the diode is said to be *reverse biased*. In considering a p–n diode in the context of VLSI devices, the forward-bias characteristics are more interesting than the reverse-bias characteristics. Therefore, we shall adopt the convention where a positive applied voltage also means a forward-bias voltage. Physically, this means the external voltage is connected such that

the p-side is biased positively relative to the n-side, as in the case illustrated in Fig. 2.12(c). The total potential drop ψ_m and the externally applied voltage V_{app} are related by

$$\psi_m = \psi_{bi} - V_{app}, \tag{2.81}$$

where $V_{app} > 0$ means the diode is forward biased and $V_{app} < 0$ means the diode is reverse biased. If Eq. (2.81) is used in Eq. (2.80), it gives the total depletion-layer width of a forward- or reverse-biased diode.

A quasineutral region has a finite resistivity determined by its dopant impurity concentration (see Fig. 2.9). When a current flows in a region of finite resistivity, there is a corresponding voltage drop, or *IR* drop, along the current path. In writing Eq. (2.81), the *IR* drops in the quasineutral regions are assumed to be negligible so that V_{app} is the same as the *voltage across the space-charge region*, V'_{app}. ***If IR drops in the quasineutral regions are not negligible, then V_{app} should be replaced by V'_{app} in Eq. (2.81).***

- *p–n diode as a rectifier.* When a diode is forward biased, the energy barrier limiting current flow is lowered, causing electrons to be injected from the n-side into the p-side and holes injected from the p-side into the n-side, resulting in a current flow through the diode. As we shall show in Section 2.2.4, the forward current increases exponentially with V'_{app} and hence can be very large. When a diode is reverse biased, the energy barrier limiting current flow is increased. There is no current flow due to electron and hole injection, only a relatively low background or leakage current. Thus a diode has rectifying current–voltage characteristics, being conducting when it is forward biased, and nonconducting when it is reverse biased. This is illustrated in Fig. 2.14. The equations governing the current–voltage characteristics of a diode will be derived in Sections 2.2.3 and 2.2.4.
- *Depletion-layer capacitance.* Consider a small change dV_{app} in the applied voltage. dV_{app} causes a charge per unit area dQ to flow into the p-side, which is equal to the change in the charge in the p-side depletion region. Since all mobile carriers are

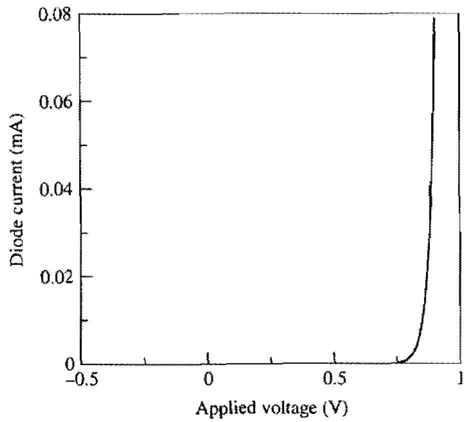

Figure 2.14. A schematic linear plot of the current of a typical silicon diode as a function of its applied voltage. On a linear plot, the reverse current is too low to be observable.

ignored in our depletion approximation, we can write the charge per unit area in the p-side depletion region as

$$Q_d(\text{p-side}) = -qN_a x_p(V_{app}), \tag{2.82}$$

where we have indicated that the p-side depletion-layer width, x_p, is a function of V_{app}. Notice that Q_d for the p-side is negative because ionized acceptors have a charge $-q$. The depletion-layer capacitance per unit area is

$$C_d \equiv \frac{dQ}{dV_{app}} = \frac{dQ_d(\text{p-side})}{dV_{app}} = \frac{\varepsilon_{si}}{W_d}. \tag{2.83}$$

That is, the depletion-layer capacitance of a diode is equivalent to a parallel-plate capacitor of separation W_d and dielectric constant ε_{si}. Physically, this is due to the fact that only the majority carriers at the edges of the depletion layer, not the space charge within the depletion region, respond to changes in the applied voltage.

- *Extending the depletion approximation to include injected current flows in the space-charge region.* When a diode is forward biased, the electrons flowing from the n-side to the p-side and the holes flowing from the p-side to the n-side add to the space charge in the transition region of the diode. To be accurate, we cannot assume the transition region to be depleted of mobile charge carriers. However, as long as the density of mobile carriers is small compared to the densities of ionized donors and acceptors, we have a well-defined space-charge region. (When the density of mobile carriers is comparable to or larger than the densities of ionized donors or acceptors, the boundaries of the space-charge region are no longer well defined. This situation will be discussed further in Section 6.3.3.2 in the context of base widening at high injection in a bipolar transistor.) For a well-defined space-charge layer of width W_d, the associated capacitance per unit area is the same as a parallel-plate capacitor, namely, Eq. (2.83). In this case, W_d can be obtained from integrating Poisson's equation, i.e., Eq. (2.44). An example of how mobile charge carriers flowing through a space-charge region affect the space-charge-region thickness is given in Section 6.3.3.1 in the context of base widening at low injection in a bipolar transistor.

2.2.2.3 One-Sided Junctions

In many applications, such as the source or drain junction of a MOSFET or the emitter–base diode of a bipolar transistor, one side of the p–n diode is degenerately doped while the other side is lightly to moderately doped. In this case, practically all the voltage drop and the depletion layer occur across the lightly doped side of the diode. That this is the case can be inferred readily from Eq. (2.78), which implies that $x_n = N_a W_d/(N_a + N_d)$ and $x_p = N_d W_d/(N_a + N_d)$. The characteristics of a one-sided p–n diode are therefore determined primarily by the properties of the lightly doped side alone. In this sub-subsection, we shall derive the equations for an n^+–p diode where the characteristics are determined by the p-side. The results can be extended straightforwardly to a p^+–n diode.

As discussed in Section 2.1.2, for a lightly to moderately doped p-type silicon, the Fermi level is given by Eq. (2.25), and for a heavily or degenerately doped n-type silicon,

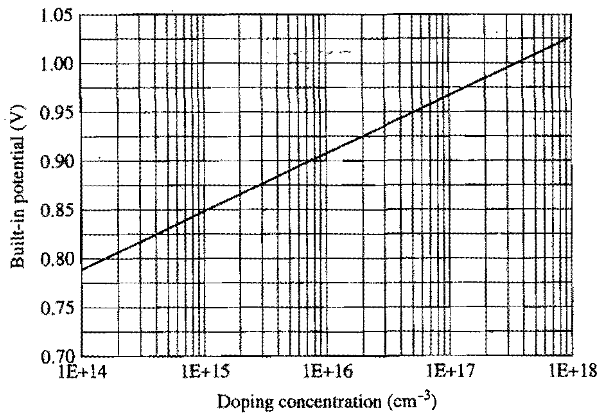

Figure 2.15. Built-in potential for a one-sided p–n junction versus the doping concentration of the lightly doped side.

it is a good approximation to assume its Fermi level to be at the conduction-band edge. Therefore, the built-in potential for an n^+–p diode, from Eqs. (2.72) and (2.73), is given by

$$\begin{aligned} q\psi_{bi} &= E_{fn} - E_{in} + kT\ln\left(\frac{N_a}{n_i}\right) \\ &\approx E_{cn} - E_{in} + kT\ln\left(\frac{N_a}{n_i}\right) \\ &\approx \frac{E_g}{2} + kT\ln\left(\frac{N_a}{n_i}\right), \end{aligned} \tag{2.84}$$

where we have made a further approximation that the intrinsic Fermi level is located half way between the conduction- and valence-band edges, E_{cn} and E_{vn}, on the n-side. [See Eq. (2.12) and the discussion that follows.] Figure 2.15 is a plot of ψ_{bi}, as approximated by Eq. (2.84), as a function of the doping concentration of the lightly doped side.

The depletion-layer width, from Eqs. (2.80) and (2.81), is

$$W_d = \sqrt{\frac{2\varepsilon_{si}(\psi_{bi} - V_{app})}{qN_a}}, \tag{2.85}$$

where $V_{app} > 0$ if the diode is forward biased and $V_{app} < 0$ if the diode is reverse biased. The depletion-layer capacitance per unit area is given by Eq. (2.83). Figure 2.16 is a plot of the depletion-layer width and capacitance as a function of doping concentration for $V_{app} = 0$. Again, V_{app} in Eq. (2.85) should be replaced by V'_{app} whenever the IR drops in the quasineutral regions are not negligible.

2.2.2.4 Thin-i-Layer p–i–n Diodes

Many modern VLSI devices operate at very high electric fields within the depletion regions of some of their p–n diodes. In fact, the junction fields are often so high that

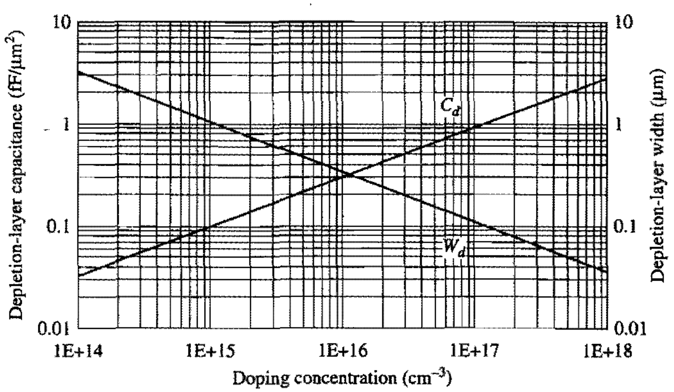

Figure 2.16. Depletion-layer width and depletion-layer capacitane, at zero bias, as a function of doping concentration of the lightly doped side of a one-sided p–n junction.

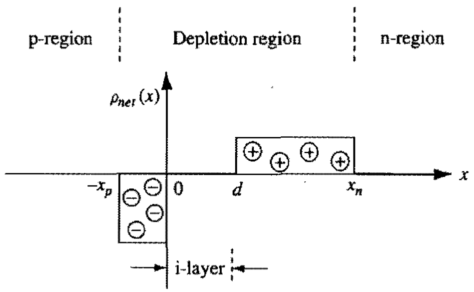

Figure 2.17. Charge distribution in a p–i–n diode.

detrimental high-field effects, such as avalanche multiplication and hot-carrier effects, limit the attainable device and circuit performance. To overcome the constraints imposed by high fields in a diode, device designers often introduce a thin but lightly doped region between the n- and the p-sides. In practice, this can be accomplished by sandwiching a lightly doped layer during epitaxial growth of the doped layers, or by grading the doping concentrations at or near the junction by ion implantation and/or diffusion. Analyses of such a diode structure become very simple if the lightly doped region is assumed to be intrinsic or undoped, i.e., if the lightly doped region is assumed to be an *i-layer*. This actually is not a bad approximation as long as the net charge concentration in the i-layer is at least several times smaller than the space-charge concentration on either side of the p–n junction, so that the contribution by the i-layer charge to the junction electric field is negligible. Figure 2.17 shows the charge distribution in such a p–i–n diode. The corresponding Poisson equation is

$$-\frac{d^2\psi_i}{dx^2}=\frac{qN_d}{\varepsilon_{si}} \quad \text{for} \quad d<x<x_n, \tag{2.86}$$

$$-\frac{d^2\psi_i}{dx^2}=0 \quad \text{for} \quad 0<x<d, \tag{2.87}$$

$$-\frac{d^2\psi_i}{dx^2}=-\frac{qN_a}{\varepsilon_{si}} \quad \text{for} \quad -x_p<x<0. \tag{2.88}$$

These equations can be solved in the same way as Eqs. (2.76) and (2.77). Thus, integrating the equations once, subject to the boundary conditions that the electric field is zero at $x=-x_p$ and at $x=x_n$, gives

$$\mathscr{E}_m=\frac{qN_ax_p}{\varepsilon_{si}}=\frac{qN_d(x_n-d)}{\varepsilon_{si}}, \tag{2.89}$$

where $\mathscr{E}_m$ is the maximum electric field which exists in the region $0\le x\le d$. Integrating the equations twice gives the total potential drop ψ_m across the junction as

$$\psi_m=\frac{\mathscr{E}_m(W_d+d)}{2}, \tag{2.90}$$

where $W_d=x_n+x_p$ is the total depletion-layer width. Eliminating $\mathscr{E}_m$ from Eqs. (2.89) and (2.90) gives

$$W_d=\sqrt{\frac{2\varepsilon_{si}(N_a+N_d)\psi_m}{qN_aN_d}+d^2}. \tag{2.91}$$

It is interesting to compare two diodes with the same externally applied voltage and the same p-side and n-side doping concentrations, one with an i-layer and one without. These two diodes have the same ψ_m. From Eq. (2.91), we can write

$$W_d=\sqrt{W_{d0}^2+d^2} \tag{2.92}$$

for the diode with an i-layer, where W_{d0} is the depletion-layer width, given by Eq. (2.80), for the diode without an i-layer. Therefore,

$$\frac{W_d}{W_{d0}}=\sqrt{1+\frac{d^2}{W_{d0}^2}}. \tag{2.93}$$

If we denote by $\mathscr{E}_{m0}$ the maximum electric field for the diode without an i-layer, then Eqs. (2.79) and (2.90) give the ratio of the electric fields as

$$\frac{\mathscr{E}_m}{\mathscr{E}_{m0}}=\frac{W_{d0}}{W_d+d}=\sqrt{1+\frac{d^2}{W_{d0}^2}}-\frac{d}{W_{d0}}. \tag{2.94}$$

Thus, introduction of a lightly doped layer between the n- and p-regions of a diode reduces the maximum electric field in the junction. The depletion-layer charge ratio for the two diodes is, by Gauss's law,

$$\frac{Q_d}{Q_{d0}} = \frac{\mathscr{E}_m}{\mathscr{E}_{m0}} = \sqrt{1+\frac{d^2}{W_{d0}^2}} - \frac{d}{W_{d0}}, \tag{2.95}$$

where Q_{d0} is the depletion-layer charge for the diode without an i-layer.

The depletion-layer capacitance per unit area can be calculated from Eq. (2.83), i.e., from $dQ_d(\text{p-side})/dV_{app}$, and the result is

$$C_d = \frac{\varepsilon_{si}}{W_d}. \tag{2.96}$$

The junction depletion-layer capacitance is related to the depletion-layer width in exactly the same way with or without an i-layer. This is expected from the physical picture of a parallel-plate capacitor where the capacitance is determined by the separation of the plates and not by any fixed charge distribution between the plates. The ratio of the capacitance with an i-layer to that without an i-layer is

$$\frac{C_d}{C_{d0}} = \frac{W_{d0}}{W_d} = \frac{1}{\sqrt{1+d^2/W_{d0}^2}}, \tag{2.97}$$

where $C_{d0} = \varepsilon_{si}/W_{d0}$ is the depletion-layer capacitance for the diode without an i-layer.

2.2.3 The Diode Equation

In considering the current–voltage characteristics of a p–n diode, it is much more convenient to work with the quasi-Fermi potentials, instead of the intrinsic Fermi potential. The current densities and the quasi-Fermi potentials are given by Eqs. (2.63) to (2.66). These are repeated here for convenience:

$$J_n(x) = -qn(x)\mu_n(x)\frac{d\phi_n(x)}{dx}, \tag{2.98}$$

$$J_p(x) = -qp(x)\mu_p(x)\frac{d\phi_p(x)}{dx}, \tag{2.99}$$

where

$$\phi_n(x) \equiv \psi_i(x) - \frac{kT}{q}\ln\left[\frac{n(x)}{n_i}\right] \tag{2.100}$$

is the quasi-Fermi potential for electrons and

$$\phi_p(x) \equiv \psi_i(x) + \frac{kT}{q}\ln\left[\frac{p(x)}{n_i}\right] \tag{2.101}$$

is the quasi-Fermi potential for holes, and ψ_i is the electrostatic potential given by Eq. (2.40). In terms of the quasi-Fermi potentials, the pn product is

$$p(x)n(x) = n_i^2 \exp\left\{\frac{q[\phi_p(x) - \phi_n(x)]}{kT}\right\}. \tag{2.102}$$

In writing Eqs. (2.98) to (2.102), we have indicated explicitly the x dependence of the variables. In theory, the current–voltage characteristics of a diode can be obtained from these coupled equations. However, simple and close-form equations relating the electron and hole densities and currents to the applied voltage can be obtained if some approximations and assumptions are made. Here we discuss the physical bases for these approximations and assumptions, which will be used in later subsections to obtain equations describing the behavior of a diode in response to an applied voltage.

- *Quasineutrality.* As discussed in Section 2.1.4.8, the majority-carrier response time is on the order of 10^{-12} s. As we shall show later [see Fig. 2.24(b)], this time is extremely short compared to typical minority-carrier lifetimes. Therefore, as minority carriers are injected into a doped silicon region, the majority carriers respond practically instantaneously to maintain quasineutrality. For instance, let us consider the p-region of a forward-biased diode. As electrons are injected from the n-side, the change in electron concentration in the p-region instantaneously induces a change in the hole concentration in the region such that $\Delta p_p(x) = \Delta n_p(x)$ to maintain quasineutrality. Similarly, for the n-region of a forward-biased diode, we have $\Delta p_n(x) = \Delta n_n(x)$.
- *High-level injection.* A forward-bias voltage is said to cause a high-level injection of minority carriers in a p–n diode if it results in the injected minority-carrier density being comparable to or larger than the majority-carrier density. For a one-sided n^+–p diode, high-level injection means $\Delta n_p(x)$ is comparable to or larger than $p_{p0}(x)$ [$= N_a(x)$ for silicon at room temperature]. When high injection occurs, there is no simple relationship between the current components and the applied voltage. Usually, Eqs. (2.98) to (2.101) are used in numerical simulation to determine the electron and hole currents at any point in the diode.
- *Low-level injection.* A forward-bias voltage is said to cause only low-level injection of minority carriers in a p–n diode if the injected minority-carrier density is small compared to the majority-carrier density. For a one-sided n^+–p diode, this means $\Delta n_p(x) \ll p_{p0}(x)$. Unless stated otherwise, ***low-level injection is assumed*** in all cases discussed in this book. In the low-level-injection approximation, the majority-carrier density is simply equal to the density of ionized dopant impurities, and Eq. (2.102) suggests that the minority-carrier density is proportional to $\exp\left\{q\left[\phi_p(x) - \phi_n(x)\right]/kT\right\}$. Therefore, we need to establish the relationship between $\phi_p(x) - \phi_n(x)$ and the externally applied voltage V_{app} in order to derive the equations governing the current–voltage characteristics of a diode. This is done in Section 2.2.3.1 below.

2.2.3.1 Spatial Variaton of Majority-Carrier Quasi-Fermi Potential and *IR* Drop in a Quasineutral Region

Consider a p-type silicon having an equilibrium hole density of $p_{p0}(x)$ connected as a resistor. A current will flow in the resistor when a voltage is applied across it. The current flow causes an incremental voltage drop dV between x and $x + dx$. From Eq. (2.60), this incremental voltage drop is

$$dV(x) = d\psi_f(x) = -\frac{J_p(x) + J_n(x)}{qp_{p0}(x)\mu_p(x) + qn_{p0}(x)\mu_n(x)}dx \quad \text{(p-resistor)}$$
$$\approx -\frac{J_p(x)}{qp_{p0}(x)\mu_p(x)}dx, \tag{2.103}$$

where we have used the fact that $J_n(x) \ll J_p(x)$ and $n_{p0} \ll p_{p0}$ in a p-type resistor. Integrating Eq. (2.103) from contact to contact we have the voltage drop across the p-type resistor.

Next, let us consider the quasineutral p-region of a forward-biased p–n diode. The forward bias causes a current to flow in the diode, and hence through the quasineutral p- and n-regions. As the hole current flows in the quasineutral p-region, it causes a drop in the hole quasi-Fermi potential according to Eq. (2.98), i.e.,

$$d\phi_p(x) = -\frac{J_p(x)}{qp_p(x)\mu_p(x)}dx \quad \text{(p-region of forward-biased diode)}. \tag{2.104}$$

If we assume the quasineutral p-region extends from $x = 0$ (depletion-layer edge) to $x = W_p$ (p-region contact), we have

$$\phi_p(W_p) - \phi_p(0) = -\int_0^{W_p} \frac{J_p(x)}{qp_p(x)\mu_p(x)}dx \quad \text{(p-region of forward-biased diode)}$$
$$= -\int_0^{W_p} \frac{J_p(x)}{q[p_{p0}(x) + \Delta p_p(x)]\mu_p(x)}dx$$
$$\approx -\int_0^{W_p} \frac{J_p(x)}{qp_{p0}(x)\mu_p(x)}dx, \tag{2.105}$$

where we have used the fact that $p_p(x) = p_{p0}(x) + \Delta p_p(x)$ and that $\Delta p_p(x) = \Delta n_p(x) \ll p_{p0}(x)$ at low injection. Comparing with Eq. (2.103), we see that Eq. (2.105) has the physical meaning of being the voltage drop caused by the hole current flowing through the p-region. Similarly, the change in ϕ_n in the quasineutral n-region between the depletion-layer edge and the n-contact is equal to the IR drop caused by the electron current flowing through the quasineutral n-region.

2.2.3.2 Change of Quasi-Fermi Potentials across the Space-Charge Region

In theory, as long as there are electrons and holes flowing across the space-change region, there are ϕ_n and ϕ_p drops across the space-charge region governed by Eqs. (2.98) and (2.99). It is shown in Appendix 4 that the behavior of ϕ_n and ϕ_p in the space-charge region in forward bias is quite different than in reverse bias.

- *Forward-biased diode.* In forward bias, the results in Appendix 4 show that the drops in ϕ_n and ϕ_p across the space-charge region are small compared to kT/q. As a result, ϕ_n

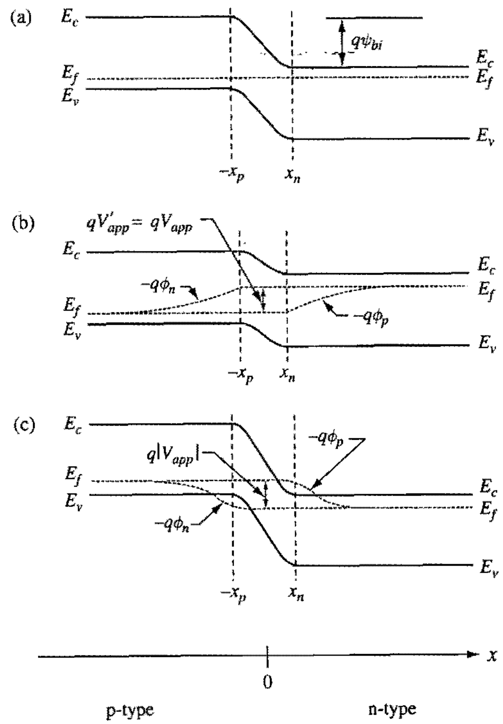

Figure 2.18. Schematics showing the variations of the quasi-Fermi potentials, ϕ_p for holes and ϕ_n for electrons, as a function of distance in a p–n diode. (a) A diode at thermal equilibrium with $V_{app} = 0$. (b) A forward-biased diode ($V_{app} > 0$) with negligible IR drops in the quasineutral regions. (c) A reverse-biased diode ($V_{app} < 0$). In the case of reverse bias, the drops in ϕ_n and ϕ_p across the space-charge region are small only at small $|V_{app}|$. At large $|V_{app}|$, the drops in ϕ_n and ϕ_p across the space-charge region increase with $|V_{app}|$.

and ϕ_p are essentially constant across the space-charge region, as illustrated schematically in Fig. 2.18(b).
- *Reverse-biased diode.* In the case of reverse bias, the results in Appendix 4 show that the drops in ϕ_n and ϕ_p across the space-charge layer are small compared to kT/q only for small reverse bias ($|V_{app}|$ less than about $4kT/q$). For larger reverse bias, the drops in ϕ_n and ϕ_p across the space-charge layer increase approximately linearly with increase in reverse bias. Therefore, ϕ_n and ϕ_p are also relatively constant across the space-charge region for the case of small reverse bias, as illustrated schematically in Fig. 2.18(c).

2.2.3.3 Relationship Between Minority-Carrier Density and Applied Voltage

The relationship between the voltage across the space-charge region, V'_{app}, and the majority-carrier quasi-Fermi potentials at the space-charge-region boundaries, $\phi_p(-x_p)$ and $\phi_n(x_n)$ is

$$
\begin{aligned}
V'_{app} &\equiv V_{app} - IR(\text{p-side}) - IR(\text{n-side}) \\
&= V_{app} - [\phi_p(\text{p-contact}) - \phi_p(-x_p)] - [\phi_n(x_n) - \phi_n(\text{n-contact})] \\
&= \phi_p(-x_p) - \phi_n(x_n).
\end{aligned} \tag{2.106}
$$

In Eq. (2.106), we have used the results discussed in Sections 2.1.4.5 and 2.1.4.6 which state that $V_{app} = [E_{fn}(\text{n-contact}) - E_{fp}(\text{p-contact})]/q = \phi_p(\text{p-contact}) - \phi_n(\text{n-contact})$. For forward bias and small reverse bias, the drops in the quasi-Fermi potentials across the space-charge region are small compared to kT/q, i.e., $\phi_p(-x_p) \approx \phi_p(x_n)$ and $\phi_n(-x_p) \approx \phi_n(x_n)$. Therefore, Eqs. (2.102) and (2.106) can be combined, for forward bias and small reverse bias, to give the electron density on the p-side at the space-charge-layer edge as

$$
\begin{aligned}
n_p(-x_p) &= \frac{n_i^2}{p_p(-x_p)} \exp\{q[\phi_p(-x_p) - \phi_n(-x_p)]/kT\} \\
&\approx \frac{n_i^2}{p_p(-x_p)} \exp\{q[\phi_p(-x_p) - \phi_n(x_n)]/kT\} \\
&= \frac{n_i^2}{p_p(-x_p)} \exp(qV'_{app}/kT) \\
&= \frac{n_{p0}(-x_p)p_{p0}(-x_p)}{p_p(-x_p)} \exp(qV'_{app}/kT) \\
&\approx n_{p0}(-x_p)\exp(qV'_{app}/kT),
\end{aligned} \tag{2.107}
$$

where we have used the low-injection approximation to write $p_p \approx p_{p0}$. (The case of high injection will be discussed later.) Similarly, we have

$$
p_n(x_n) \approx p_{n0}(x_n)\exp(qV'_{app}/kT) \quad \text{forward bias and small reverse bias} \tag{2.108}
$$

is the hole density at the space-charge-layer edge on the n-side. ***Equations (2.107) and (2.108) are the most important boundary conditions governing a p–n diode.*** They relate the minority-carrier concentrations at the space-charge-region boundaries of the quasi-neutral regions to their thermal-equilibrium values and to the voltage across the space-charge region. For a forward-biased diode ($V'_{app} > 0$), we have an excess of minority carriers at the boundaries of the quasineutral regions. For a reverse-biased diode ($V'_{app} < 0$), we have a depletion of minority carriers at the boundaries of the quasineutral regions. Equations (2.107) and (2.108) are often referred to as the *Shockley diode equations* (Shockley, 1950).

The fact that Eqs. (2.107) and (2.108) are valid only for small reverse bias is often overlooked in the literature. It is shown in Appendix 4 that for reverse biases more negative than about $-4kT/q$, Eqs. (2.107) and (2.108) overestimate the degree of minority-carrier depletion at the quasineutral-region boundaries. However, it is also shown in Appendix 4 that once the depletion of minority carriers at the boundaries has reached 90%, corresponding to $|V_{app}| \sim 3\,kT/q$, further depletion of minority carriers has little effect on the diode current. In other words, using Eqs. (2.107) and (2.108) for $|V_{app}| > 3kT/q$ does not lead to any significant error in the calculated reverse-biased diode currents. Therefore, Eqs. (2.107) and (2.108) can be used to describe the transport properties in a reverse-biased diode as if they are valid for arbitrary reverse biases.

The distinction between V'_{app} and V_{app} is important whenever there is significant parasitic series resistance in a forward-biased diode, for instance, in the forward-biased emitter–base diode of a bipolar transistor. In most cases, the parasitic resistance can be modeled as a lump resistor in series with the diode, allowing us to quantify the difference between V'_{app} and V_{app} readily. For simplicity in writing the equations, we shall not make the distinction between V_{app} and V'_{app} when we use Eqs. (2.107) and (2.108) to derive the equations for the current–voltage characteristics. The distinction between V_{app} and V'_{app} will be pointed out wherever it is important to do so.

2.2.3.4 Diode Equation at High Minority-Carrier Injection

As stated in the derivation of Eqs. (2.107) and (2.108), these equations are valid at low injection. If the low-injection condition is not met, these equations are not valid and Eq. (2.102) should be used instead. At sufficiently large forward biases, the injected minority-carrier concentration, particularly on the lightly doped side of the diode, can be so large that, in order to maintain quasineutrality, the electron and hole concentrations become approximately equal. In this case, Eq. (2.102) gives $n \approx p \approx n_i \exp(qV'_{app}/2kT)$. At such high levels of minority-carrier injection, the concept of a well-defined transition region is no longer valid, and the quasi-Fermi potentials do not have simple behavior in any region of the diode (Gummel, 1967). The effect of high minority-carrier injection on the measured current–voltage characteristics of a diode will be discussed further in Section 2.2.4.10. An example of how the "boundary" of a p–n junction can be "relocated" at high minority-carrier injection can be found in Section 6.3.3 in connection with the discussion of base-widening effects in a bipolar transistor.

2.2.4 Current–Voltage Characteristics

As discussed in Section 2.2.1, at thermal equilibrium, the drift component of the current caused by the electric field in the space-charge region is exactly balanced out by the diffusion component of the current caused by the electron and hole concentration gradients across the junction, resulting in zero current flow in the diode. When an external voltage is applied, this current component balance is upset, and current will flow in the diode. If carriers are generated by light or some other means, thermal equilibrium is disturbed, and current can also flow in the diode. Here only the current flow in a diode as a result of an externally applied voltage is discussed. We first consider the current–voltage characteristics of an ideal diode governed by the Shockley diode equations (2.107) and (2.108). The space-charge-region current will be added later in Section 2.2.4.10 when we consider the deviation of a practical diode from ideal behavior.

Consider a forward-biased p–n diode. Electrons are injected from the n-side into the p-side, and holes are injected from the p-side into the n-side. Since space-charge-region current is ignored, the hole current leaving the p-side is the same as the hole current entering the n-side. Similarly, the electron current leaving the n-side is equal to the electron current entering the p-side. To determine the total current flowing in the diode, all we need to do is to determine the hole current entering the n-side and the electron current entering the p-side.

The starting point for describing the current–voltage characteristics is the continuity equations. For electrons, it is given by Eq. (2.68) which is repeated here:

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - R_n + G_n, \tag{2.109}$$

where R_n and G_n are the electron recombination and generation rates, respectively. (A detailed discussion of generation and recombination processes is given in Appendix 5.) Equation (2.109) can be rewritten as

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{n - n_0}{\tau_n}, \tag{2.110}$$

where

$$\tau_n \equiv \frac{n - n_0}{R_n - G_n} \tag{2.111}$$

is the *electron lifetime*, and n_0 is the electron concentration at thermal equilibrium. Substituting Eq. (2.54) for J_n into Eq. (2.110) gives

$$\frac{\partial n}{\partial t} = n\mu_n\frac{\partial \mathscr{E}}{\partial x} + \mu_n\mathscr{E}\frac{\partial n}{\partial x} + D_n\frac{\partial^2 n}{\partial x^2} - \frac{n - n_0}{\tau_n}. \tag{2.112}$$

2.2.4.1 Diodes with Uniformly Doped Regions

Let us consider electrons in the p-region of a p–n diode. For simplicity, we assume the p-region to be uniformly doped so that at low electron injection currents the hole density is uniform in the p-region. As will be shown in Section 6.1.2, the electric field is zero for a region where the majority-carrier concentration is uniform. Thus, for the p-region under discussion, $\partial\mathscr{E}/\partial x = 0$ and $\mathscr{E} = 0$. For electrons in this p-region, Eq. (2.112) reduces to

$$\frac{\partial n_p}{\partial t} = D_n\frac{\partial^2 n_p}{\partial x^2} - \frac{n_p - n_{p0}}{\tau_n}. \tag{2.113}$$

At steady state, Eq. (2.113) becomes

$$\frac{\partial n_p}{\partial t} = 0 = D_n\frac{\partial^2 n_p}{\partial x^2} - \frac{n_p - n_{p0}}{\tau_n}, \tag{2.114}$$

which can be rewritten as

$$\frac{d^2 n_p}{dx^2} - \frac{n_p - n_{p0}}{L_n^2} = 0, \tag{2.115}$$

where

$$L_n \equiv \sqrt{\tau_n D_n} = \sqrt{\frac{kT\mu_n\tau_n}{q}} \tag{2.116}$$

is the *electron diffusion length* in the p-region. It should be noted that ***the quantities in Eq. (2.116) are all for minority carriers, not majority carriers.***

In deriving the equations for minority-carrier transport, we can focus on minority electrons or minority holes. As we shall show later, the current–voltage characteristics of a one-sided diode are determined primarily by the transport of minority carriers in the lightly doped side. Forward-biased diodes are usually found in the operation of bipolar transistors. High-speed bipolar transistors are n–p–n type, instead of p–n–p type. That is, most commonly encountered one-sided forward-biased diodes are of the n^+–p type, instead of p^+–n type. Therefore, we choose to focus on minority electrons in deriving the transport equations. Also, we like to make some rearrangement to simplify the algebra in deriving the current equations. Earlier in this chapter, the physical junction of a p–n diode is assumed to be located at $x=0$ with the p-silicon to the left side of the junction and the p-side depletion-layer edge located at $x = -x_p$. The n-silicon is located to the right side of the junction. The excess electrons in the p-region of the diode are injected from the n-side. These excess electrons will then move further into the p-region, contributing to electron current and becoming recombined along the way. That is, the p-side space-charge-region boundary is really the starting location for considering the distribution and transport of the excess electrons in the p-region. For considering the transport of electrons in the p-region, the algebra is simpler if we flip the p–n diode in Fig. 2.18 such that the n-region is on the left and the p-region is on the right, resulting in electrons flowing in the x-direction. The algebra can be further simplified if we shift the origin such that the quasineutral region of the p-side starts at $x=0$ and ends at $x = W_p$. Note that in this arrangement, the electron current in the p-region has a negative sign (negative charges flowing in the x-direction). This is illustrated in Fig. 2.19 for an n^+–p diode.

In this rearranged coordinate system, the electron density at $x=0$ is given by the Shockley diode equation, i.e., Eq. (2.107), while the electron density at $x = W_p$ is equal to n_{p0}, i.e.,

$$n_p(0) = n_{p0}\exp(qV_{app}/kT) \tag{2.117}$$

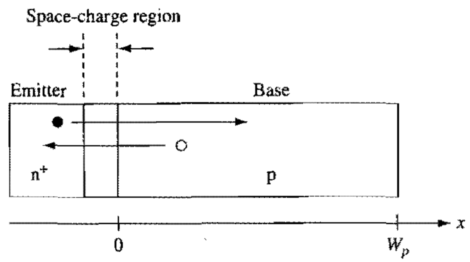

Figure 2.19. Schematic showing the coordinates used to develop the transport equations for a p–n diode. An n^+–p diode is assumed with the quasineutral p-region starting at $x=0$ and ending at $x = W_p$.

and

$$n_p(W_p) = n_{p0}. \tag{2.118}$$

[To be accurate, V_{app} should be replaced by V'_{app} in Eq. (2.117). For simplicity in writing the equations, we are not making the distinction between V_{app} and V'_{app} unless there is confusion.] Solving Eq. (2.115) subject to these boundary conditions gives

$$n_p(x) - n_{p0} = n_{p0}\left[\exp\left(\frac{qV_{app}}{kT}\right) - 1\right]\frac{\sinh[(W_p - x)/L_n]}{\sinh(W_p/L_n)}. \tag{2.119}$$

Since there is no electric field in the quasineutral p-region, there is no electron drift-current component, only an electron diffusion-current component. The electron current density entering the p-region is

$$\begin{aligned} J_n(0) &= qD_n\left(\frac{dn_p}{dx}\right)_{x=0} \\ &= -\frac{qD_n n_{p0}[\exp(qV_{app}/kT) - 1]}{L_n \tanh(W_p/L_n)} \\ &= -\frac{qD_n n_i^2[\exp(qV_{app}/kT) - 1]}{p_{p0}L_n \tanh(W_p/L_n)}, \end{aligned} \tag{2.120}$$

where in writing the last equation we have used the fact that $n_{p0}p_{p0} = n_i^2$. Equations (2.119) and (2.120) are valid for a p-region of arbitrary width W_p. Note that J_n is negative in sign because electrons have a charge $-q$ and are flowing in the x-direction.

The hole density in the n-region and the hole current density entering the n-side have the same forms as Eq. (2.119) and Eq. (2.120), respectively, and can be derived in an analogous manner (cf. Exercise 2.16). ***The total current flowing through a p–n diode is the sum of the electron current on the p-side and the hole current on the n-side.*** That is, the diode current density is

$$\begin{aligned} J_{diode} &= -\frac{qD_n n_i^2[\exp(qV_{app}/kT) - 1]}{p_{p0}L_n \tanh(W_p/L_n)} - \frac{qD_p n_i^2[\exp(qV_{app}/kT) - 1]}{n_{n0}L_p \tanh(W_n/L_p)} \\ &= -\left[\frac{qL_n n_i^2}{N_a\tau_n \tanh(W_p/L_n)} + \frac{qL_p n_i^2}{N_d\tau_p \tanh(W_n/L_p)}\right]\left[\exp\left(\frac{qV_{app}}{kT}\right) - 1\right], \end{aligned} \tag{2.121}$$

where we have assumed that all the dopants are ionized so that $p_{p0} = N_a$ and $n_{n0} = N_d$. [The diode current represented by Eq. (2.121) is due to the diffusion of minority carriers in the quasineutral regions. It does not include the generation-recombination current in the space-charge region, which will be discussed in Section 2.2.4.10. The total diode current is the sum of the diffusion current and the generation-recombination current.] The negative sign in Eq. (2.121) is due to the fact that we placed the p-region to the right of the n-region, causing electrons to flow in the $+x$ direction and holes to flow in the $-x$ direction. The negative sign will not be there if we place the p-region to the left of the n-region. Ignoring the sign, Eq. (2.121) is often referred to as the *Shockley diode current equation*, or simply the Shockley diode equation. It is applicable to both forward bias ($V_{app} > 0$) and reverse bias ($V_{app} < 0$). Figure 2.20 is a semi-log plot of the diode current

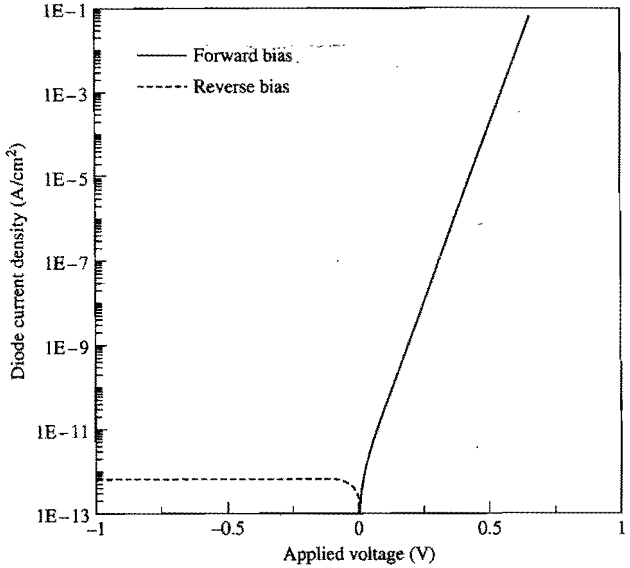

Figure 2.20. The current density of an ideal diode as given by Eq. (2.121). We assume $N_a = N_d = 10^{17}\ \text{cm}^{-3}$, and use the corresponding values for L and τ in Fig. 2.24. W/L is assumed to be large so that $\tanh(W/L) = 1$.

density given by Eq. (2.121) as a function applied voltage. It represents the I–V characteristics of an ideal diode. It is of interest to contrast Fig. 2.20 with Fig. 2.14. On a linear plot (Fig. 2.14), the rectifying characteristics of the diode current are evident, with a turn-on voltage of about 0.8 V. On a semi-log plot (Fig. 2.20), only the exponential dependence of the forward-bias current on voltage and a low-level reverse-bias background current are obvious. Deviations of a practical diode from an ideal case are usually observable in a semi-log plot, but not in a linear plot, and will be discussed later in Section 2.2.4.10. At sufficiently large reverse bias, the diode will break down (not shown in Fig. 2.20). High-field effects, including avalanche breakdown of a p–n diode, will be covered in Section 2.5.

2.2.4.2 Emitter and Base of a Diode

Equation (2.121) shows that the minority-carrier current is inversely proportional to the doping concentration. Thus, in a one-sided diode, the minority-carrier current in the lightly doped side is much larger than that in the heavily doped side. The diode current is dominated by the flow of minority carriers in the lightly doped side of the diode, while minority-carrier current in the heavily doped side usually can be neglected in comparison. (The effect of heavy doping can increase the minority-current flowing in the heavily doped region substantially. Heavy-doping effect is particularly important in bipolar devices, and will be covered in Chapter 6. The effect of heavy doping on the magnitudes of the currents in a diode will be discussed as exercises.) The lightly doped side is often referred to as the *base* of the diode. The heavily doped side is often referred to as the *emitter* of the diode, since the minority carriers entering the base are emitted from it.

In discussing the current–voltage characteristics of a diode, often only the minority-carrier current flow in the base is considered, since the minority-carrier current flow in the emitter is small in comparison. (However, if the width of an emitter is not larger than its minority-carrier diffusion length, the minority-carrier current flow in the emitter may not be negligible. Diodes with such emitters will be discussed further in Section 2.2.4.9.) As a result, unless stated explicitly, the region of the diode under discussion is assumed to be the base. That is, only the term in Eq. (2.121) corresponding to the base is kept. Whenever the emitter term is not negligible, both terms in Eq. (2.121) should be kept. In the following subsections, we examine in detail the current–voltage characteristics of one-sided n^+–p diodes. The equations derived can be modified readily to describe p^+–n diodes by changing the parameters for electrons in p-silicon to parameters for holes in n-silicon.

2.2.4.3 Forward-Biased n^+–p Diodes

We first consider the case where the n^+–p diode is moderately forward biased, i.e., $V_{app} > 0$, and $qV_{app} / kT \gg 1$. In this case, Eqs. (2.119) and (2.120) become

$$n_p(x) - n_{p0} = n_{p0} \exp(qV_{app}/kT) \frac{\sinh[(W_p - x)/L_n]}{\sinh(W_p/L_n)} \quad \text{(forward biased)} \qquad (2.122)$$

and

$$J_n(0) = -\frac{qD_n n_i^2 \exp(qV_{app}/kT)}{p_{p0} L_n \tanh(W_p/L_n)} \quad \text{(forward biased)}. \qquad (2.123)$$

That is, both ***the excess minority-carrier concentration and the minority-carrier current increase exponentially with the applied voltage*** (see Fig. 2.20).

2.2.4.4 Reverse-Biased n^+–p Diodes

Next we consider the case where the n^+–p diode is reverse-biased, i.e., $V_{app} < 0$, and $|qV_{app}| \gg kT$. In this case, Eqs. (2.119) and (2.120) become

$$n_p(x) - n_{p0} = -n_{p0} \frac{\sinh[(W_p - x)/L_n]}{\sinh(W_p/L_n)} \quad \text{(reverse biased)} \qquad (2.124)$$

and

$$J_n(0) = \frac{qD_n n_i^2}{p_{p0} L_n \tanh(W_p/L_n)} \quad \text{(reverse biased)}. \qquad (2.125)$$

Notice that $n_p(x) - n_{p0}$ is negative, and J_n is positive. The reverse bias causes a gradual depletion of electrons in the p-region near the depletion-region boundary, and this electron concentration gradient causes an electron current to flow from the quasineutral p-region towards the depletion region (in $-x$ direction according to our coordinates). This

is the electron diffusion component of the leakage current in a reverse-biased diode. It is also referred to as the electron *saturation current* of a diode.

The hole saturation current can be inferred from Eq. (2.121). The total diffusion leakage current in a diode is the sum of the electron and hole saturation currents. Notice that ***the diffusion leakage current is independent of the applied voltage.***

2.2.4.5 Wide-Base n^+–p Diodes

A diode is *wide-base* if its base width is large compared to the minority-carrier diffusion length in the base. For an n^+–p diode, this means $W_p/L_n \gg 1$. For a forward-biased wide-base diode, Eqs. (2.122) and (2.123) reduce to

$$n_p(x) - n_{p0} = n_{p0} \exp(qV_{app}/kT) \exp(-x/L_n) \quad \text{(forward, wide base)} \qquad (2.126)$$

and

$$J_n(0) = -\frac{qD_n n_i^2}{p_{p0} L_n} \exp\left(\frac{qV_{app}}{kT}\right) \quad \text{(forward, wide base)}. \qquad (2.127)$$

Thus, for a forward-biased wide-base diode, the excess minority-carrier concentration decreases exponentially with distance from the depletion-region boundary, and the minority-carrier current is independent of the base width.

For a reverse-biased wide-base diode, Eqs. (2.124) and (2.125) reduce to

$$n_p(x) - n_{p0} = -n_{p0} \exp(-x/L_n) \quad \text{(reverse, wide base)} \qquad (2.128)$$

and

$$J_n(0) = \frac{qD_n n_i^2}{p_{p0} L_n} \quad \text{(reverse, wide base)}. \qquad (2.129)$$

That is, the minority-carrier electrons in the base within a diffusion length of the depletion-region boundary diffuse towards the depletion region, with a saturation current density given by Eq. (2.129) which is independent of the base width.

2.2.4.6 Narrow-Base n^+–p Diodes

A diode is called *narrow-base* if its base width is small compared to the minority-carrier diffusion length in the base. In this case, this means $W_p/L_n \ll 1$. For a forward-biased narrow-base diode, Eqs. (2.122) and (2.123) reduce to

$$n_p(x) - n_{p0} = n_{p0} \exp\left(\frac{qV_{app}}{kT}\right)\left(1 - \frac{x}{W_p}\right) \quad \text{(forward, narrow base)} \qquad (2.130)$$

and

$$J_n(0) = -\frac{qD_n n_i^2}{p_{p0} W_p} \exp\left(\frac{qV_{app}}{kT}\right) \quad \text{(forward, narrow base)}. \qquad (2.131)$$

For a reverse-biased narrow-base diode, the corresponding equations are

$$n_p(x) - n_{p0} = -n_{p0}\left(1 - \frac{x}{W_p}\right) \quad \text{(reverse, narrow base)} \qquad (2.132)$$

and

$$J_n(0) = \frac{qD_n n_i^2}{p_{p0} W_p} \quad \text{(reverse, narrow base).} \tag{2.133}$$

For both forward and reverse biases, the minority-carrier current density in a narrow-base n^+–p diode increases as $1/W_p$. That is, for a narrow-base diode, the base current increases rapidly as the base width is reduced.

2.2.4.7 Spatial Distribution of Excess Minority Carriers

It can be seen from Eqs. (2.122) and (2.124) that both a forward-biased diode and a reverse-biased diode have the same sinh $[(W-x)/L]$ spatial dependence for the distribution of excess minority carriers (actually depletion of minority carriers in a reverse-biased diode). Figure 2.21 is a plot of the relative magnitude of the excess minority-carrier density as a function of x/L with W/L as a parameter. The $\exp(-x/L)$ distribution is for the case of $W/L = \infty$. It shows that a diode behaves like a wide-base diode for $W/L > 2$. For $W/L < 2$, the diode behavior depends strongly on W. For $W/L < 1$, the distribution can be approximated by the $1 - x/W$ dependence of a narrow-base diode.

2.2.4.8 Dependence of Minority-Carrier Current on Base Width

Figure 2.22 is a plot of the minority-carrier current density given by Eq. (2.120), normalized to its wide-base value. It shows that when $W/L < 1$, the minority-carrier current increases very reapidly as the diode base width decreases.

2.2.4.9 Shallow-Junction or Shallow-Emitter Diodes

Thus far, we have assumed the minority-carrier current in the emitter to be negligible compared to that in the base. A diode has a *shallow emitter* if the minority-carrier diffusion length in the emitter is comparable to or smaller than the width of the emitter region. The width of the emitter region of a p–n diode is also referred to as the junction depth. Therefore, a shallow-emitter diode is also a p–n junction having an electrically *shallow junction*. Figure 2.22 applies to the emitter region as well. Thus, we see from Fig. 2.22 that when $W/L < 1$ in the emitter, the minority-carrier current in the emitter increases very rapidly as the emitter depth decreases.

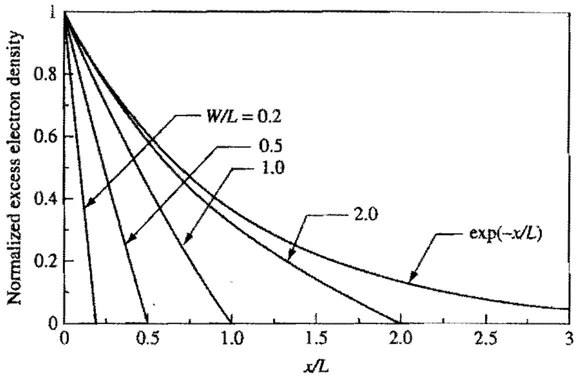

Figure 2.21. Relative magnitude of the excess minority-carrier concentration in the base of a diode as a function of distance from the base depletion-layer edge, with W/L as a parameter, where L is the minority-carrier diffusion length in the base and W is the base-region width. The case of $W/L = \infty$ is given by $\exp(-x/L)$.

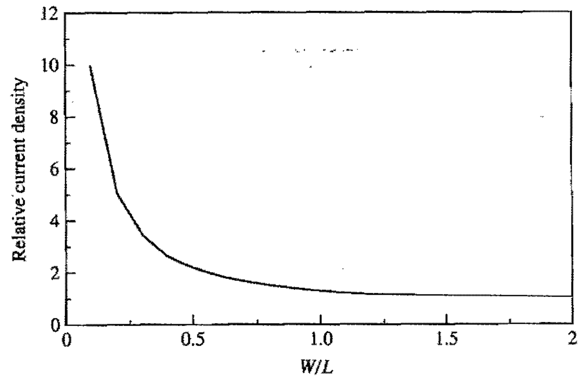

Figure 2.22. Relative maganitude of the minority-carrier current density in the base region of a diode as a function of W/L, normalized to the current at $W/L = \infty$. Here L is the minority-carrier diffusion length in the base, and W is the width of the base region.

As can be inferred from Fig. 2.24(c), to be developed later in Section 2.2.4.12, the minority-carrier diffusion length is about 0.3 μm for a doping concentration of $1 \times 10^{20}\ \text{cm}^{-3}$, and much larger for lower doping concentrations. This length is larger than the emitter depth of a typical one-sided p–n diode in a modern VLSI device (e.g., the emitter of a bipolar transistor and the source and/or drain of a CMOS device). That is, ***typical p–n diodes in modern VLSI devices should be treated as shallow-junction diodes.***

There are effective means for reducing the minority-carrier current in a shallow-emitter diode. For instance, a shallow emitter can be contacted using a doped polysilicon layer instead of a metal or metal silicide layer. The physics of minority-carrier transport in a shallow emitter will be covered in detail in Chapters 6 and 7 in the context of modern bipolar transistors.

2.2.4.10 Space-Charge-Region Current and Ideality Factor of a Diode

Thus far, we have neglected the current originating from the generation and recombination of electrons and holes within the space-charge region. In practical silicon diodes, the space-charge region current can be larger than the Shockley diode current at reverse bias and at low forward bias. It is shown in Appendix 5 that the space-charge-region current can be written in the form

$$I_{SC}(V_{app}) = I_{SC0}[\exp(qV_{app}/2kT) - 1], \tag{2.134}$$

with

$$I_{SC0} = \frac{A_{diode} q n_i W_d}{\tau_n + \tau_p}, \tag{2.135}$$

where W_d is the width of the space-charge region, A_{diode} is the cross-sectional area of the diode, and τ_n and τ_p are the electron and hole lifetimes, respectively. Equation (2.134) is often referred to as the *Sah–Noyce–Shockley diode equation* (Sah *et al.*, 1957; Sah, 1991).

From Eq. (2.121), we can write the Shockley diode current in the form

$$I_{diode} = I_0[\exp(qV_{app}/kT) - 1], \tag{2.136}$$

with

$$I_0 = A_{diode} q n_i^2 \left[\frac{D_n}{p_{p0} L_n \tanh(W_p/L_n)} + \frac{D_p}{n_{n0} L_p \tanh(W_n/L_p)} \right]. \tag{2.137}$$

As discussed in Section 2.2.3.4, ***Eq. (2.136) is valid only at low injection levels***. For an n^+–p diode, high injection occurs when n_p approaches N_a where N_a is the acceptor concentration of the p-side. At high injection, *IR* drops in the quasineutral regions can be significant. Also, I_{diode} changes to an $\exp(qV'_{app}/2kT)$ dependence (see the discussion in Section 2.2.3.4). The onset of high injection can be pushed to higher voltage by increasing N_a.

The current measured at the diode terminals is

$$I_{total} = I_{diode} + I_{SC}. \tag{2.138}$$

Figure 2.23 is a schematic semi-log plot of a diode current as a function of its forward-bias terminal voltage, with series resistances neglected. A semi-log current–voltage plot

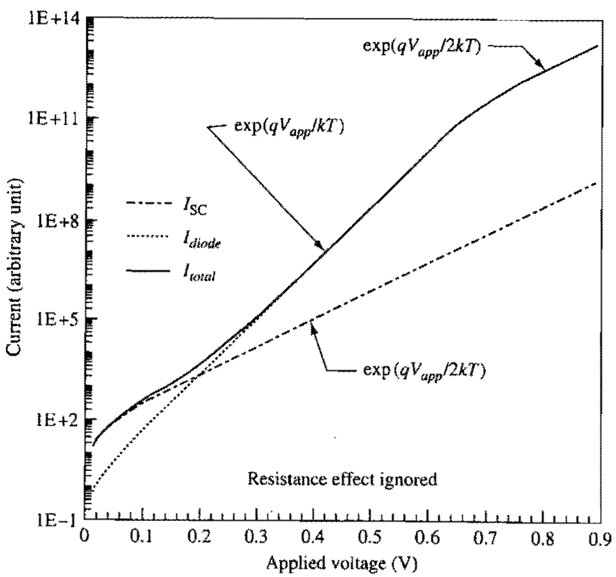

Figure 2.23. A schematic Gummel plot of the forward-bias current of a p–n diode. Series resistance effects are ignored. I_{diode} is the Shockley diode current. I_{SC} is the space-charge-region current.

for a diode or a bipolar transistor is called a *Gummel plot*. The slope in a Gummel plot is often used to infer the ideality of a diode. That is, the forward diode current is often expressed in the form

$$I_{total}(\text{forward}) \sim \exp(qV_{app}/mkT), \tag{2.139}$$

where m is called the *ideality factor*. Note that ***it is the diode terminal voltage V_{app}, not V'_{app} across the space-charge region that is in Eq. (2.139). The difference between V_{app} and V'_{app} is contained in the ideality factor.*** When m is unity, the current is considered "ideal." Figure 2.23 suggests that a forward diode current is ideal except at very small and very large forward biases. The nonideality at small forward bias is caused by the space-charge-region current. Space-charge-region current leads to $m \sim 2$ [see Eq. (2.134)]. The nonideality with $m \sim 2$ at very large forward bias is due to high-injection effect in the Shockley diode current (see the discussion in Section 2.2.3.4). At intermediate voltages, we have $1 < m < 2$.

Finite resistivity of the p- and n-regions results in voltage drops between the ohmic contacts and the junction. Finite resistivity effect is important only at very large forward biases. On a Gummel plot, finite resistivity effect can lead to m being very large. In general, when $1 < m < 2$ at large forward bias, it is not easy to clearly tell if the nonideality is caused by series resistance or by high injection. It may be a combination of both. However, when $m > 2$, we know that the series resistance effect dominates because the high injection effect by itself has an ideality factor of no larger than 2.

Series resistance effects can be reduced by increasing the diode doping concentrations, particularly the doping concentration of the base side of the diode. As discussed earlier, increasing doping concentration also delays the onset of high injection. Practical silicon diodes usually can be designed such that it appears quite ideal for forward biases of up to 0.8 V or slightly higher. Degradation in ideality factor is usually observable only at low forward biases and only in diodes having significant amounts of generation-recombination centers in the space-charge region. (An example of how the ideality factor changes with forward bias can be seen in the base current of a modern bipolar transistor shown in Fig. 6.13.)

2.2.4.11 Temperature Dependence and Magnitude of Diode Leakage Currents

For a reverse-biased diode, the total leakage current is the sum of the space-charge-region saturation current I_{SC0} and the diffusion saturation current I_0 given by Eqs. (2.135) and (2.137), respectively. The temperature dependence of I_0 is dominated by the temperature dependence of the n_i^2 factor, which, as shown in Eq. (2.13), is proportional to $\exp(-E_g/kT)$ where E_g is the bandgap energy. The space-charge-region leakage current I_{SC0}, being proportional to n_i, has a temperature dependence of $\exp(-E_g/2kT)$. In other words, the diffusion leakage current has an activation energy of about 1.1 eV while the generation-recombination leakage current has an activation energy of about 0.5 eV. This difference in activation energy can be used to distinguish the sources of the observed leakage current (Grove and Fitzgerald, 1966). [The diffusion leakage current is independent of reverse-bias voltage. The space-charge-region current is proportional to the space-charge-layer width which increases with reverse-bias voltage. In some devices, this difference in voltage dependence may be used to distinguish the two leakage current sources.]

For an n^+–p diode with a base doping concentration of 1×10^{17} cm^{-3}, using the values of $D_n/L_n = L_n/\tau_n$ in Fig. 2.24 (to be derived later), Eq. (2.129) gives an electron diffusion leakage current density of about 8×10^{-13} A/cm^2 at room temperature. The observed diffusion leakage current in a typical n^+–p diode (which is the sum of the electron and hole diffusion currents) is comparable to the space-charge-region generation current, both being on the order of 10^{-13} A/cm^2 at room temperature (Kircher, 1975). However, the diffusion leakage current, due to its larger activation energy, is usually larger than the generation-recombination leakage current at elevated temperatures.

2.2.4.12 Minority-Carrier Mobility, Lifetime, and Diffusion Length

There have been many attempts to measure the minority-carrier lifetimes, mobilities, and diffusion lengths. For doping concentrations greater than about 1×10^{19} cm^{-3}, the experiments are quite difficult, since the minority-carrier concentrations are too small, and as a result there is quite a bit of spread in the reported data (Dziewior and Silber, 1979: Dziewior and Schmid, 1977: del Alamo *et al.*, 1985a, b). For purposes of device modeling, the following empirical equations have been proposed for minority-carrier electrons (Swirhun *et al.*, 1986) and minority-carrier holes (del Alamo *et al.*, 1985a, b):

$$\mu_n = 232 + \frac{1180}{1 + (N_a/8 \times 10^{16})^{0.9}} \text{cm}^2\text{-V-s}^{-1} \tag{2.140}$$

$$\mu_p = 130 + \frac{370}{1 + (N_d/8 \times 10^{17})^{1.25}} \text{cm}^2\text{-V-s}^{-1} \tag{2.141}$$

$$\frac{1}{\tau_n} = 3.45 \times 10^{-12} N_a + 0.95 \times 10^{-31} N_a^2 \,\text{s}^{-1} \tag{2.142}$$

$$\frac{1}{\tau_p} = 7.8 \times 10^{-13} N_d + 1.8 \times 10^{-31} N_d^2 \,\text{s}^{-1}. \tag{2.143}$$

The minority-carrier mobilities, lifetimes, and diffusion lengths are plotted as a function of doping concentration in Fig. 2.24(a), (b), and (c), respectively. The diffusion lengths are calculated from the mobilities and lifetimes using the relation $L = (kT\mu\tau/q)^{1/2}$.

There are more recent models of minority-carrier mobilities. In particular, there is a physics-based model that describes the mobilities of both majority and minority carriers in a consistent manner (Klaassen, 1990). For minority electrons, the Klaassen model is about the same as that in Fig. 2.24(a). For minority holes, Klaassen's model gives about the same mobilities as in Fig. 2.24(a) for high ($>2 \times 10^{18}$cm^{-3}) doping concentrations and about 30% lower mobilities at low ($<1 \times 10^{18}$cm^{-3}) doping concentrations (Klaassen *et al.*, 1992).

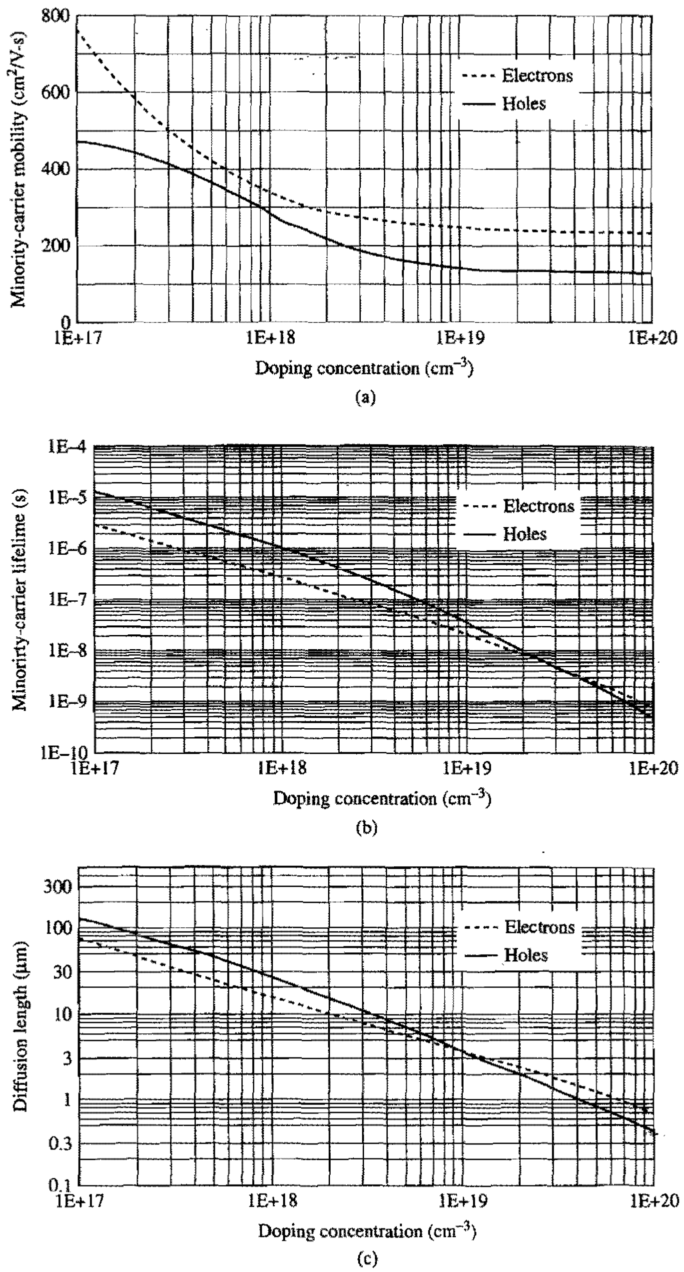

Figure 2.24. Minority-carrier (a) mobilities, (b) lifetimes, and (c) diffusion lengths as a function of doping concentration, calculated using the empirical equations (2.140) to (2.143).

2.2.5 Time-Dependent and Switching Characteristics

As discussed in Section 2.2.2, there is a capacitance associated with the depletion layer of a diode. As the diode is switched from off (zero-biased or reverse-biased) to on (forward-biased), it takes some time before the diode is turned on and reaches the steady state. This time is associated with charging up the depletion-layer capacitor and filling up the p- and n-regions with excess minority carriers. Similarly, when a diode is switched from the on state to the off state, it takes some time before the diode is turned off. This time is associated with discharging the depletion-layer capacitor and discharging the excess minority carriers stored in the p- and n-regions. The majority-carrier response time, or dielectric relaxation time, is negligibly short, on the order of 10^{-12} s, as shown in Section 2.1.4.8.

Consider the time needed to charge and discharge the depletion-layer capacitor. From Fig. 2.16, the depletion-layer capacitance C_d is typically on the order of 1 fF/μm^2. To turn a diode from off to on, and from on to off, the voltage swing V is typically about 1 V. If the diode is connected so that it carries a current density J of 1 mA/μm^2, then the time associated with charging and discharging the deplection-layer capacitor is on the order of C_dV/J, which is on the order of 10^{-12} s. Of course, this time changes in proportion to the current density J. However, as we shall show below, the time needed to charge and discharge the depletion-layer capacitor is usually very short compared with the time associated with charging and discharging the p- and n-regions of their minority carriers.

2.2.5.1 Excess Minority Carriers in the Base and Base Charging Time

Consider an n^+–p diode with a p-region base width W. When a forward bias is applied to it, minority-carrier electrons are injected into the base. As discussed in Section 2.2.4, for a wide-base diode, the minority-carrier density decreases exponentially with increasing distance, and practically all the minority carriers recombine before they reach the minority-carrier sink at $x = W$. For a narrow-base diode, on the other hand, practically all the minority carriers can travel across the base region without recombining.

The total *excess minority-carrier charge* (electrons) per unit area in the p-type base region is

$$Q_B = -q\int_0^W (n_p - n_{p0})\,dx. \tag{2.144}$$

For a wide-base diode, substituting Eqs. (2.126) and (2.127) into Eq. (2.144), we obtain

$$\begin{aligned} Q_B\,(\text{wide base}) &= -q(n_p - n_{p0})_{x=0}\,L_n \\ &= J_n(x = 0)\tau_n, \end{aligned} \tag{2.145}$$

where we have used $\tau_n = L_n^2/D_n$ from Eq. (2.116).

For a narrow-base diode, substituting Eqs. (2.130) and (2.131) into Eq. (2.144), we obtain

$$\begin{aligned} Q_B\,(\text{narrow base}) &= -q(n_p - n_{p0})_{x=0}\left(\frac{W}{2}\right) \\ &= J_n\,(x = 0)t_B, \end{aligned} \tag{2.146}$$

where the *base-transit time* t_B is defined by

$$\begin{aligned} t_B &\equiv \frac{Q_B\,(\text{narrow base})}{J_n\,(x = 0)} \\ &= \frac{W^2}{2D_n}. \end{aligned} \tag{2.147}$$

As will be shown below, t_B is also equal to the average time for the minority carriers to traverse the narrow base region.

In a wide-base diode, it takes a time equal to the minority-carrier lifetime to fill the base with minority carriers. In a narrow-base diode, it takes a time equal to the base-transit time to fill the base with minority carriers. It should be noted that the charging current, $J_n(x=0)$, is different for wide-base and narrow-base diodes. The dependence of $J_n(x=0)$ on base width is shown in Fig. 2.22.

2.2.5.2 Average Time for Traversing a Narrow Base

From Eq. (2.130), the excess electron concentration at any point x in the narrow p-type base region is

$$n_p - n_{p0} = (n_p - n_{p0})_{x=0}\left(1 - \frac{x}{W}\right). \tag{2.148}$$

Let $v(x)$ be the *apparent velocity* of these excess carriers at point x. The current density due to those excess carriers at x is then

$$J_n(x) = -qv(x)(n_p - n_{p0})_x = -qv(x)(n_p - n_{p0})_{x=0}\left(1 - \frac{x}{W}\right). \tag{2.149}$$

The electron current density at $x=0$ is given by Eq. (2.131), i.e.,

$$J_n\,(x = 0) = -\frac{q\,D_n}{W}\,(n_p - n_{p0})_{x=0}. \tag{2.150}$$

Assuming negligible recombination in the narrow base region, then current continuity requires $J_n(x)$ to be independent of x, i.e.,

$$v(x) = \frac{D_n}{W - x}. \tag{2.151}$$

The average time for traversing the base is thus given by

$$t_{avg} \equiv \int_0^W \frac{dx}{v(x)} = \frac{W^2}{2\,D_n}. \tag{2.152}$$

Comparison of Eqs. (2.147) and (2.152) shows that the base-transit time is equal to the average time for the minority carriers to traverse the narrow base.

It is instructive to esimate the magnitude of t_B. Modern n–p–n bipolar transistors typically have base widths of about 0.1 μm, and a peak base doping concentration of about 2×10^{18} cm^{-3} (Nakamura and Nishizawa, 1995). The corresponding minority electron mobility, from Fig. 2.24, is about 300 cm^2/V-s. The base-transit time is therefore less than 1×10^{-11}s, which is extremely short compared with the corresponding

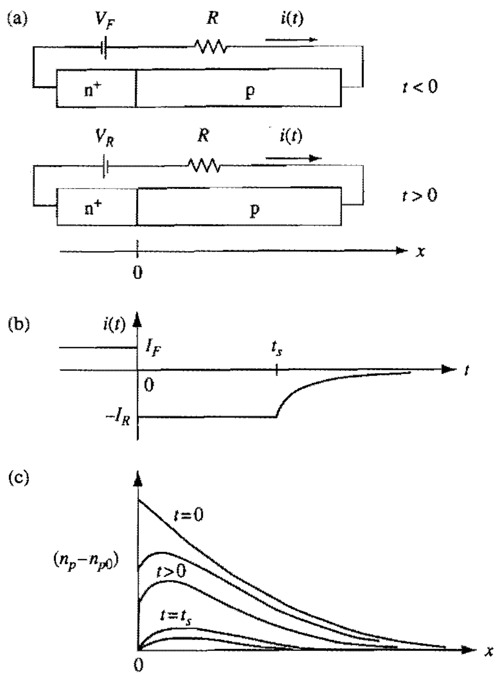

Figure 2.25. Schematics showing the switching of an n$^+$–p diode from forward bias to reverse bias: (a) the circuit schematics, (b) the diode current as a function of time, and (c) the excess-electron distribution in the base for different times.

minority-carrier lifetime on the order of 1×10^{-7}s. ***Recombination is negligible in the base layers of modern bipolar transistors.***

2.2.5.3 Discharge Time of a Forward-Biased Diode

Consider an n$^+$–p diode in a circuit configuration shown in Fig. 2.25(a). For simplicity, let us assume the external voltage, V_F or V_R, driving the circuit to be large compared to the internal junction voltage, i.e., the voltage immediately across the diode depletion layer, which is typically less than 1.0 V. At $t < 0$, there is a forward current of $I_F \approx V_F / R$ as illustrated in Fig. 2.25(b), and an excess electron distribution in the base region as illustrated in Fig. 2.25(c).

At time $t = 0$, the external bias is switched to a reverse voltage of V_R. The excess electrons in the base start to diffuse back towards the depletion region of the diode. Those electrons at the edge of the depletion region are swept away by the electric field in the depletion region towards the n$^+$ emitter at a saturated velocity of about 10^7 cm/s. As shown in Fig. 2.16, the depletion-layer width is typically on the order of 0.1 μm. The transit time across the depletion region is typically on the order of 10^{-12}s. As we shall see later, except for diodes of very narrow base widths, this time is extremely short compared to the total time for emptying the excess electrons out of the base region. Thus, as long as

there are sufficient excess electrons in the base region, the reverse current is limited not by the diffusion of excess electrons but by the external resistor and has a value of $I_R \approx V_R / R$, and the slope $(dn_p / dx)_{x=0}$, being proportional to I_R, is approximately constant.

As the excess electrons are discharged, part of the external voltage starts to appear across the p–n junction, and the junction becomes less forward biased. However, as long as there is still an appreciable amount of excess electrons stored in the base, the amount of external voltage appearing across the p–n junction remains very small. This is evident from Eq. (2.107), which indicates that even after the excess-electron concentration at the edge of the depletion layer has decreased by a factor of 10, the junction voltage has changed by only $2.3kT / q$, or 60mV. This is consistent with our assumptions that the reverse current remains essentially constant. During this time, the diode remains in the on condition.

At time $t = t_s$, the excess electrons have been depleted to the point that the reverse current is limited by the diffusion of electrons instead of by the external resistor. The rate of voltage change across the junction increases. Finally, when all the excess electrons are removed, the p–n diode is completely off. The external reverse-bias voltage appears entirely across the junction, and the reverse current is limited by the diode leakage current.

The time needed to switch off a forward-biased diode can be estimated from a charge-control analysis (Kuno, 1964). For simplicity, we shall estimate only the time during which the reverse current is approximately constant, and during which the diode remains in the on condition. Since the junction voltage remains approximately constant during this time, charging and discharging of the depletion-layer capacitance of the junction can be ignored. Let us consider the change in the amount of charge within the p-type base region. From Eq. (2.110), the continuity equation governing the electron concentration in the base region is

$$A_{diode}\,\frac{\partial(n_p - n_{p0})}{\partial t} = \frac{1}{q}\,\frac{\partial i_n(t)}{\partial x} - A_{diode}\,\frac{n_p - n_{p0}}{\tau_n}, \tag{2.153}$$

where $i_n(t)$ is the time-dependent electron current in the base region and A_{diode} is the cross-sectional area of the diode. Multiplying Eq. (2.153) by $-q$ and integrating over the base region, we have

$$\begin{aligned} -A_{diode}\,q\,\frac{\partial}{\partial t}\int_0^W (n_p - n_{p0})\,dx = &-\int_0^W \frac{\partial i_n(t)}{\partial x}\,dx \\ &+ A_{diode}\,\frac{q}{\tau_n}\int_0^W (n_p - n_{p0})\,dx\,, \end{aligned} \tag{2.154}$$

or

$$i_n(0,t) = A_{diode}\,\frac{dQ_B(t)}{dt} + A_{diode}\,\frac{Q_B(t)}{\tau_n} + i_n(W,t), \tag{2.155}$$

where $Q_B(t)$ is the excess minority charge per unit area stored in the base region, given by Eq. (2.144). Equation (2.155) is simply the continuity equation for the base region stated in the charge-control form. $i_n(0, t)$ is the electron current entering the base region, and

$i_n(W, t)$ is the electron current leaving the base region. At $x = W$, the electrons represented by $i_n(W, t)$ can simply exit the base region and continue on as an electron current outside the base, which is the case for electrons exiting the base of an n–p–n bipolar transistor (to be discussed in Chapter 6). Alternatively, the electrons represented by $i_n(W, t)$ can recombine with holes at the base ohmic contact located at $x = W$. The recombination gives rise to a current equal to $i_n(W, t)$ outside the base region. In either case, the current $i_n(W, t)$ is continuous across the base boundary at $x = W$, as required by charge conservation. The current flowing through the external resistor R is $i_n(0, t)$.

It is tempting to equate the current difference $i_n(0, t) - i_n(W, t)$ to the resistor current, but that is an inaccurate picture of current continuity. To see this, let us consider the steady-state situation when $dQ_B(t)/dt = 0$, and the special situation where recombination within the base is negligible ($\tau_n \to \infty$). In this case, Eq. (2.155) gives $i_n(0) - i_n(W) = 0$. However, the resistor current is not zero. The resistor current is $i_n(0)$, which is equal to $i_n(W)$ in this special case.

Consider a forward-biased diode being discharged. In this case, $i_n(0, t)$ is due to electrons diffusing back towards the n$^+$ emitter. For the coordinates system used here (see Section 2.2.4.1), these electrons travel in the $-x$ direction. Therefore, $i_n(0, t)$ is a positive quantity, and Eq. (2.155) gives

$$A_{diode}\frac{dQ_B(t)}{dt} + A_{diode}\frac{Q_B(t)}{\tau_n} + i_n(W,t) = I_R. \qquad (2.156)$$

Equation (2.156) is the continuity equation stated in the charge-control form for the base region of a diode at the initial stage of being discharged.

- *Discharge time for a wide-base diode.* For a wide-base diode, $i_n(W, t) = 0$ and the solution for Eq. (2.156) is

$$A_{diode}Q_B(t) = I_R\tau_n + [A_{diode}Q_B(0) - I_R\tau_n]\exp(-t/\tau_n), \qquad (2.157)$$

or

$$\frac{Q_B(t)}{Q_B(0)} = \frac{I_R\tau_n}{A_{diode}Q_B(0)}[1 - \exp(-t/\tau_n)] + \exp(-t/\tau_n), \qquad (2.158)$$

where $Q_B(0)$ is the excess minority charge per unit area just after the diode is switched from forward bias to reverse bias. For a wide-base diode, Eq. (2.145) gives $A_{diode}Q_B(0) = -\tau_n I_F$. (Note the negative sign in Q_B, since Q_B is negative for electrons.) Therefore, Eq. (2.158) gives

$$\frac{Q_B(t)}{Q_B(0)} = -\frac{I_R}{I_F}[1 - \exp(-t/\tau_n)] + \exp(-t/\tau_n). \qquad (2.159)$$

Figure 2.26 is a plot of the charge ratio $Q_B(t)/Q_B(0)$ as a function of t/τ_n with the current ratio I_R/I_F as a parameter. It shows that a forward-biased wide-base diode discharges with a time constant approximately equal to the minority-carrier lifetime, unless the reverse discharge current is much larger than the forward charging current. Even for $I_R/I_F = 10$, the diode discharges in a time of approximately $\tau_n/10$, which, as can be seen

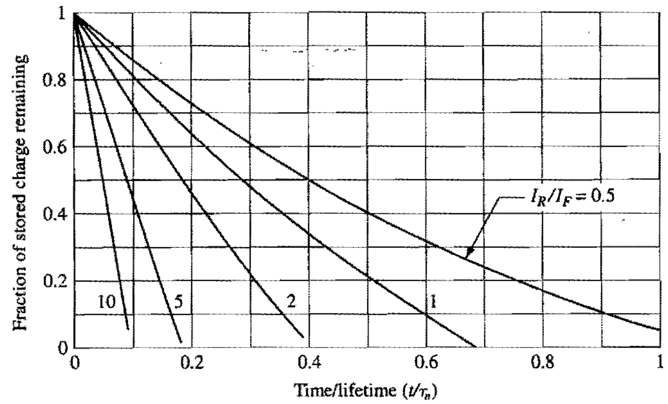

Figure 2.26. Plot of charges ratio $Q_B(t)/Q_B(0)$ as a function of t/τ_n during the discharge of a forward-biased diode, with the ratio of discharge current to charging current, I_R/I_F, as a parameter.

from Fig. 2.24(b), is larger than 10^{-8} s for most diodes of practical doping concentrations. This time is very long compared to the typical switching delays of VLSI circuits. The important point is that ***it takes a long time to drain off the excess minority carriers stored in a wide-base diode and turn it off.*** It is important to minimize excess minority carriers stored in forward-biased diodes if these diodes are to be switched off fast.

- *Discharge time for a narrow-base diode.* For a narrow-base diode, recombination can be ignored. Therefore, we have $|i_n(0)| = |i_n(W)| = I_F$ while the diode is in forward bias, and the distribution of excess electrons in the base has a constant gradient given by Eq. (2.130). At $t > 0$, after the diode has been switched from forward bias to reverse bias, electrons continue to flow towards and recombine at the base contact, i.e., $|i_n(W, t)| > 0$. As we shall show below, a narrow-base diode discharges in a time very small compared to τ_n. That is, during the discharge of a narrow-base diode, the recombination term $Q_B(t)/\tau_n$ can be neglected, and the minority electrons are discharged only through back diffusion towards the n$^+$ emitter and recombination at the base contact. With this approximation, Eq. (2.156) reduces to

$$A_{diode}\frac{dQ_B(t)}{dt} = I_R - i_n(W,t). \qquad (2.160)$$

To get an idea of how fast a narrow-base diode can be discharged, let us assume that the gradient of the electron distribution at $x = W$ remains about the same for a short time immediately after the diode is switched to reverse bias as during forward bias. That is, for a short time after switching from forward bias to reverse bias, we have $i_n(W, t) \sim -I_F$. [Note the negative sign for $i_n(W, t)$. Electrons flowing in the x-direction lead to a negative current.] Also, we note that Eq. (2.146) gives $A_{diode}Q_B(0) = -t_B I_F$ for a narrow-base diode, where t_B is the base transit time. With these assumptions, Eq. (2.160) gives

$$\frac{Q_B(t)}{Q_B(0)} \approx 1 - \frac{(I_R + I_F)t}{t_B I_F} \qquad (2.161)$$

for a short time after switching from on to off. Equation (2.161) shows that the discharge time for a narrow-base diode lasts approximately $t_B I_F/(I_R + I_F)$ which, for a large I_R/I_F ratio, can be much shorter than the base transit time. A complex but closed-form solution can be obtained in the large I_R/I_F limit (Lindmayer and Wrigley, 1965), which shows that most of the charge has come out by about $t_B/3$. The important point is that ***a forward-biased narrow-base diode can be switched off fast.***

2.2.6 Diffusion Capacitance

For a forward-biased diode, in addition to the capacitance associated with the space-charge layer, there is an important capacitance component associated with the rearrangement of the excess minority carriers in the diode in response to a change in the applied voltage. This minority-carrier capacitance is called *diffusion capacitance* C_D.

Consider an n^+–p wide-emitter narrow-base diode, i.e., a diode where the depth or width of the n^+ emitter region is large compared to its hole diffusion length and the width of the p-type base region is small compared to its electron diffusion length. (This diode is of interest because it represents the emitter–base diode of an n–p–n bipolar transistor.) When a voltage V_{app} is applied across the diode, an electron current of magnitude I_n is injected from the emitter into the base and a hole current of magnitude I_p is injected from the base into the emitter. The diode current is $I_n + I_p$. Both I_n and I_p are proportional to $\exp(qV_{app}/kT)$.

- *Quasisteady state.* In a quasisteady state, the voltage is assumed to vary slowly in time such that the *minority charge distribution can respond to the applied voltage fully* without any delay. In this case, the excess electron charge in the base is given by Eq. (2.146), i.e.,

$$A_{diode}\,|Q_n(V_{app})| = I_n(V_{app})t_B \quad \text{(narrow base)}, \tag{2.162}$$

which in turn gives

$$\begin{aligned} A_{diode}\frac{\Delta|Q_n(V_{app})|}{\Delta V_{app}} &= \frac{q}{kT}I_n(V_{app})t_B \\ &= \frac{q}{kT}I_n(V_{app})\left(\frac{W_B^2}{2D_{nB}}\right) \quad \text{(narrow base)}, \end{aligned} \tag{2.163}$$

where we have used Eq. (2.147) for the base transit time t_B. In Eq. (2.163), W_B is the base width and D_{nB} is the electron diffusion coefficient in the base. Similarly, using Eq. (2.145) and Eq. (2.116), we have

$$\begin{aligned} A_{diode}\frac{\Delta|Q_p(V_{app})|}{\Delta V_{app}} &= \frac{q}{kT}I_p(V_{app})\tau_{pE} \\ &= \frac{q}{kT}I_p(V_{app})\left(\frac{L_{pE}^2}{D_{pE}}\right) \quad \text{(wide emitter)} \end{aligned} \tag{2.164}$$

for the stored holes in the n^+ emitter, where L_{pE}, D_{pE}, and τ_{pE} are the diffusion length, diffusion coefficient, and lifetime, respectively, of holes in the emitter. Equations (2.163) and (2.164) relate the change in the stored charge caused by a change in the voltage across the diode in a quasisteady state. However they do *not* represent the true diffusion capacitance components of a forward-biased diode, which we shall discuss next.

- *Diffusion capacitance components.* Consider the discharge of a forward-biased base region illustrated schematically in Fig. 2.25(c). When the diode is forward biased, the electron distribution is represented by the $t=0$ curve. When the forward bias is reduced, or when the diode is switched to reverse bias, the electron distribution evolves as a function of time, as indicated by the $t>0$ curves. Part of the excess electrons diffuses to the left (back towards the emitter) and part of them diffuses to the right. The opposing electron currents suggest that the *net* charge moved through the external circuit in the discharge process is less than the total stored charge represented by the $t=0$ curve in Fig. 2.25(c). When an ac voltage is applied across the diode, only those electrons located sufficiently close to the depletion-region boundaries can keep up with the signal and get into and out of the base. The exact amount depends on the signal frequency. These signal-following electrons give rise to $i_n(0, t)$, the time-dependent electron current at the emitter end of the quasineutral base region. As discussed in Section 2.2.5.3, $i_n(0, t)$ is the electron component of the current in the external circuit. Similarly, if we consider the stored holes in the emitter, the signal-following holes in the n^+ emitter give rise to a hole current component $i_p(0, t)$ at the base end of the emitter region and in the external circuit. These signal-following stored charges are responsible for the diffusion capacitance.

The exact diffusion capacitance components can be derived from a frequency-dependent small-signal analysis of the current through a diode starting from the differential equations governing the transport of minority carriers (Shockley, 1949; Lindmayer and Wrigley, 1965; Pritchard, 1967). This is done for a wide-emitter narrow-base diode in Appendix 6. Here we simply state the results.

For a wide-emitter and narrow-base n^+–p diode, the low-frequency diffusion capacitance due to the excess electrons in the base is

$$C_{Dn} = \frac{qI_n(V_{app})}{kT}\left(\frac{W_B^2}{3D_{nB}}\right) = \frac{2}{3}\frac{q}{kT}I_n(V_{app})t_B \quad \text{(narrow base)}, \tag{2.165}$$

and that due to the excess holes in the emitter is

$$C_{Dp} = \frac{qI_p(V_{app})}{kT}\left(\frac{L_{pE}^2}{2D_{pE}}\right) = \frac{1}{2}\frac{q}{kT}I_p(V_{app})\tau_{pE} \quad \text{(wide emitter)}. \tag{2.166}$$

The total diffusion capacitance is

$$C_D = C_{Dn} + C_{Dp}. \tag{2.167}$$

Comparison with Eqs. (2.163) and (2.164) shows that 2/3 of the stored charge in the narrow base and 1/2 of the stored charge in the wide emitter contribute to the diffusion capacitance of a forward-biased diode. [In the case of a narrow base, a closed-form solution can be obtained in the large-discharge-current limit for the transient discharge current. Integration of the transient discharge current shows that 2/3 of the total stored charge in the narrow base diffuses back to the emitter when the base region is discharged. This fraction is the same as the fraction of total stored charge in the base contributing to the diffusion capacitance. In other words, one can think of the diffusion capacitance as coming from the portion of the stored minority charge that is "reclaimable" in the form of an ac current as the diode responds to an ac signal (Lindmayer and Wrigley, 1965).]

It is instructive to examine the relative magnitude of the two capacitance components C_{Dn} and C_{Dp}. Using the hole equivalent of Eq. (2.127) for hole current and Eq. (2.131) for electron current, and the relationship in Eq. (2.116), we have

$$\frac{C_{Dn}\,(\text{narrow base})}{C_{Dp}\,(\text{wide emitter})} = \frac{2}{3}\frac{N_E}{N_B}\frac{W_B}{L_{pE}}. \tag{2.168}$$

The ratio N_E/N_B is typically about 100 for an n^+–p diode. For an emitter with $N_E = 1 \times 10^{20} \text{cm}^{-3}$, L_{pE} is about 0.3 μm (see Fig. 2.24). Therefore, for practical one-sided diodes where the base width is larger than 0.03 μm, the ratio C_{Dn}/C_{Dp} is much larger than unity. That is, the diffusion capacitance of a one-sided p–n diode is dominated by the minority charge stored in the base of the diode. The diffusion capacitance due to the minority charge stored in the emitter is small in comparison. The effect of heavy doping, when included, will increase the amount of stored charge and hence the diffusion capacitance. Since the heavy-doping effect is larger in the more heavily doped emitter than in the base, it will make the ratio C_{Dn}/C_{Dp} smaller than that given by Eq. (2.168) (See Exercise 2.18).

2.3 MOS Capacitors

The metal–oxide–semiconductor (MOS) structure is the basis of CMOS technology. The Si–SiO_2 MOS system has been studied extensively (Nicollian and Brews, 1982) because it is directly related to most planar devices and integrated circuits. In this section, we review the fundamental properties of MOS capacitors and the basic equations that govern their operation. The effects of charges in the oxide layer and at the oxide–silicon interface are discussed in Section 2.3.6.

2.3.1 Surface Potential: Accumulation, Depletion, and Inversion

2.3.1.1 Energy-Band Diagram of an MOS System

The cross section of an MOS capacitor is shown in Fig. 2.27. It consists of a conducting gate electrode (metal or heavily doped polysilicon) on top of a thin layer of silicon dioxide grown on a silicon substrate. The energy band diagrams of the three components when separate are shown in Figure. 2.28. Before we discuss the energy band diagram of an MOS device, it is necessary to first introduce the concept of *free electron level* and *work function* which play key roles in the relative energy band placement when two different materials are brought into contact. Figure 2.28(c) shows the band diagram of a

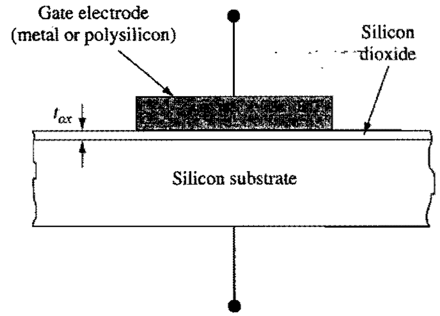

Figure 2.27. Schematic cross section of an MOS capacitor.

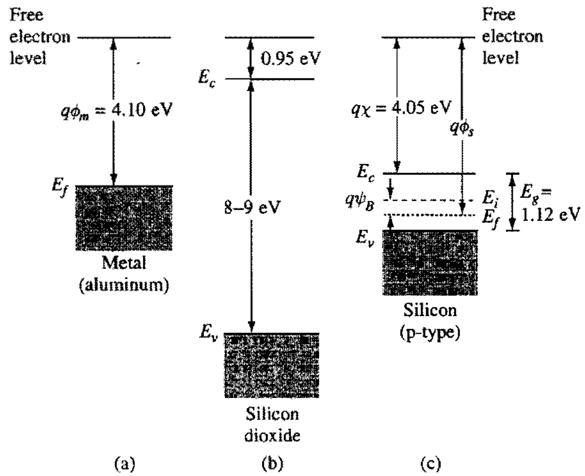

Figure 2.28. Energy-band diagram of the three components of an MOS capacitor: (a) metal (aluminum), (b) silicon dioxide, and (c) p-type silicon.

p-type silicon with the addition of the free electron level at some energy above the conduction band. The free electron level is defined as the energy level above which the electron is free, i.e., no longer bonded to the lattice.[3] In silicon, the free electron level is 4.05 eV above the conduction band edge, as shown in Figure. 2.28(c). In other words, an electron at the conduction band edge must gain an additional energy of 4.05 eV (called the *electron affinity*, $q\chi$) in order to break loose from the crystal field of silicon. Figure 2.28(b) shows the band diagram of silicon dioxide – an insulator with a large energy gap in the range of 8–9 eV. The free electron level in silicon dioxide is 0.95 eV above its conduction band.

[3] In other texts, the free electron level is often referred to as the *vacuum level*. Here we use a different term to avoid the implication that the vacuum level is universal.

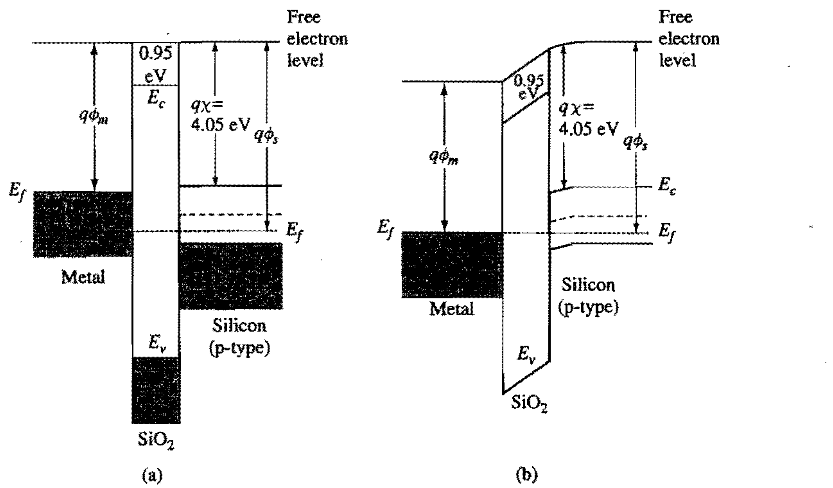

Figure 2.29. Band diagrams of an MOS system under (a) the flatband condition, and (b) zero gate-voltage condition.

Work function is defined as the energy difference between the free electron level and the Fermi level. For the p-type silicon example in Fig. 2.28(c), the work function, $q\phi_s$, can be expressed as:

$$q\phi_s = q\chi + \frac{E_g}{2} + q\psi_B. \tag{2.169}$$

Here ψ_B is the difference between the Fermi potential and the intrinsic potential given by Eq. (2.48). The same definition of work function, $q\phi_m$, applies to metals (remember that the conduction band is half filled in metals), as shown in Fig. 2.28(a). It means that an electron at the Fermi level needs to receive an energy equal to $q\phi_m$ to be free from the metal. Different metals have different work functions.

When two different materials are brought into contact, they must share the same free electron level at the interface, i.e., the free electron level is continuous from one material to the next. This is because at the interface of two materials, an electron that is free from the crystal field of one material is also free from the crystal field of the other material. Figure 2.29(a) shows the band diagram of an MOS system under the *flatband condition* in which there is no field in all three materials. Since for this example the metal work function is less than the silicon work function, the flatband condition is reached by applying a negative gate voltage, $-(\phi_s - \phi_m) \equiv \phi_{ms}$, called the *flatband voltage*, with respect to the silicon substrate. This is seen in Fig. 2.29(a) as the displacement between the two Fermi levels. In general, the flatband voltage of an MOS device is given by

$$V_{fb} = (\phi_m - \phi_s) - \frac{Q_{ox}}{C_{ox}} = \phi_{ms} - \frac{Q_{ox}}{C_{ox}}, \tag{2.170}$$

where Q_{ox} is the equivalent oxide charge per unit area at the oxide–silicon interface (defined in Section 2.3.7), and C_{ox} is the oxide capacitance per unit area,

ε_{ox} (2.171)

for an oxide film of thickness t_{ox} and permittivity ε_{ox}. In modern VLSI technologies, Q_{ox}/q at the Si–SiO_2 interface can be controlled to below 10^{10} cm^{-2} (positive) for $\langle 100 \rangle$-oriented surfaces. Its contribution to the flatband voltage is less than 50 mV for thin gate oxides used in 1-μm technology and below ($t_{ox} \leq 20$ nm). Therefore, the flatband voltage is mainly determined by the work function difference ϕ_{ms}.

At zero gate voltage when the Fermi levels line up, electric fields are developed in both the oxide and silicon, as shown in Fig. 2.29(b). One should note that the electron affinity, $q\chi$, is a material property which depends only on the type of the semiconductor and does not change with either the location or the doping type. When there is band bending as in Fig. 2.29(b) or, e.g., as in the depletion region of a p–n junction, the free electron level would bend in parallel with the conduction band such that the distance between the free electron level and the conduction band remains constant, much like the energy gap. While in this case the free electron level is not flat within one material, it must still be continuous between adjacent materials. Because of the common free electron level at the interface, ***the electron energy barrier is $q\phi_{ox}$ = 4.05 eV − 0.95 eV = 3.1 eV between the conduction bands of silicon and silicon dioxide.*** This figure has important significance when discussing the reliability of Si–SiO_2 systems (Section 2.5.3). For simplicity, the free electron level is mostly omitted in subsequent MOS band diagrams. One should keep in mind its essential role in setting the relative band placement between different materials.

The same principle of continuity of free-electron level at interface applies to metal–semiconductor junctions (Schottky diodes discussed in Section 2.4.1) and heterojunction devices with different bandgaps in establishing their band alignment relationships.

2.3.1.2 Gate Voltage and Surface Potential

Figure 2.30 shows the band diagrams of an MOS device when different gate voltages are applied. Free electron levels are omitted. The top level in the oxide region represents the

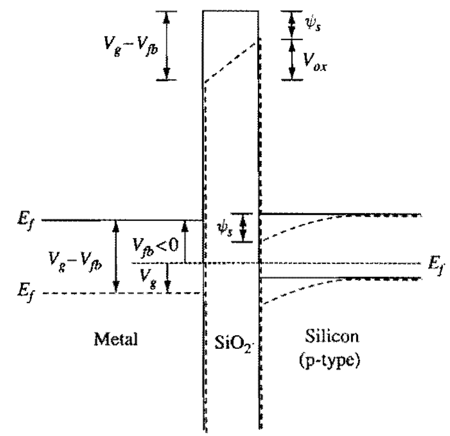

Figure 2.30. Energy band and potential diagrams of an MOS capacitor showing how the bands change under different gate bias conditions. The solid lines represent the flatband condition. The dashed lines represent the condition when a positive gate voltage is applied. Note that in the diagram the electron

conduction band. The same energy band diagrams are also used as potential diagrams to show the relationship between the applied voltage and the potential drop in different regions of the device. The applied gate voltage is referenced with respect to the Fermi level of the p-type substrate. In the flatband condition depicted by the solid lines in Fig. 2.30, $V_{fb} < 0$ is applied to the gate, as in the case of Fig. 2.29(a). When a positive gate voltage, V_g, is applied, as depicted by the dashed lines in Fig. 2.30, the metal Fermi level is displaced downward from that of the flatband condition by a total amount of $V_g - V_{fb}$. Because of the fixed band relationship between the metal and oxide, the oxide conduction band on the metal side is also displaced downward by the same amount. This causes a field to develop in the oxide and, at the same time, a downward bending of the bands in the p-type silicon near the surface. The amount of band bending in silicon is defined as the *surface potential*, ψ_s, i.e., the potential at the silicon surface relative to that in the bulk substrate. Because of the fixed band relationship between the oxide and silicon, it is clear that

$$V_g - V_{fb} = \psi_s + V_{ox}, \tag{2.172}$$

where V_{ox} is the potential drop across the oxide, as indicated in Fig. 2.30.

How $V_g - V_{fb}$ is partitioned into ψ_s and V_{ox} depends on both the oxide thickness and the doping concentration of the p-type silicon. Based on the dielectric boundary conditions discussed in Section 2.1.4.2, a field relationship exists at the silicon–oxide interface,

$$\varepsilon_{ox}\mathscr{E}_{ox} = \varepsilon_{si}\mathscr{E}_s, \tag{2.173}$$

or $\mathscr{E}_{ox} \approx 3\mathscr{E}_s$, assuming negligible trapped charge at the interface. Note that the above equation applies to both the magnitude and the direction of the fields. In most cases, there is negligible net charge in the oxide and Poisson's equation becomes $d\mathscr{E}/dx = 0$. Therefore, the field in the oxide is constant,[4] and $V_{ox} = \mathscr{E}_{ox}t_{ox}$.

2.3.1.3 Accumulation, Depletion, and Inversion

Figure 2.31 shows the band diagrams of p-type ((a)–(d)) and n-type ((e)–(h)) MOS capacitors under different gate bias voltages with respect to the flatband voltage. For simplicity, the flatband voltage is taken to be zero for all cases. The flatband condition for p-type MOS discussed before is shown in Fig. 2.31(a). There is no charge, no field, and the carrier concentration equals the ionized acceptor concentration throughout the silicon. Now consider the case when a negative voltage is applied to the gate of a p-type MOS capacitor, as shown in Fig. 2.31(b). This raises the metal Fermi level (i.e., electron energy) with respect to the silicon Fermi level and creates an electric field in the oxide that would accelerate a negative charge toward the silicon substrate. A field is also induced at the silicon surface in the same direction as the oxide field. Because of the low carrier concentration in silicon (compared with metal), the bands bend upward toward the oxide interface. ***The Fermi level stays flat within the silicon, since there is no net flow of conduction current***, as was discussed in Section 2.1.4.5. Due to the band bending, the

[4] If the field in the oxide is not constant, $\mathscr{E}_{ox}$ in Eq. (2.173) is defined as the oxide field at the oxide–silicon interface.

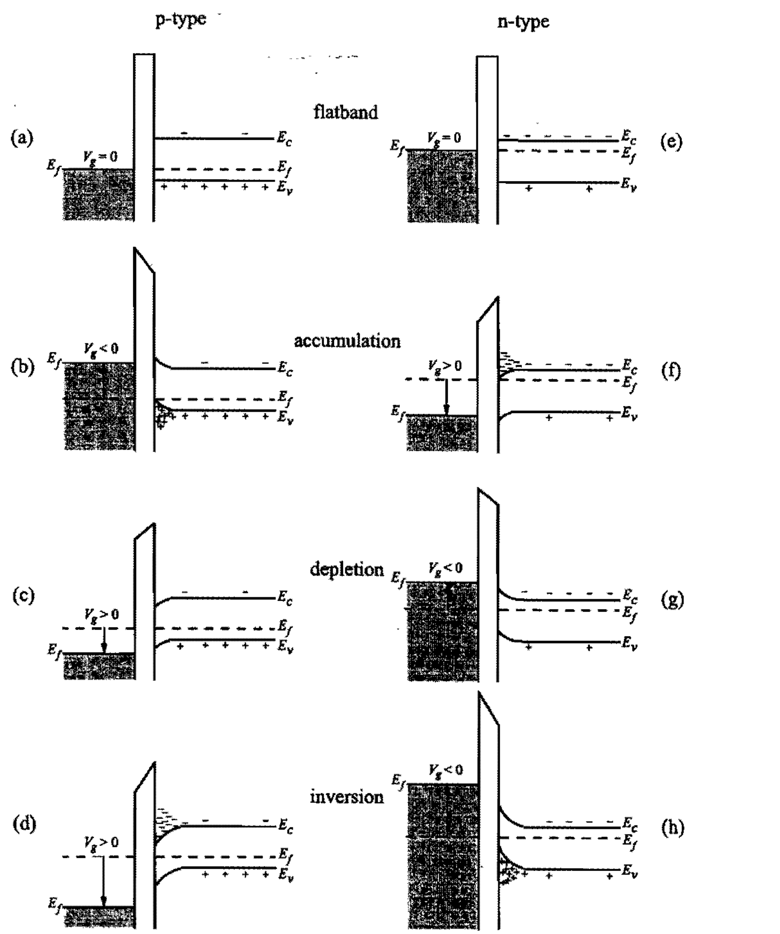

Figure 2.31. Energy-band diagrams for ideal (zero flatband voltage) (a)–(d) p-type and (e)–(h) n-type MOS capacitors under different bias conditions: (a), (e), flat band; (b), (f), accumulation; (c), (g), depletion; (d), (h), inversion. (After Sze, 1981.)

valence band at the surface is much closer to the Fermi level than is the valence band in the bulk silicon. This results in a hole concentration much higher at the surface than the equilibrium hole concentration in the bulk. Since excess holes are accumulated at the surface, this is referred to as the *accumulation* condition. One can think of the excess holes as being attracted toward the surface by the negative gate voltage. An equal amount of negative charge appears on the metal side of the MOS capacitor, as required for charge neutrality.

On the other hand, if a positive voltage is applied to the gate of a p-type MOS capacitor, the metal Fermi level moves downward, which creates an oxide field in the direction of accelerating a negative charge toward the metal electrode. A similar field is induced in the silicon, which causes the bands to bend downward toward the surface, as shown in Fig. 2.31(c). Since the valence band at the surface is now farther away from the Fermi level than is the valence band in the bulk, the hole concentration at the surface is lower than the concentration in the bulk. This is referred to as the *depletion* condition. One can think of the holes as being repelled away from the surface by the positive gate voltage. The situation is similar to the depletion layer in a p–n junction discussed in Section 2.2.2. The depletion of holes at the surface leaves the region with a net negative charge arising from the unbalanced acceptor ions. An equal amount of positive charge appears on the metal side of the capacitor.

As the positive gate voltage increases, the band bending also increases, resulting in a wider depletion region and more (negative) depletion charge. This goes on until the bands bend downward so much that at the surface, the conduction band is closer to the Fermi level than the valence band is, as shown in Fig. 2.31(d). When this happens, not only are the holes depleted from the surface, but the surface potential is such that it is energetically favorable for electrons to populate the conduction band. In other words, the surface behaves like n-type material with an electron concentration given by Eq. (2.49). ***Note that this n-type surface is formed not by doping, but instead by inverting the original p-type substrate with an applied electric field.*** This condition is called *inversion*. The negative charge in the silicon consists of both the ionized acceptors and the thermally generated electrons in the conduction band. Again, it is balanced by an equal amount of positive charge on the metal gate. The surface is inverted as soon as $E_i = (E_c + E_v)/2$ crosses E_f. This is called *weak inversion* because the electron concentration remains small until E_i is considerably below E_f. If the gate voltage is increased further, the concentration of electrons at the surface will be equal to, and then exceed, the hole concentration in the substrate. This condition is called *strong inversion*.

So far we have discussed the band bending for accumulation, depletion, and inversion of silicon surface in a p-type MOS capacitor. Similar conditions hold true in an n-type MOS capacitor, except that the polarities of voltage, charge, and band bending are reversed, and the roles of electrons and holes are interchanged. The band diagrams for flatband, accumulation, depletion, and inversion conditions of an n-type MOS capacitor are shown in Fig. 2.31(e)–(h), where the metal work function per electron charge ϕ_m is assumed to be equal to that of the n-type silicon, given by

$$\phi_s = \chi + \frac{E_g}{2q} - \psi_B, \tag{2.174}$$

instead of Eq. (2.169). Accumulation occurs when a positive voltage is applied to the metal gate and the silicon bands bend downward at the surface. Depletion and inversion occur when the gate voltage is negative and the bands bend upward toward the surface.

2.3.2 Electrostatic Potential and Charge Distribution in Silicon

2.3.2.1 Solving Poisson's Equation

In this sub-subsection, the relations among the surface potential, charge, and electric field are derived by solving Poisson's equation in the surface region of silicon. A more detailed band diagram at the surface of a p-type silicon is shown in Fig. 2.32. The potential $\psi(x) =$

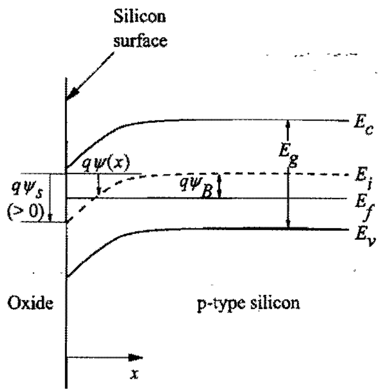

Figure 2.32. Energy-band diagram near the silicon surface of a p-type MOS device. The band bending ψ is defined as positive when the bands bend downward with respect to the bulk. Accumulation occurs when $\psi_s < 0$. Depletion and inversion occur when $\psi_s > 0$.

$\psi_i(x) - \psi_i(x = \infty)$ is defined as the amount of band bending at position x, where $x = 0$ is at the silicon surface and $\psi_i(x = \infty)$ is the intrinsic potential in the bulk silicon. Remember that $\psi(x)$ is positive when the bands bend downward. The boundary conditions are $\psi = 0$ in the bulk silicon, and $\psi = \psi(0) = \psi_s$ at the surface. The surface potential ψ_s depends on the applied gate voltage, as discussed in Section 2.3.1.2. Poisson's equation, Eq. (2.44), is

$$\frac{d^2\psi}{dx^2} = -\frac{d\mathscr{E}}{dx} = -\frac{q}{\varepsilon_{si}}\left[p(x) - n(x) + N_d^+(x) - N_a^-(x)\right]. \tag{2.175}$$

For a uniformly doped p-type substrate of acceptor concentration N_a with complete ionization, $N_d^+(x) - N_a^-(x) = -N_a$, independent of x. Charge neutrality condition deep in the bulk substrate requires

$$N_d^+(x) - N_a^-(x) = -N_a = -p_0 + \frac{n_i^2}{p_0}, \tag{2.176}$$

where $p_0 = p(x = \infty)$ and $n_i^2/p_0 = n(x = \infty)$ are the majority (holes) and minority (electrons) carrier densities in the bulk substrate, respectively. In general, $p(x)$ and $n(x)$ are given by Eq. (2.50) and Eq. (2.49), which can be expressed in terms of $\psi(x)$ using $\psi(x) = \psi_i(x) - \psi_i(x = \infty)$ and $\psi_B = \psi_f - \psi_i(x = \infty)$ as defined in Fig. 2.32:

$$p(x) = n_i e^{q[\psi_f - \psi_i(x)]/kT} = n_i e^{q[\psi_B - \psi(x)]/kT} = p_0 e^{-q\psi(x)/kT} \tag{2.177}$$

and

$$n(x) = n_i e^{q[\psi_i(x) - \psi_f]/kT} = n_i e^{q[\psi(x) - \psi_B]/kT} = \frac{n_i^2}{p_0} e^{q\psi(x)/kT}. \tag{2.178}$$

Note that ψ_f is independent of x because there is no net current flow perpendicular to the surface and the Fermi level stays flat. Also note that $p_0 = p(x = \infty) = n_i \exp(q\psi_B/kT)$ and $n_i^2/p_0 = n(x = \infty) = n_i\exp(-q\psi_B/kT)$ in the last step of the above equations.

In practice, $N_a \gg n_i$, and $p_0 \approx N_a$ from Eq. (2.176). Substituting the last three equations into Eq. (2.175) and replacing p_0 by N_a yield

$$\frac{d^2\psi}{dx^2} = -\frac{q}{\varepsilon_{si}}\left[N_a\left(e^{-q\psi/kT}-1\right) - \frac{n_i^2}{N_a}\left(e^{q\psi/kT}-1\right)\right]. \tag{2.179}$$

Multiplying $(d\psi/dx)\,dx$ on both sides of Eq. (2.179) and integrating from the bulk ($\psi = 0$, $d\psi/dx = 0$) toward the surface, one obtains

$$\begin{aligned}&\int_0^{d\psi/dx}\frac{d\psi}{dx}\,d\!\left(\frac{d\psi}{dx}\right)\\ &= -\frac{q}{\varepsilon_{si}}\int_0^{\psi}\left[N_a\left(e^{-q\psi/kT}-1\right) - \frac{n_i^2}{N_a}\left(e^{q\psi/kT}-1\right)\right]d\psi,\end{aligned} \tag{2.180}$$

which gives the electric field at x, $\mathscr{E} = -d\psi/dx$, in terms of ψ:

$$\begin{aligned}\mathscr{E}^2(x) = \left(\frac{d\psi}{dx}\right)^2 = \frac{2kTN_a}{\varepsilon_{si}}&\left[\left(e^{-q\psi/kT}+\frac{q\psi}{kT}-1\right)\right.\\ &\left.+\frac{n_i^2}{N_a^2}\left(e^{q\psi/kT}-\frac{q\psi}{kT}-1\right)\right].\end{aligned} \tag{2.181}$$

At $x = 0$, we let $\psi = \psi_s$ and $\mathscr{E} = \mathscr{E}_s$. From Gauss's law, Eq. (2.43), the total charge per unit area induced in the silicon (equal and opposite to the charge on the metal gate) is

$$\begin{aligned}Q_s = -\varepsilon_{si}\mathscr{E}_s = \pm\sqrt{2\varepsilon_{si}kTN_a}&\left[\left(e^{-q\psi_s/kT}+\frac{q\psi_s}{kT}-1\right)\right.\\ &\left.+\frac{n_i^2}{N_a^2}\left(e^{q\psi_s/kT}-\frac{q\psi_s}{kT}-1\right)\right]^{1/2}.\end{aligned} \tag{2.182}$$

This function is plotted in Fig. 2.33. At the flat-band condition, $\psi_s = 0$ and $Q_s = 0$. In accumulation, $\psi_s < 0$ (bands bending upward) and the first term in the square brackets dominates once $-q\psi_s/kT > 1$. The accumulation charge density is then proportional to $\exp(-q\psi_s/2kT)$ as indicated in the Fig 2.33.[5] In depletion, $\psi_s > 0$ and $q\psi_s/kT > 1$, but $\exp(q\psi_s/kT)$ is not large enough to make the n_i^2/N_a^2 term appreciable. Therefore, the $q\psi_s/kT$ term in the square brackets dominates and the negative depletion charge density (from ionized acceptor atoms) is proportional to $\psi_s^{1/2}$. When ψ_s increases further, the $(n_i^2/N_a^2)\exp(q\psi_s/kT)$ term eventually becomes larger than the $q\psi_s/kT$ term and dominates the square bracket. This is when inversion occurs. The negative inversion charge density is proportional to $\exp(q\psi_s/2kT)$ as indicated in Fig. 2.33.

A popular criterion for the onset of strong inversion is for the surface potential to reach a value such that $(n_i^2/N_a^2)\exp(q\psi_s/kT) = 1$, i.e.,

$$\psi_s(\text{inv}) = 2\psi_B = 2\frac{kT}{q}\ln\left(\frac{N_a}{n_i}\right). \tag{2.183}$$

[5] These results stem from the Maxwell–Boltzmann approximation made in Section 2.1.1.3, which overestimates the occupancy of electron states near and below the Fermi level. For accumulation and inversion layers with carrier densities in the degenerate range, i.e., when $\psi_s < -0.2$ V or > 0.9 V where the Fermi level goes into the valence or the conduction band, the more exact Fermi–Dirac distribution gives a less steep rise of the sheet charge density with the surface potential.

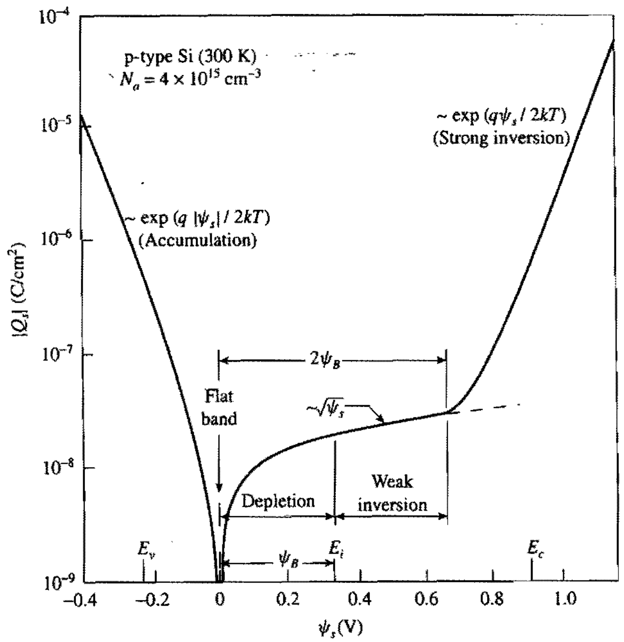

Figure 2.33. Variation of total charge density (fixed plus mobile) in silicon as a function of surface potential ψ_s for a p-type MOS device. The labels E_v, E_i, E_c indicate the surface potential values where the valence band, the intrinsic level, and the conduction band cross the Fermi level. (After Sze, 1981.)

Under this condition, the electron concentration given by Eq. (2.178) at the surface becomes equal to the depletion charge density N_a. ***After inversion takes place, even a slight increase in the surface potential results in a large buildup of electron density at the surface.*** The inversion layer effectively shields the silicon from further penetration of the gate field. Since almost all of the incremental charge is taken up by electrons, there is no further increase of either the depletion charge or the depletion-layer width. The expression in Eq. (2.183) is a rather weak function of the substrate doping concentration. For typical values of $N_a = 10^{16}$–10^{18} cm^{-3}, $2\psi_B$ varies only slightly, from 0.70 to 0.94 V.

2.3.2.2 Depletion Approximation

In general, Eq. (2.181) must be solved numerically to obtain $\psi(x)$. In particular cases, approximations can be made to allow the integral to be carried out analytically. For example, in the depletion region where $2\psi_B > \psi > kT/q$, only the $q\psi/kT$ term in the square bracket needs to be kept and

$$\frac{d\psi}{dx} = -\sqrt{\frac{2qN_a\psi}{\varepsilon_{si}}}. \tag{2.184}$$

One can then rearrange the factors and integrate:

$$\int_{\psi_s}^{\psi} \frac{d\psi}{\sqrt{\psi}} = -\int_0^x \sqrt{\frac{2qN_a}{\varepsilon_{si}}}\, dx, \tag{2.185}$$

where ψ_s is the surface potential at $x = 0$ as assumed before. Therefore,

$$\psi = \psi_s \left(1 - \sqrt{\frac{qN_a}{2\varepsilon_{si}\psi_s}}\, x\right)^2, \tag{2.186}$$

which can be written as

$$\psi = \psi_s \left(1 - \frac{x}{W_d}\right)^2. \tag{2.187}$$

This is a parabolic equation with the vertex at $\psi = 0$, $x = W_d$, where

$$W_d = \sqrt{\frac{2\varepsilon_{si}\psi_s}{qN_a}} \tag{2.188}$$

is the depletion-layer width defined as the distance to which the band bending extends. The total depletion charge density in silicon, Q_d, is equal to the charge per unit area of ionized acceptors in the depletion region:

$$Q_d = -qN_a W_d = -\sqrt{2\varepsilon_{si} q N_a \psi_s}. \tag{2.189}$$

These results are very similar to those of the one-sided abrupt p–n junction under the depletion approximation, discussed in Section 2.2.2. ***In the MOS case, however, W_d reaches a maximum value W_{dm} at the onset of strong inversion when $\psi_s = 2\psi_B$.*** Substituting Eq. (2.183) into Eq. (2.188) gives the maximum depletion width:

$$W_{dm} = \sqrt{\frac{4\varepsilon_{si} kT \ln(N_a/n_i)}{q^2 N_a}}. \tag{2.190}$$

2.3.2.3 Strong Inversion

Beyond strong inversion, the $(n_i^2/N_a^2)\exp(q\psi/kT)$ term representing the inversion charge in Eq. (2.181) becomes appreciable and must be kept, together with the depletion charge term:

$$\frac{d\psi}{dx} = -\sqrt{\frac{2kTN_a}{\varepsilon_{si}}\left(\frac{q\psi}{kT} + \frac{n_i^2}{N_a^2} e^{q\psi/kT}\right)}. \tag{2.191}$$

This equation can only be integrated numerically. The boundary condition is $\psi = \psi_s$ at $x = 0$. After $\psi(x)$ is solved, the electron distribution $n(x)$ in the inversion layer can be calculated from Eq. (2.178). Examples of numerically calculated $n(x)$ are plotted in Fig. 2.34 for two values of ψ_s with $N_a = 10^{16}$ cm^{-3}. ***The electrons are distributed extremely close to the surface with an inversion-layer width less than 50 Å.*** A higher surface potential or field tends to confine the electrons even closer to the surface. In

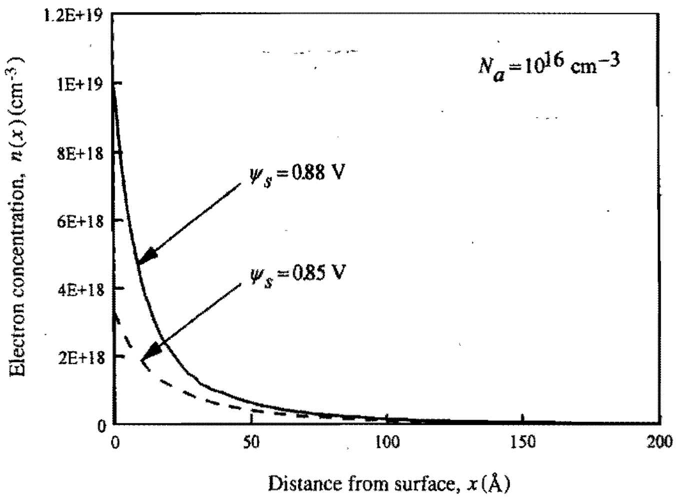

Figure 2.34. Electron concentration versus distance in the inversion layer of a p-type MOS device.

general, electrons in the inversion layer must be treated quantum-mechanically as a 2-D gas (Stern and Howard, 1967). According to the quantum-mechanical model, inversion-layer electrons occupy discrete energy bands and have a peak distribution 10–20 Å away from the surface. More details will be discussed in Section 4.2.4.

When the inversion charge density per unit area, Q_i, is much greater than the depletion charge density, Eq. (2.182) can be approximated by

$$Q_i = -\sqrt{\frac{2\varepsilon_{si} kT n_i^2}{N_a}}\, e^{q\psi_s/2kT}. \tag{2.192}$$

Since the electron concentration at the surface is

$$n(0) = \frac{n_i^2}{N_a} e^{q\psi_s/kT}, \tag{2.193}$$

one can write

$$|Q_i| = \sqrt{2\varepsilon_{si} kT n(0)}. \tag{2.194}$$

The effective inversion-layer thickness (classical model) can be estimated from $Q_i/qn(0) = 2\varepsilon_{si}kT/qQ_i$, which is inversely proportional to Q_i. Similar expressions also hold true for the surface charge density of extra holes under accumulation, except that the factor n_i^2/N_a is replaced by N_a.

2.3.2.4 Surface Potential and Charge Density as a Function of Gate Voltage

In Section 2.3.2.1, charge and potential distributions in silicon were solved in terms of the surface potential ψ_s as a boundary condition. ψ_s is not directly measurable, but is controlled by and can be determined from the applied gate voltage. The gate voltage

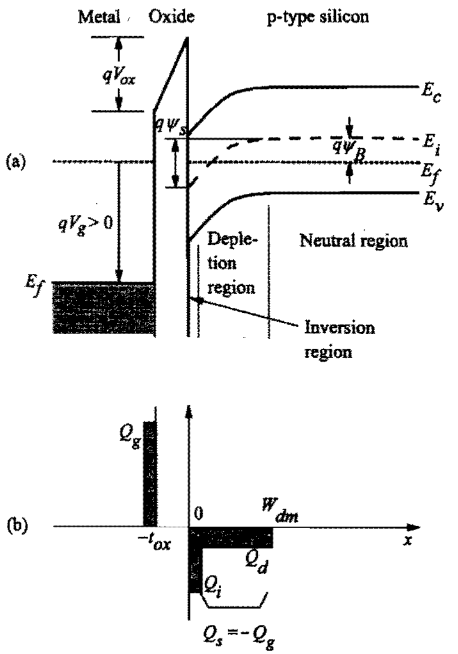

Figure 2.35. (a) Band diagram of a p-type MOS capacitor with a positive voltage applied to the gate ($V_{fb} = 0$). (b) Charge distribution under inversion condition.

equation, Eq. (2.172), relates the potential drop V_{ox} across the oxide and the band bending ψ_s in silicon to the departure from the flatband condition due to the applied gate voltage V_g (Fig. 2.35(a)). Assuming negligible fixed charges in the oxide, the potential drop V_{ox} can be expressed as $\mathscr{E}_{ox}t_{ox}$, which equals $(\varepsilon_{si}/\varepsilon_{ox})\mathscr{E}_s t_{ox}$ based on the boundary condition, Eq. (2.173). Applying Gauss's law, $Q_s = -\varepsilon_{si}\mathscr{E}_s$, ***the gate bias equation becomes***

$$V_g - V_{fb} = V_{ox} + \psi_s = \frac{-Q_s}{C_{ox}} + \psi_s, \tag{2.195}$$

where Q_s is the total charge per unit area induced in the silicon, and $C_{ox} = \varepsilon_{ox}/t_{ox}$ is the oxide capacitance per unit area for an oxide of thickness t_{ox}. There is a negative sign in front of Q_s in Eq. (2.195) because the charge on the metal gate is always equal but opposite to the charge in silicon, i.e., Q_s is negative when $V_g - V_{fb}$ is positive and vice versa. The charge distribution in an MOS capacitor is shown schematically in Fig. 2.35 (b), where the total charge Q_s may include both depletion and inversion components. For simplicity of discussion, oxide and interface trapped charges are ignored here. They will be discussed in detail in Sections 2.3.6 and 2.3.7.

In general, Q_s is a function of ψ_s given by Eq. (2.182), and plotted in Fig. 2.33. Equation (2.195) is then an implicit equation that can be solved for ψ_s as a function of V_g. An example of the numerical solution is shown in Fig. 2.36. Below the condition for strong inversion, $\psi_s = 2\psi_B$, ψ_s increases more or less linearly with V_g. Beyond $\psi_s = 2\psi_B$, ψ_s nearly saturates –

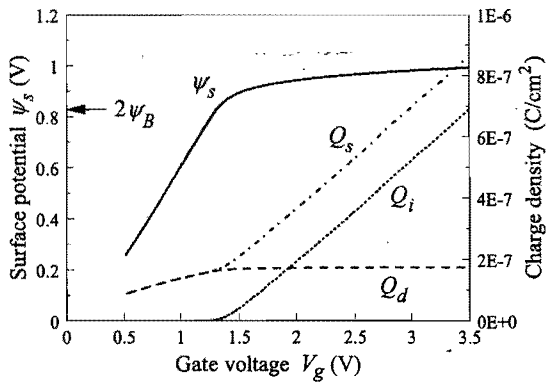

Figure 2.36. Numerical solutions of surface potential, total silicon charge density, inversion charge density, and depletion charge density from the gate voltage equation (2.195) coupled to Eq. (2.182). The MOS device parameters are $N_a = 10^{17}$ cm^{-3}, $t_{ox} = 10$ nm, and $V_{fb} = 0$.

increasing by less than 0.2 V while V_g increases by 2 V. After ψ_s is solved, Q_s is calculated and plotted as a function of V_g in Fig. 2.36. By numerically evaluating the integrals in Exercise 2.6, Q_s is separated into its two components, the depletion charge density Q_d and the inversion charge density Q_i, which are also plotted in Fig. 2.36. ***It is clear that before the $\psi_s = 2\psi_B$ condition, the charge in the silicon is predominantly of the depletion type.*** Under such depletion conditions, $Q_s(\psi_s) = Q_d(\psi_s)$, an analytical expression for $\psi_s(V_g)$ can be derived by solving a quadratic equation (see Eq. (2.202)). After $\psi_s = 2\psi_B$, the depletion charge no longer increases with V_g because of shielding by the inversion layer discussed before. ***Almost all of the increase of Q_s beyond $\psi_s = 2\psi_B$ is taken up by Q_i with a slope $dQ_i/dV_g \approx C_{ox}$.*** While on the linear scale it appears that Q_i is zero below the $\psi_s = 2\psi_B$ threshold, on the log scale it can be seen that Q_i actually remains finite and decreases exponentially with V_g. It is the source of the subthreshold leakage current in MOSFETs – an important design consideration further addressed in detail in Section 3.1.3.2. Under extreme accumulation and inversion conditions, $-Q_s \approx C_{ox}(V_g - V_{fb})$, since both V_g and V_{ox} can be much larger than the silicon bandgap, $E_g/q = 1.12$ V (for CMOS technologies with $V_{dd} \gg 1$ V), while ψ_s is at most comparable to E_g/q (surface potential pinned to either the valence band or the conduction band edge).

2.3.3 Capacitances in an MOS Structure

2.3.3.1 Definition of Small-Signal Capacitances

We now consider the capacitances in an MOS structure. In most cases, MOS capacitances are defined as small-signal differential of charge with respect to voltage or potential. They can easily be measured by applying a small ac voltage on top of a dc bias across the device and sensing the out-of-phase ac current at the same frequency (the in-phase component gives the small-signal conductance). The total MOS capacitance per unit area is

$$C_g = \frac{d(-Q_s)}{dV_g}. \tag{2.196}$$

If we differentiate Eq. (2.195) with respect to $-Q_s$ and define the silicon part of the capacitance as

$$C_{si} = \frac{d(-Q_s)}{d\psi_s}, \tag{2.197}$$

we obtain

$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{d\psi_s}{d(-Q_s)} = \frac{1}{C_{ox}} + \frac{1}{C_{si}}. \tag{2.198}$$

In other words, the total capacitance equals the oxide capacitance and the silicon capacitance connected in series. The capacitances are defined in such a way that they are all positive quantities. An equivalent circuit is shown in Fig. 2.37(a). In reality, there is also an interface trap capacitance in parallel with C_{si}. It arises from charging and discharging of Si–SiO_2 interface traps and will be discussed in more detail in Section 2.3.7.

2.3.3.2 Capacitance–Voltage Characteristics: Accumulation

A typical capacitance-versus-gate-voltage (C–V) curve of a p-type MOS capacitor is plotted in Fig. 2.38, assuming zero flatband voltage. In fact, there are several different curves, depending on the frequency of the applied ac signal. We start with the "low-frequency" or *quasistatic* C–V curve. When the gate voltage is negative (by more than a few kT/q) with respect to the flatband voltage, the p-type MOS capacitor is in accumulation and $Q_s \propto \exp(-q\psi_s/2kT)$, as shown in Fig. 2.33. Therefore, $C_{si} = -dQ_s/d\psi_s = (q/2kT)Q_s = (q/2kT)C_{ox}|V_g - V_{fb} - \psi_s|$, and the MOS capacitance per unit area is given by

$$\frac{1}{C_g} = \frac{1}{C_{ox}}\left[1 + \frac{2kT/q}{|V_g - V_{fb} - \psi_s|}\right]. \tag{2.199}$$

Since $2kT/q \approx 0.052$ V and ψ_s is limited to 0.1 to 0.3 V in accumulation, the MOS capacitance rapidly approaches C_{ox} when the gate voltage is 1–2 V more negative than the flat-band voltage.[6]

2.3.3.3 Capacitance at Flat Band

When the gate bias is zero in Fig. 2.38, the MOS is near the flat-band condition; therefore, $q\psi_s/kT \ll 1$. The inversion charge term in Eq. (2.182) can be neglected and the first exponential term can be expanded into a power series. Keeping only the first three terms of the series, one obtains $Q_s = -(\varepsilon_{si}q^2N_a/kT)^{1/2}\psi_s$. From Eq. (2.198), the flatband capacitance per unit area is given by

$$\frac{1}{C_{fb}} = \frac{1}{C_{ox}} + \sqrt{\frac{kT}{\varepsilon_{si}q^2N_a}} = \frac{1}{C_{ox}} + \frac{L_D}{\varepsilon_{si}}, \tag{2.200}$$

[6] Actually, C_g approaches C_{ox} slower than that depicted by Eq. (2.199) because of the Fermi–Dirac distribution at degenerate carrier densities.

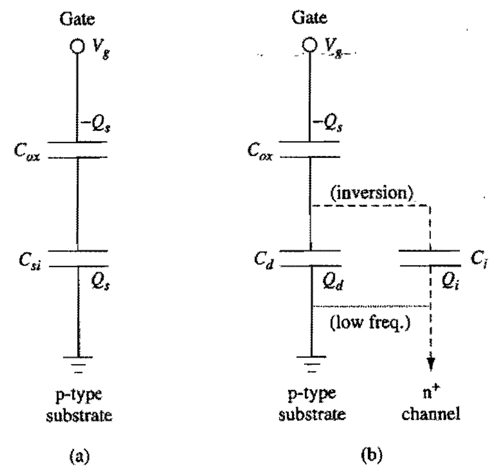

Figure 2.37. Equivalent circuits of an MOS capacitor. (a) All the silicon capacitances are lumped into C_{si}. (b) C_{si} is broken up into a depletion charge capacitance C_d and an inversion-layer capacitance C_i. C_d arises from the majority carriers, which can respond to high-frequency as well as low-frequency signals. C_i arises from the minority carriers, which can only respond to low-frequency signals, unless the surface inversion channel is connected to a reservoir of minority carriers as in a gated diode configuration. The thin dotted connection in (b) is effective only at low frequencies where minority carriers can respond.

where L_D is the Debye length defined in Eq. (2.53). In most cases, C_{fb} is somewhat less than C_{ox}. For very thin oxides and low substrate doping, C_{fb} can be much smaller than C_{ox}.

2.3.3.4 Capacitance–Voltage Characteristics: Depletion

When the gate voltage is slightly higher than the flatband voltage in a p-type MOS capacitor, the surface starts to be depleted of holes; $1/C_{si}$ becomes appreciable and the capacitance decreases. Using the depletion approximation, one can find an analytical expression for C_g in this case. From Eq. (2.188) and Eq. (2.189),

$$C_d = \frac{d(-Q_d)}{d\psi_s} = \sqrt{\frac{\varepsilon_{si}qN_a}{2\psi_s}} = \frac{\varepsilon_{si}}{W_d}. \tag{2.201}$$

The last expression is identical to the depletion-layer capacitance per unit area in the p–n junction case discussed in Section 2.2.2. The bias equation (2.195) becomes

$$V_g - V_{fb} = \frac{qN_aW_d}{C_{ox}} + \psi_s = \frac{\sqrt{2\varepsilon_{si}qN_a\psi_s}}{C_{ox}} + \psi_s. \tag{2.202}$$

Substituting C_d from Eq. (2.201) for C_{si} in Eq. (2.198), and eliminating ψ_s using Eq. (2.202), one obtains

$$C_g = \frac{C_{ox}}{\sqrt{1 + [2C_{ox}^2(V_g - V_{fb})/(\varepsilon_{si}qN_a)]}}. \tag{2.203}$$

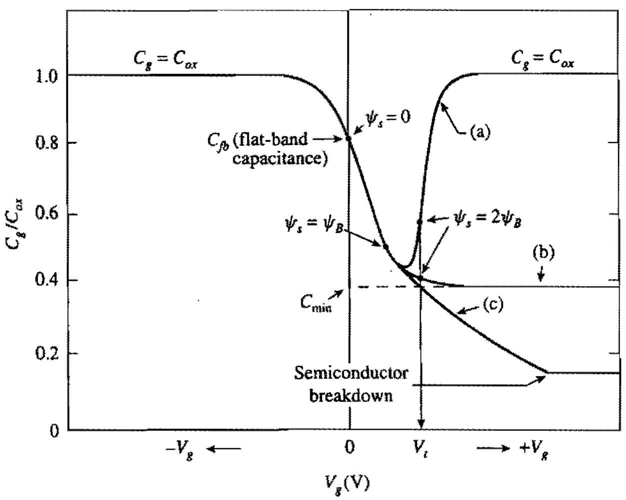

Figure 2.38. MOS capacitance-voltage curves: (a) low frequency, (b) high frequency, (c) deep depletion. $V_{fb} = 0$ is assumed. (After Size, 1981.)

This equation shows how the MOS capacitance decreases with increasing V_g under the depletion condition. It serves as a good approximation to the middle portions of the C–V curves in Fig. 2.38, provided that the MOS capacitor is not biased near the flat-band or the inversion condition.

2.3.3.5 Low-Frequency *C–V* Characteristics: Inversion

As the gate voltage increases further, however, the capacitance stops decreasing when $\psi_s = 2\psi_B$ [Eq. (2.183)] is reached and inversion occurs. ***Once the inversion layer forms, the capacitance starts to increase, since C_{si} is now given by the variation of the inversion charge with respect to ψ_s, which is much larger than the depletion capacitance.*** Assuming that the silicon charge is dominated by the inversion charge, one can carry out an approximation as in the accumulation case and show that the MOS capacitance in strong inversion is also given by Eq. (2.199). One difference is that ψ_s at inversion is in the range of 0.7 to 1.0 V, significantly higher than that at accumulation. In any case, the capacitance rapidly increases back to C_{ox} when the gate voltage is more than 2 to 3 V beyond the flat-band voltage, as shown in the low-frequency C–V curve (a) in Fig. 2.38.

2.3.3.6 High-Frequency Capacitance–Voltage Characteristics

The above discussion of the low-frequency MOS capacitance assumes that the minority carrier, i.e., the inversion charge, is able to follow the applied ac signal. This is true only if the frequency of the applied signal is lower than the reciprocal of the minority-carrier response time. The minority-carrier response time can be estimated from the generation–recombination current density, $J_R = qn_iW_d/\tau$, where τ is the minority-carrier lifetime discussed in Section 2.1.4. The time it takes to generate enough minority carriers to

replace something comparable to the depletion charge, $Q_d = qN_aW_d$, is on the order of $Q_d/J_R = (N_a/n_i)\tau$ (Jund and Poirier, 1966). This is typically 0.1–10 s. Therefore, ***for frequencies higher than 100 Hz or so, the inversion charge cannot respond to the applied ac signal.*** Only the depletion charge (majority carriers) can respond to the signal, which means that the silicon capacitance is given by C_d of Eq. (2.201) with W_d equal to its maximum value, W_{dm}, in Eq. (2.190). The high-frequency capacitance per unit area thus approaches a constant minimum value, C_{min}, at inversion given by

$$\frac{1}{C_{min}} = \frac{1}{C_{ox}} + \sqrt{\frac{4kT\ln(N_a/n_i)}{\varepsilon_{si}q^2N_a}}. \tag{2.204}$$

This is shown in the high-frequency C–V curve (b) in Fig. 2.38.

Typically, C–V curves are traced by applying a slow-varying ramp voltage to the gate with a small ac signal superimposed on it. However, if the ramp rate is fast enough that the ramping time is shorter than the minority-carrier response time, then there is insufficient time for the inversion layer to form, and the MOS capacitor is biased into deep depletion as shown by curve (c) in Fig. 2.38. In this case, the depletion width can exceed the maximum value given by Eq. (2.190), and the MOS capacitance decreases further below C_{min} until impact ionization takes place (Sze, 1981). Note that deep depletion is not a steady-state condition. If an MOS capacitor is held under such bias conditions, its capacitance will gradually increase toward C_{min} as the thermally generated minority charge builds up in the inversion layer until an equilibrium state is established. The time it takes for an MOS capacitor to recover from deep depletion and return to equilibrium is referred to as the *retention time*. It is a good indicator of the defect density in the silicon wafer and is often used to qualify processing tools in a facility.

It is possible to obtain *low-frequency-like* C–V curves at high measurement frequencies. One way is to expose the MOS capacitor to intense illumination, which generates a large number of minority carriers in the silicon. Another commonly used technique is to form an n^+ region adjacent to the MOS device and connect it electrically to the p-type substrate (Grove, 1967). The n^+ region then acts like a reservoir of electrons which can exchange minority carriers freely with the inversion layer. In other words, the n^+ region is connected to the *surface channel* of the inverted MOS device. This structure is similar to that of a gated diode, to be discussed in Section 2.3.5. Based on the equivalent circuit in Fig. 2.37(b), the total MOS capacitance per unit area is given by

$$C_g = \frac{C_{ox}(C_d + C_i)}{C_{ox} + C_d + C_i}. \tag{2.205}$$

When the MOS device is biased well into strong inversion, the *inversion-layer capacitance* C_i can be approximated by

$$C_i = \frac{d(-Q_i)}{d\psi_s} = \frac{|Q_i|}{2kT/q} \tag{2.206}$$

using Eq. (2.192).

The majority and minority carrier contributions to the total capacitance can be separately measured in a *split* C–V setup shown in Fig. 2.39(a) (Sodini *et al.*, 1982).

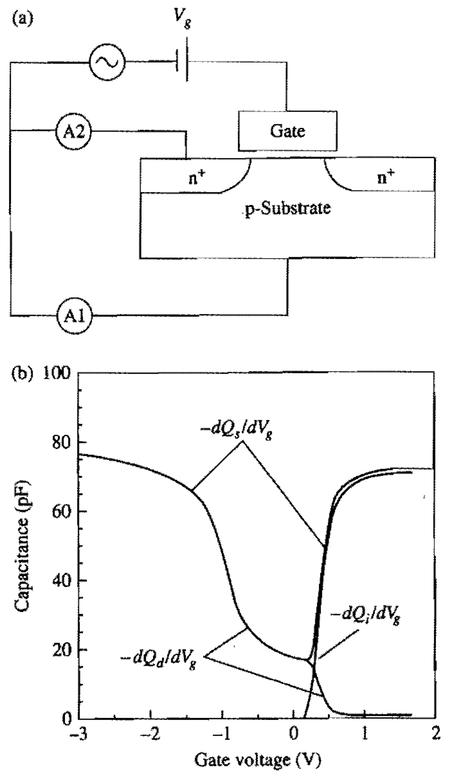

Figure 2.39. (a) Setup of the split C–V measurement. Both the dc bias and the small-signal ac voltage are applied to the gate. Small signal ac currents are measured by two ammeters, A1 and A2, connected separately as shown. (b) Measured C–V curves where the $-dQ_d/dV_g$ component is obtained from A1, and the $-dQ_i/dV_g$ component is obtained from A2. The sum is the total capacitance per unit area, $-dQ_s/dV_g$.

With a small signal ac voltage applied to the gate, the out-of-phase ac currents are sensed by two ammeters: one (A1) connected to the p-type substrate for the hole current, and another (A2) connected to the n$^+$ region for the electron current. Typical measured results are shown in Fig. 2.39(b). The hole contribution to the capacitance measured by A1 is

$$-\frac{dQ_d}{dV_g} = \frac{C_{ox}C_d}{C_{ox} + C_d + C_i}. \tag{2.207}$$

And the electron contribution to the capacitance measured by A2 is

$$-\frac{dQ_i}{dV_g} = \frac{C_{ox}C_i}{C_{ox} + C_d + C_i}. \tag{2.208}$$

They add up to the total capacitance per unit area, $C_g = -dQ_s/dV_g$. Note that the $-dQ_d/dV_g$ curve decreases to zero soon after strong inversion when C_i (Eq. (2.206)) becomes

dominant ($\gg C_{ox}$). To put it in another way, the highly conductive inversion channel shields the majority carriers in the bulk silicon so they do not respond to the modulation of gate field. The $-dQ_i/dV_g$ curve can be integrated to yield the inversion charge density as a function of the gate voltage. It is used, for example, in channel mobility measurements where the inversion charge density must be determined accurately.

2.3.4 Polysilicon-Gate Work Function and Depletion Effects

2.3.4.1 Work Function and Flatband Voltage of Polysilicon Gates

In the mainstream CMOS VLSI technology thus far, n$^+$-polysilicon gate has been used for nMOSFET and p$^+$-polysilicon gate for pMOSFET to obtain threshold voltages of low magnitude in both devices. The Fermi level of heavily doped n$^+$ polysilicon is near the conduction band edge, so its work function is given by the electron affinity, $q\chi$. From Eq. (2.169), the work function difference for an n$^+$ polysilicon gate on a p-type substrate of doping concentration N_a is

$$\phi_{ms} = -\frac{E_g}{2q} - \psi_B = -0.56 - \frac{kT}{q}\ln\left(\frac{N_a}{n_i}\right) \tag{2.209}$$

in volts. Similarly, the work function difference for a p$^+$ polysilicon gate on an n-type substrate of doping concentration N_d is

$$\phi_{ms} = \frac{E_g}{2q} + \psi_B = 0.56 + \frac{kT}{q}\ln\left(\frac{N_d}{n_i}\right), \tag{2.210}$$

which is symmetric to Eq. (2.209). These relations give rise to flatband voltages with key implications on the scalability of MOSFET devices, as will be discussed in Chapter 4.

The band diagram of an n$^+$-polysilicon-gated p-type MOS capacitor at zero gate voltage is shown in Fig. 2.40(a), where the Fermi levels line up and the free electron level of the bulk p-type silicon is higher in electron energy than the free electron level of the n$^+$ polysilicon gate. This sets up an oxide field in the direction of accelerating electrons toward the gate, and at the same time a downward bending of the silicon bands (depletion) toward the surface to produce a field in the same direction. The flatband condition is reached by applying a negative voltage equal to the work function difference to the gate, as shown in Fig. 2.40(b).

2.3.4.2 Polysilicon-Gate Depletion Effects

The use of polysilicon gates is a key advance in modern CMOS technology, since it allows the source and drain regions to be self-aligned to the gate, thus eliminating parasitics from overlay errors (Kerwin *et al.*, 1969). However, if the polysilicon gate is not doped heavily enough, problems can arise from depletion of the gate itself. This is especially a concern with the dual n$^+$–p$^+$polysilicon-gate process in which the gates are doped by ion implantation (Wong *et al.*, 1988). Gate depletion results in an additional capacitance in series with the oxide capacitance, which in turn leads to a reduced inversion-layer charge density and degradation of the MOSFET transconductance.

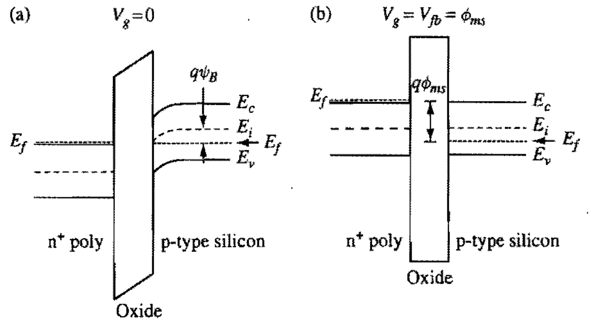

Figure 2.40. Band diagram of an n$^+$-polysilicon-gated p-type MOS capacitor biased at (a) zero gate voltage and (b) flatband condition.

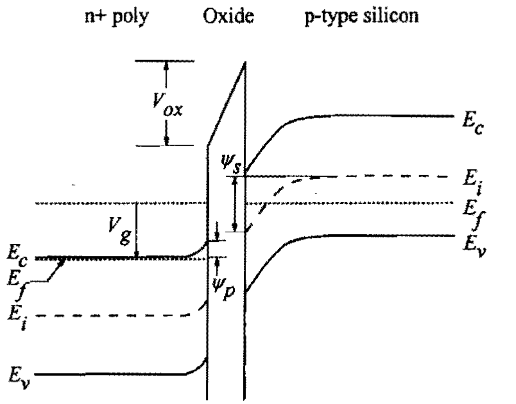

Figure 2.41. Band and potential diagram showing polysilicon-gate depletion effects when a positive voltage is applied to the n$^+$ polysilicon gate of a p-type MOS capacitor.

The analysis of MOS capacitance in the last subsection can be extended to include the polysilicon depletion effect and quantify certain observed features in the *C–V* characteristics. Consider the band diagram of an n$^+$-polysilicon-gated p-type MOS capacitor biased into inversion as shown in Fig. 2.41. Since the oxide field points in the direction of accelerating a negative charge toward the gate, the bands in the n$^+$ polysilicon bend slightly upward toward the oxide interface. This depletes the surface of electrons and forms a thin space-charge region in the polysilicon layer, which lowers the total capacitance.

2.3.4.3 Effect of Polysilicon Doping Concentration on *C–V* Characteristics

Typical low-frequency *C–V* curves in the presence of gate depletion effects are shown in Fig. 2.42 (Rios and Arora, 1994). A distinct feature is that the capacitance at inversion does not return to the full oxide capacitance as in Fig. 2.38. Instead, the inversion capacitance exhibits a maximum value somewhat less than C_{ox}, depending on the

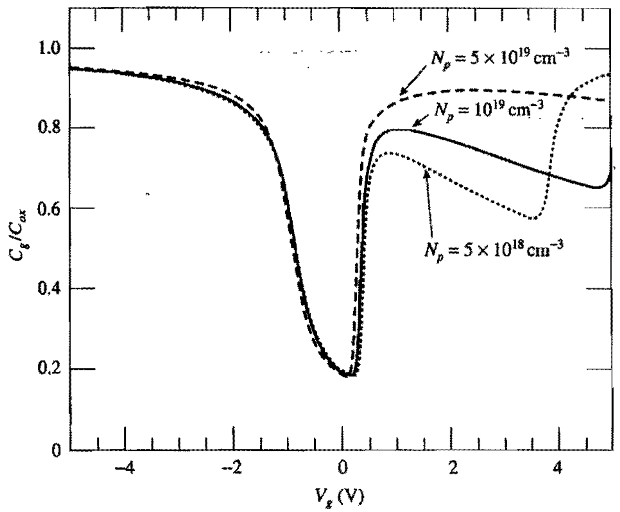

Figure 2.42. Low-frequency *C–V* curves of a p-type MOS capacitor with n$^+$ polysilicon gate doped at several different concentrations. (After Rios and Arora, 1994.)

effective doping concentration of the polysilicon gate. The higher the doping concentration, the less the gate depletion effect is and the closer the maximum capacitance is to the oxide capacitance.

The existence of a local maximum in the low-frequency *C–V* curve can be understood semiquantitatively as follows. In Fig. 2.41, we assume ψ_s to be the amount of band bending in the bulk silicon and ψ_p to be that in the n$^+$polysilicon. From charge neutrality, the total charge density Q_g of the ionized donors in the depletion region of the n$^+$ polysilicon gate is equal and opposite to the combined inversion and depletion charge density Q_s in the silicon substrate, i.e., $Q_g = -Q_s$. The gate bias equation for an applied voltage V_g is obtained by adding an additional term, ψ_p, for the band bending in the polysilicon gate to Eq. (2.195):

$$V_g = V_{fb} + \psi_s + \psi_p - \frac{Q_s}{C_{ox}}. \tag{2.211}$$

Differentiating Eq. (2.211) with respect to $-Q_s$ and using the capacitance definitions (2.196) and (2.197), we obtain

$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si}} + \frac{1}{C_p}, \tag{2.212}$$

where $C_p = -dQ_s/d\psi_p = dQ_g/d\psi_p$ is the capacitance of the polysilicon depletion region.

When a p-type MOS device is biased well into strong inversion such that Q_s is dominated by the inversion charge, the low-frequency capacitance is approximately given by Eq. (2.206), $C_{si} = |Q_s|/(2kT/q) = Q_g/(2kT/q)$. Based on the depletion approximation, Eqs. (2.189) and (2.201), the polysilicon depletion capacitance can be expressed in terms of the depletion charge density Q_g as $C_p = \varepsilon_{si}qN_p/Q_g$, where N_p is the doping concentration of the polysilicon gate. Substituting these expressions into Eq. (2.212) yields

$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{2kT/q}{Q_g} + \frac{Q_g}{\varepsilon_{si}qN_p}. \tag{2.213}$$

As V_g becomes more positive, C_{si} ($\propto Q_g$) increases but C_p ($\propto 1/Q_g$) decreases. This results in a local maximum of the low-frequency capacitance as observed in Fig. 2.42. For $N_p < 10^{19}$ cm^{-3}, another abrupt rise in the MOS capacitance may be observed at a much higher gate voltage, as shown in Fig. 2.42. This is due to the onset of inversion (to p$^+$) at the n$^+$ polysilicon surface.

In practice, polysilicon gates cannot be doped much higher than 1×10^{20} cm^{-3}. Modern CMOS devices often operate at a maximum oxide field of 5 MV/cm, corresponding to a sheet electron density of $Q_g/q = 10^{13}$ cm^{-2}. For these values, the polysilicon depletion capacitance has the equivalent effect as adding 3–4 Å to the gate oxide thickness. The above discussion applies to p$^+$ polysilicon gates on n-type silicon as well as to n$^+$ polysilicon gates on p-type silicon. Note that for n$^+$ polysilicon gates on n-type silicon and p$^+$ polysilicon gates on p-type silicon, gate depletion occurs when the substrate is accumulated.

2.3.5 MOS under Nonequilibrium and Gated Diodes

An important building block of VLSI devices is the gated diode, or gate-controlled diode. Consider an MOS capacitor where there is an n$^+$ region adjacent to the gated p-type region (Grove, 1967). The n$^+$ region and the p-type region form an n$^+$–p diode. This structure is shown schematically in Fig. 2.43 and was mentioned briefly at the end of Section 2.3.3.

2.3.5.1 Inversion Condition of an MOS Under Nonequilibrium

As discussed in Section 2.2.1, when both the n$^+$ region and the p-type substrate are connected to the same potential (grounded), the p–n junction is in equilibrium and the Fermi level is constant across the p–n junction. If the gate voltage is large enough to invert the p-type surface, which occurs for a surface potential bending of $\psi_s(\text{inv}) = 2\psi_B$, the inverted channel is connected to the n$^+$ region and has the same potential as the n$^+$ region. In other words, ***the electron quasi-Fermi level in the channel region is the same as the Fermi level in the n$^+$ region*** as well as the Fermi level in the p-type substrate. The depletion region now extends from the p–n junction to the region under the gate between the inverted channel and the substrate, as shown in Fig. 2.43(b).

If, on the other hand, the p–n junction is reverse-biased at a voltage V_R as shown in Fig. 2.44, the MOS is in a nonequilibrium condition in which $np \neq n_i^2$. From Eq. (2.107), the electron concentration on the p-type side of the junction is

$$n = \frac{n_i^2}{N_a} e^{-qV_R/kT}, \tag{2.214}$$

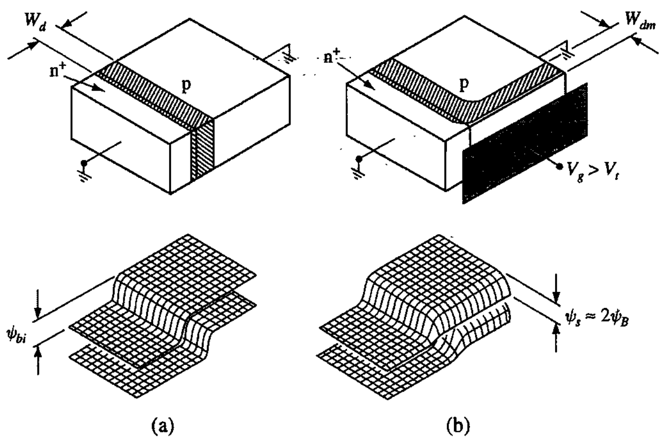

Figure 2.43. Gated diode or p-type MOS with adjacent n$^+$ region in equilibrium (zero voltage across the p–n junction). The gate is biased at (a) flatband and (b) inversion conditions. (After Grove, 1967.)

since the Fermi level of the n$^+$ region is now qV_R lower than the Fermi level in the p-type substrate.[7] Now consider the case when a positive voltage large enough to bend the bands by $2\psi_B$ is applied to the gate, as shown in Fig. 2.44(b). This brings the conduction band at the surface $2\psi_B$ closer to the electron quasi-Fermi level. From Eq. (2.178), the electron concentration at the surface is increased by a factor of $\exp(2q\psi_B / kT) = (N_a/n_i)^2$ over that of Eq. (2.214), i.e.,

$$n = \frac{n_i^2}{N_a} e^{2q\psi_B/kT} e^{-qV_R/kT} = N_a e^{-qV_R/kT}. \tag{2.215}$$

Since this is much lower than the depletion charge density N_a, the surface remains depleted. Even though the positive voltage is sufficient to invert the surface in the equilibrium case, it is not enough to cause inversion in the reverse-biased case. This is because ***the reverse bias lowers the quasi-Fermi level of electrons so that even if the bands at the surface are bent as much as in the equilibrium case in Fig. 2.43(b), the conduction band is still not close enough to the quasi-Fermi level of electrons for inversion to occur.***

To reach inversion in the nonequilibrium case, a much larger gate voltage, sufficient to bend the bands by $2\psi_B + V_R$, must be applied. This is the case shown in Fig. 2.44(c), where the electron concentration at the surface is now

$$n = \frac{n_i^2}{N_a} e^{q(2\psi_B+V_R)/kT} e^{-qV_R/kT} = N_a, \tag{2.216}$$

the same as the condition for inversion introduced in Section 2.3.2. Notice that the surface depletion layer is much wider than in the equilibrium case, just as in a reverse-biased p–n junction.

[7] While this is not exactly true, it will not affect later results. Further discussions follow Eq. (2.216).

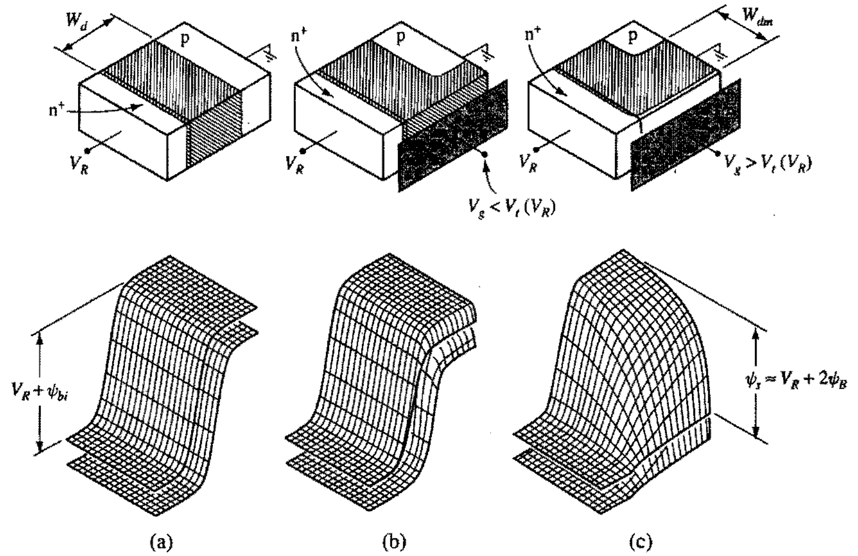

Figure 2.44. Gated diode or p-type MOS with adjacent n^+ region under nonequilibrium (reverse bias across the p–n junction). The gate is biased at (a) flat-band, (b) depletion, and (c) inversion conditions. (After Grove, 1967.)

To be more exact, Eq. (2.107), on which Eq. (2.214) is based, is not valid under a large reverse bias V_R. It is discussed in Appendix 4 that at large reverse biases, the electron quasi-Fermi level on the p-type side of the depletion region (at $x = 0$ in Fig. A4.5) is higher than the Fermi level of the n-type region (to the left of $x = -W_d$). In other words, the electron quasi-Fermi level on the p-type side of the depletion region is displaced downward from the Fermi level of the p-type region by less than qV_R. While this is true at the flatband condition in the gated diode depicted in Fig. 2.44(a), once a positive gate voltage is applied to bend down the p-type bands as in Fig. 2.44(c), the electron concentration at the p-type surface increases and the electron quasi-Fermi level there moves down toward the Fermi level of the n-type region. As far as the bands and the electron concentration at the p-type surface are concerned, the positive gate voltage has a similar effect as moving from the p-type side of the depletion region (at $x = 0$) toward the n-type side of the depletion region (at $x = -W_d$) in Fig. A4.5. When the threshold condition is reached, the electron quasi-Fermi level at the p-type surface would be the same as the Fermi level of the n-type region, i.e., displaced downward from the Fermi level of the p-type region by exactly qV_R. Another way to see this is that, in a region where the electron concentration is appreciable, electron quasi-Fermi level must remain essentially flat since the leakage current in a reverse-biased diode, $J_n \propto n\, d\phi_n/dx$, is negligibly small. In any case, the band bending requirement for threshold depicted in Fig. 2.44(c), $\psi_s = V_R + 2\psi_B$, is always valid.[8]

[8] This condition applies to the neutral p-type region. Less band bending is needed to reach the threshold for a point at the surface but within the depletion region of the reverse biased p–n junction, as is evident in Fig. 2.44(c).

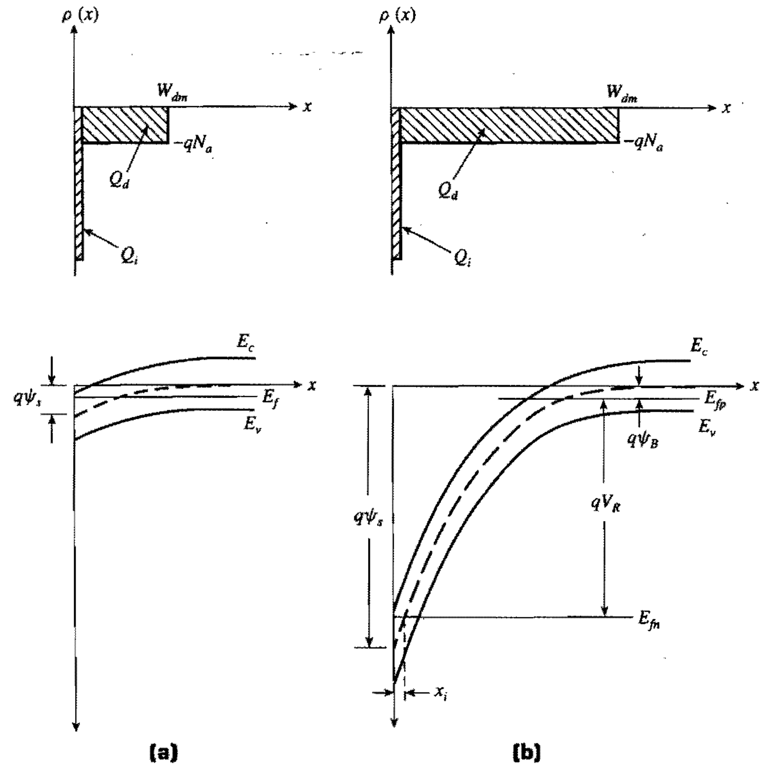

Figure 2.45. Comparison of charge distribution and energy-band variation of an inverted p-type region for (a) the equillibrium case and (b) the nonequillibrium case. (After Grove, 1967.)

2.3.5.2 Band Bending and Charge Distribution of an MOS Under Nonequilibrium

The above discussions are further illustrated in Fig. 2.45, where the charge distribution and band bending in a cross section perpendicular to the gate through the neutral p-type region are shown for both the equilibrium and the nonequilibrium cases. The equilibrium case is the same as that discussed in Section 2.3.2. In the nonequilibrium case, the hole quasi-Fermi level is the same as the Fermi level in the bulk p-type silicon, but the electron quasi-Fermi level is dictated by the Fermi level in the n^+ region (not shown in Fig. 2.45), which is qV_R lower than the p-type Fermi level. As a result, surface inversion occurs at a band bending

$$\psi_s(\text{inv}) = V_R + 2\psi_B, \tag{2.217}$$

and the maximum depletion width is a function of the reverse bias V_R,

$$W_{dm} = \sqrt{\frac{2\varepsilon_{si}(V_R + 2\psi_B)}{qN_a}}, \tag{2.218}$$

from Eq. (2.188).

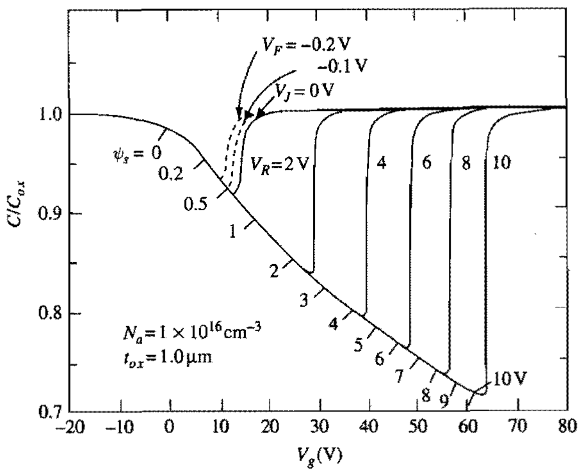

Figure 2.46. Normalized C–V characteristics of a p-type MOS capacitor with an adjacent n^+ region as that shown in Fig. 2.44(c). V_g is the voltage applied to the gate with respect to the p-type substrate. A series of C–V curves are shown for a range of V_R: the reverse bias voltage applied to the n^+/p junction.

When the surface is depleted, the gated diode behaves like an n^+–p diode with depletion of the p-region extending to underneath the gate electrode. When the surface is inverted, the gated diode behaves like an n^+–p diode with both the n^+ region and depletion of the p-region extending to underneath the gate electrode. High-field effects in gated diodes are discussed in Section 2.5.5

For a p-type MOS capacitor, the effect of an adjacent reverse biased n^+ region on the C–V characteristics is shown in Fig. 2.46. With increasing V_g, the surface potential ψ_s also increases, as labeled under the curve. At $V_R = 0$, the C–V curve resembles a regular low-frequency C–V curve (curve (a) in Fig. 2.38) with inversion (sharp rise of the capacitance to C_{ox}) taking place at $V_g \approx 13$ V where $\psi_s \approx 0.7$ V ($2\psi_B$). Note that as V_R increases, the onset of inversion shifts to increasingly more positive gate voltages as the MOS goes into deeper and deeper depletion (lower C_{min}). It can be seen that the value of surface potential at inversion increases by approximately V_R, consistent with Eq. (2.217). The decrease of C_{min}, the serial combination of C_{ox} and $C_d = \varepsilon_{si}/W_{dm}$, with V_R follows from Eq. (2.218).

2.3.6 Charge in Silicon Dioxide and at the Silicon–Oxide Interface

It is often said that the real magic in silicon technology lies not in the silicon crystalline material but in silicon dioxide. Silicon dioxide forms critical components of silicon devices, serves as insulation and passivation layers, and is often used as an effective masking and/or diffusion-barrier layer in device fabrication.

Thus far we have treated silicon dioxide as an ideal insulator, with no space charge in or associated with it, and no charge exchange between it and the silicon it covers. The

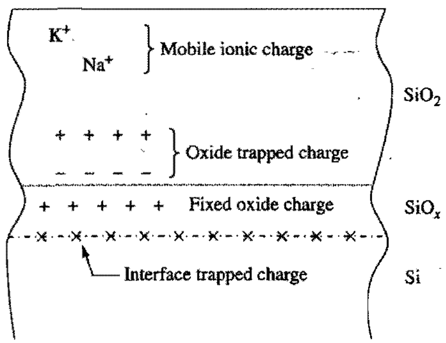

Figure 2.47. Charges and their location in thermally oxidized silicon. (After Deal, 1980.)

silicon dioxide and the oxide–silicon interface in real devices are never completely electrically neutral. There can be mobile ionic charges, electrons, or holes trapped in the oxide layer. There can also be fabrication-process-induced fixed oxide charges near the oxide–silicon interface, and charges trapped at the so-called *surface states* at the oxide–silicon interface. Electrons and holes can make transitions from the crystalline states near the oxide–silicon interface to the surface states, and vice versa. Since every device has some regions that are covered by silicon dioxide, the electrical characteristics of a device are very sensitive to the density and properties of the charges inside its oxide regions and at its silicon–oxide interface.

The nomenclature for describing the charges associated with the silicon dioxide in real devices was standardized in 1978 (Deal, 1980). The net charge per unit area is denoted by Q. Thus, Q_m denotes the *mobile charge* per unit area, Q_{ot} denotes the *oxide trapped charge* per unit area, Q_f denotes the *fixed oxide charge* per unit area, and Q_{it} denotes the *interface trapped charge* per unit area. The names and locations of these charges are illustrated in Fig. 2.47. The properties and characteristics of these charges are discussed further below.

2.3.6.1 Surface States and Interface Trapped Charge

At the Si–SiO_2 interface, the lattice of bulk silicon and all the properties associated with its periodicity terminate. As a result, localized states with energy in the forbidden energy gap of silicon are introduced at or very near the Si–SiO_2 interface (Many *et al.*, 1965). These localized surface states are illustrated schematically in Fig. 2.48. Interface trapped charges are electrons or holes trapped in these states.

Just like impurity energy levels in bulk silicon discussed in Section 2.1.2, the probability of occupation of a surface state by an electron or by a hole is determined by the surface-state energy relative to the Fermi level. Thus, as the surface potential is changed, the energy level of a surface state, which is fixed relative to the energy-band edges at the surface, moves with it. This change relative to the Fermi level causes a change in the probability of occupation of the surface state by an electron. For instance, referring to Fig. 2.32, as the bands are bent downward, or as the surface potential is increased,

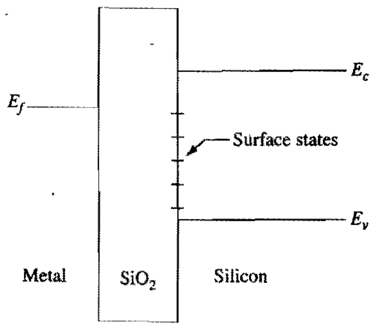

Figure 2.48. Schematic energy-band diagram of an MOS structure, illustrating the presence of surface states.

more surface states move below the Fermi level and hence become occupied by electrons. This change of interface trapped charge with a change in the surface potential gives rise to an additional silicon capacitance component, which will be discussed further in Section 2.3.7.

Electrons in silicon but near an oxide–silicon interface can make transitions between the conduction-band states and the surface states. An electron in the conduction band can contribute readily to electrical conduction current, while an electron in a surface state, i.e., an interface trapped electron, does not contribute readily to electrical conduction current, except by *hopping* among the surface states or by first making a transition to the conduction band. Similarly, holes in silicon but near an oxide–silicon interface can make transitions between the valence-band states and the surface states, and trapped interface holes do not contribute readily to electrical conduction. By trapping electrons and holes, surface states can reduce the conduction current in MOSFETs. Furthermore, the trapped electrons and holes can act like charged scattering centers, located at the interface, for the mobile carriers in a surface channel, and thus lower their mobility (Sah *et al.*, 1972).

Surface states can also act like localized generation–recombination centers. Depending on the surface potential, a surface state can first capture an electron from the conduction band, or a hole from the valence band. This captured electron can subsequently recombine with a hole from the valence band, or the captured hole can recombine with an electron from the conduction band. In this way, the surface state acts like a recombination center. Similarly, a surface state can act like a generation center by first emitting an electron followed by emitting a hole, or by first emitting a hole followed by emitting an electron. Thus, the presence of surface states can lead to surface generation–recombination leakage currents.

The density of surface states, and hence the density of interface traps, is a function of silicon substrate orientation and a strong function of the device fabrication process (EMIS, 1988; Razouk and Deal, 1979). In general, for a given device fabrication process, the dependence of the interface trap density on substrate orientation is $\langle 100 \rangle < \langle 110 \rangle < \langle 111 \rangle$. Also, a postmetallization or "final" anneal in hydrogen, or in a hydrogen-containing ambient, at temperatures around 400 °C is quite effective in minimizing the density of interface traps. Consequently, $\langle 100 \rangle$ silicon and postmetallization anneal in hydrogen are commonly used in modern VLSI device fabrication.

2.3.6.2 Fixed Oxide Charge

Fixed oxide charges are positive charges located in the oxide layer very close to the $Si–SiO_2$ interface. In fact, for modeling purposes, the fixed oxide charges are usually assumed to be located at the $Si–SiO_2$ interface. They are primarily due to excess silicon species introduced during oxidation and during postoxidation heat treatment (Deal *et al.*, 1967). The dependence of the density of fixed oxide charges on substrate orientation is the same as that of interface traps, namely $\langle 100 \rangle < \langle 110 \rangle < \langle 111 \rangle$.

The presence of fixed oxide charges at the oxide-silicon interface affects the potential in the silicon, which will be discussed in the next subsection. In addition, the fixed oxide charges act as charged scattering centers and thus reduce the mobility of the carriers in a surface inversion channel (Sah *et al.*, 1972).

2.3.6.3 Mobile Ionic Charge

Mobile ionic charges in SiO_2 are usually due to sodium or potassium contamination introduced during device fabrication. Unlike fixed oxide charges, which are not mobile, Na^+ and K^+ ions are quite mobile in SiO_2 and can be moved from one end of the oxide layer to the other when an electric field is applied across the oxide layer, particularly at somewhat elevated temperatures (>200 °C) (Hillen and Verwey, 1986). As these positively charged ions drift close to the $Si–SiO_2$ interface, they repel holes from, and attract electrons to, the silicon surface, often causing unwanted surface electron current to flow among n^+ diffusion regions in a p-type substrate or well. Also, when these positively charged ions come close to the silicon surface, they can act as charged scattering centers for the carriers in the surface inversion channel, thus reducing their mobility.

In VLSI fabrication processes, ***mobile-ion contamination problems must be avoided.*** This is accomplished by a combination of proper passivation, usually using phosphosilicate glass, and "clean" fabrication technology (Hillen and Verwey, 1986).

2.3.6.4 Oxide Trapped Charge

If electron–hole pairs are generated in an oxide layer, e.g., by ionizing radiation, some of these electrons and holes can be subsequently trapped in the oxide. Also, if electrons or holes are injected into an oxide layer, by tunneling or by hot-carrier injection, some of them can be trapped in the oxide.

Electron and hole traps in SiO_2 can easily be introduced by bombardment with high-energy photons or particles (Bourgoin, 1989). Since bombardment by high-energy particles and photons is involved in many steps in the fabrication of modern VLSI devices (during ion implantation, plasma or reactive-ion etching, sputtering deposition, electron-beam evaporation of metal, electron-beam and x-ray lithography, etc.), electron and hole traps are often introduced in the oxide during device fabrication. Fortunately, most of these traps can be eliminated with subsequent anneals at temperatures above 550 °C (Ning, 1978).

Also, depending on the oxidation condition, electron traps can be introduced during the oxide growth process itself (EMIS, 1988). For example, oxide growth in moisture-containing ambient is known to introduce electron traps (Nicollian *et al.*, 1971).

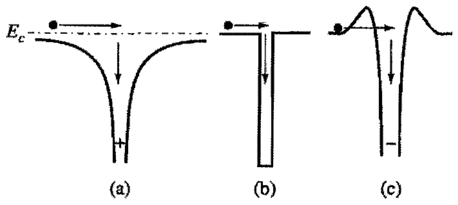

Figure 2.49. Schematics illustrating the potential wells of electron traps in silicon dioxide: (a) Coulomb-attractive trap, (b) neutral trap, and (c) Coulomb-repulsive trap.

- *Capture cross section.* Traps are usually characterized by their *capture cross sections*. Electron traps with cross sections in the range of 10^{-14}–10^{-12} cm^2 are usually *Coulomb-attractive* traps, i.e., the trap centers are positively charged prior to electron capture (Ning *et al.*, 1975; Lax, 1960). Electron traps with cross sections in the 10^{-18}–10^{-14}-cm^2 range are usually due to neutral traps (Lax, 1960), and those with cross sections smaller than 10^{-18} cm^2 are usually associated with *Coulomb-repulsive* traps, i.e., the trap centers are already negatively charged prior to electron capture (Balland and Barbottin, 1989). The potential wells representing these electron traps are illustrated in Fig. 2.49. Since the Coulomb-attractive and neutral centers have the largest capture cross sections, they are also the most important to include when considering the effects of electron traps on device characteristics. Hole traps have not been studied in as much detail as electron traps. This may be due to the fact that holes are very readily trapped when they are injected into an oxide layer (Goodman, 1966). This is consistent with the measured hole capture cross section of about 3×10^{-13} cm^2, which is as large as the largest electron traps in SiO_2 (Ning, 1976a).
- *Temperature dependence.* Consider the capture of a mobile electron into an electron trap. The trapping process has two competing components, namely, capturing the electron into some initial high-energy state of the trap center, and reemitting that same electron from the initial captured state by thermal excitation. If an electron in an initial captured state has a higher probability of cascading down towards the ground state of the trap center than of being reemitted, the electron becomes trapped. On the other hand, if the probability of reemission by thermal excitation from the initial captured state is high enough, trapping will not occur (Lax, 1960). The capture cross section, therefore, decreases with increasing temperature, since the probability for thermal reemission increases with temperature (Lax, 1960).
- *Field dependence.* If an electric field is applied across an oxide layer, it has the effect of increasing the energy of the carriers moving in the oxide layer. As these carriers gain energy from the oxide field, the probability of their being captured in some initial trap state is lowered, since the carriers now must lose more energy in the initial capture process. At the same time, an oxide field has the effect of lowering the energy barriers for the carriers trapped in a potential well, thus increasing the probability for reemitting them from their initial captured states (Lax, 1960). As a result, the capture cross section decreases with increasing oxide field (Ning, 1976b, 1978). ***The commonly used method of injecting carriers into SiO_2 by tunneling at high oxide fields tends to underestimate the amount of traps*** in SiO_2, since the capture cross sections at such high oxide fields are much smaller than those at normal device operation.

2.3.7 Effect of Interface Traps and Oxide Charge on Device Characteristics

The presence of oxide charges and interface traps has three major effects on the characteristics of devices. First, the charge in the oxide, or in the interface traps, interacts with the charge in the silicon near the surface and thus changes the silicon charge distribution and the surface potential. Second, as the density of interface trapped charge changes with changes in the surface potential, it gives rise to an additional capacitance component in parallel with the silicon capacitance C_{si} discussed in Section 2.3.3. Third, the interface traps can act as generation–recombination centers, or assist in the band-to-band tunneling process, and thus contribute to the leakage current in a gated-diode structure. These effects are discussed more quantitatively below.

2.3.7.1 Effect of Oxide Charge on Surface Potential

As discussed in Section 2.3.2, the charge distribution in silicon is a function of the surface potential. Thus, the effect of oxide charge on the charge distribution in silicon can be described in terms of its effect on surface potential. In the case of an MOS structure, the effect of oxide charge is usually described in terms of the change in gate voltage, which is a readily measurable parameter, necessary to counter the effect of the oxide charge or to restore the surface potential to that of zero oxide charge.

For simplicity of illustration, let us consider an MOS structure biased at flat-band condition. Let us assume that a sheet of oxide charge Q per unit area is placed at a distance x from the gate electrode, and a gate voltage δV_g has been applied to restore the MOS structure to its original, i.e., flat-band, condition. With the surface potential restored to its original value, the sheet of oxide charge has induced no change in the charge distribution in the silicon, which is a function of the surface potential, but a charge of magnitude $-Q$ per unit area on the gate electrode. This is illustrated in Fig. 2.50. Gauss's law in Eq. (2.43) implies that the electric field in the oxide between 0 and x due to the sheet of oxide charge and its image charge on the gate electrode is $-Q / \varepsilon_{ox}$ (see Exercise 2.5). This is also illustrated in Fig. 2.50. The potential difference supporting this electric field is $-xQ / \varepsilon_{ox}$, which is provided by the applied gate voltage. Therefore,

$$\delta V_g = -\frac{xQ}{\varepsilon_{ox}}. \tag{2.219}$$

The gate voltage necessary to offset the effect of an arbitrary oxide charge distribution can be obtained by superposition of individual elements of the charge distribution and applying Eq. (2.219) to each element. For an oxide charge distribution of $\rho_{net}(x, \psi_s) = \rho_{net}(x) + Q_{it}(\psi_s)\delta(x - t_{ox})$, which consists of an arbitrary distribution $\rho_{net}(x)$ that is independent of the surface potential and a delta-function distribution of the interface trap charge $Q_{it}(\psi_s)$ located at $x = t_{ox}$, the gate voltage necessary to offset it is

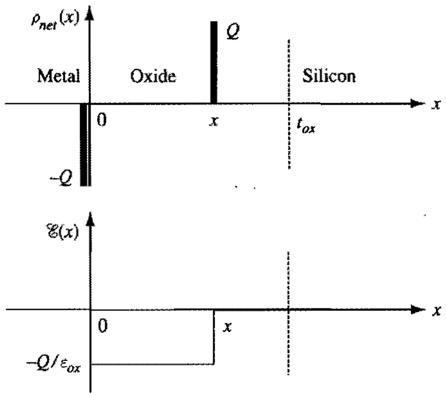

Figure 2.50. Schematic illustrating the effect of a sheet charge of areal density Q within the oxide layer of an MOS capacitor biased at flat-band condition.

$$\begin{aligned}\Delta V_g = \Delta V_g(\psi_s) &= -\frac{1}{\varepsilon_{ox}}\int_0^{t_{ox}} x\rho_{net}(x,\psi_s)dx \\ &= -\frac{1}{\varepsilon_{ox}}\left(\int_0^{t_{ox}} x\rho_{net}(x)dx + Q_{it}(\psi_s)t_{ox}\right).\end{aligned} \tag{2.220}$$

According to the charge nomenclature discussed in Subsection 2.3.6, $\rho_{net}(x)$ includes the mobile charge, the oxide trapped charge, and the fixed oxide charge. It is a common practice to define an *equivalent oxide charge per unit area*, Q_{ox}, by

$$\begin{aligned}Q_{ox} = Q_{ox}(\psi_s) &\equiv \int_0^{t_{ox}} \frac{x}{t_{ox}}\rho_{net}(x,\psi_s)dx \\ &= \int_0^{t_{ox}} \frac{x}{t_{ox}}\rho_{net}(x)dx + Q_{it}(\psi_s).\end{aligned} \tag{2.221}$$

Equation (2.220) can then be rewritten in the simple form of

$$\Delta V_g(\psi_s) = -\frac{Q_{ox}(\psi_s)}{C_{ox}}, \tag{2.222}$$

where $C_{ox} = \varepsilon_{ox}/t_{ox}$ is the oxide capacitance per unit area introduced in Eq. (2.195). Equation (2.222) states that the effect of an arbitrary oxide charge distribution is equivalent to an oxide sheet charge of areal density $Q_{ox}(\psi_s)$ located at the oxide–silicon interface.

2.3.7.2 Interface-Trap Capacitance

In Section 2.3.3, the silicon part of the capacitance is defined without including any interface trapped charge. As the interface traps are filled and emptied in response to changes in the surface potential, they give rise to an *interface-trap capacitance per unit area*, C_{it}, defined by

$$C_{it}(\psi_s) \equiv -\frac{dQ_{it}(\psi_s)}{d\psi_s}. \tag{2.223}$$

Just as in Eq. (2.197), the − sign in Eq. (2.223) is inserted to ensure that the capacitance is always a positive quantity. In Eq. (2.223), we have indicated explicitly that the interface-trap capacitance is a function of surface potential.

To include the effect of Q_{ox} in the operation of an MOS capacitor, Eq. (2.195) should be modified by adding to its right-hand side a ΔV_g term due to Q_{ox}. That is, Eq. (2.195) becomes

$$\begin{aligned}V_g - V_{fb} &= \Delta V_g(\psi_s) - \frac{Q_s(\psi_s)}{C_{ox}} + \psi_s \\ &= -\frac{Q_s(\psi_s) + Q_{ox}(\psi_s)}{C_{ox}} + \psi_s.\end{aligned} \tag{2.224}$$

The total charge on the gate electrode is now $Q_s(\psi_s) + Q_{ox}(\psi_s)$. Equation (2.196) then becomes

$$C_g = -\frac{d[Q_s(\psi_s) + Q_{ox}(\psi_s)]}{dV_g}, \tag{2.225}$$

and Eq. (2.198) becomes

$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si} + C_{it}}. \tag{2.226}$$

That is, the interface-trap capacitance is in parallel with the silicon capacitance.

As discussed in the previous subsection, the probability of a surface state being filled with an electron is governed by its energy level relative to the Fermi level. ***Only those interface traps that can be filled and emptied at a rate faster than the capacitance-measurement signal can contribute to C_{it}.*** Traps too slow to follow the capacitance-measurement signal will not contribute to C_{it}.

2.3.7.3 C–V Curves as a Monitoring and Diagnostic Tool for Oxide and Interface Quality

As discussed in Section 2.3.6, the amount of oxide charge and surface states is a function of the silicon substrate orientation and a strong function of the device fabrication process. For modern MOS devices, by the time a device fabrication process is ready for manufacturing, the amount of oxide charge and surface states is usually quite low, with Q_{ox}/q typically about 10^{11} cm^{-2} or less. For an MOS device having an oxide thickness of 10 nm, the corresponding gate voltage shift according to Eq. (2.222) is only 46 mV. At such low surface-state densities, the MOS C–V characteristics are quite ideal in that the measured C–V curves match well with the calculated ones (see Section 2.3.3 and Fig. 2.38).

However, many experimental fabrication processes, particularly those involving high-energy plasma, reactive-ion etching, and electron or ion beams, can generate significant oxide charge and surface states in MOS devices. Unless these *oxide damages* can be removed by post-process thermal annealing, the amount of residual oxide charge and surface states can be appreciable. High-field stress of silicon MOS devices can also

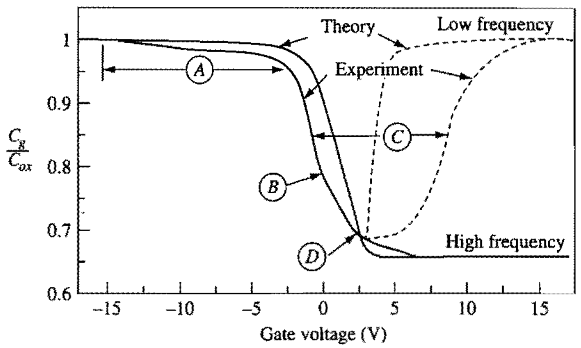

Figure 2.51. Comparison of experimental and theoretical high-frequency and low-frequency C–V curves, showing typical distortion caused by interface traps. The MOS capacitor has $N_a = 10^{16}$ cm^{-3} and $t_{ox} = 200$ nm. The symbols are explained in the text. (After Deal *et al.*, 1969.)

generate oxide charge and surface states. (High-field effects will be discussed in Section 2.5.) The presence of oxide charge and surface states can cause the measured C–V curve to appear distorted compared to the ideal C–V curve. There are two contributions to this distortion. First, Q_{it} is a part of Q_{ox} which shifts the C–V curve along the gate voltage axis according to Eq. (2.224). The shift is distorted because Q_{it} is a function of surface potential, which in turn is a function of gate voltage. Second, the additional capacitance due to the interface traps also distorts the measured C–V curve because C_{it} is a function of surface potential, which in turn is a function of gate voltage.

If the amount of oxide charge and surface states is large, the distortions in the C–V curve can be quite prominent, as illustrated in Fig. 2.51 (Deal *et al.*, 1969). The physical mechanisms responsible for the various distorted regions can be understood as follows. The distortion labeled A is where the MOS capacitor is normally in accumulation. In this gate voltage region, the valence-band edge at the silicon–oxide interface approaches or crosses the Fermi level (see Fig. 2.31). As a result, the interface states near the valence band become ionized and positively charged. (The interface states near the valence band are called donor states. They are neutral when they lie below the Fermi level and become positively charged by donating electrons when they lie above the Fermi level.) As the donor interface states become ionized, they contribute to a build up of positive interface trap charge which shifts the gate voltage in the negative direction according to Eq. (2.224). The distortion near the label B is related to interface states near the midgap, since it occurs at a gate voltage range where the MOS capacitor is between flatband and weak-inversion conditions (see Figs 2.31 and 2.38). The distortion labeled D is where the MOS capacitor is near weak inversion. To the right side of D, the capacitor is in inversion where the conduction-band edge at the silicon–oxide interface approaches or crosses the Fermi level. In this gate voltage range, the interface states near the conduction band become ionized and negatively charged. (The interface states near the conduction band are called acceptor states. They are neutral when they lie above the Fermi level and become negatively charged by accepting electrons when they lie below the Fermi level.) As the acceptor interface states become ionized, they contribute to a build up of negative interface trap charge which shifts

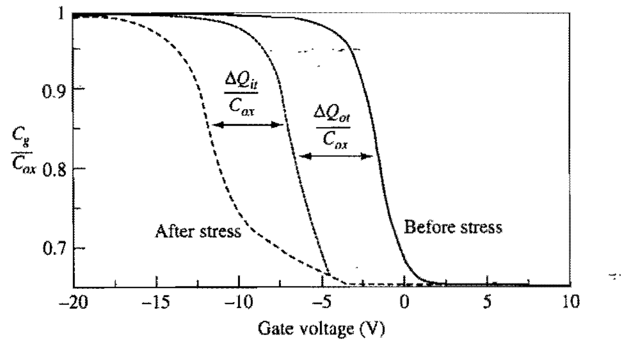

Figure 2.52. Typical high-frequency C–V plot of an MOS capacitor showing the distortion due to interface traps. The MOS capacitor has $N_a = 10^{16}$ cm^{-3} and $t_{ox} = 200$ nm. The oxide trapped charge and interface trapped charge are caused by subjecting the capacitor to a negative bias stress of 2MV/cm at 400 °C for 2 minutes. (After Deal *et al.*, 1967.)

the gate voltage in the positive direction according to Eq. (2.224). This causes the low-frequency C–V curve to shift to the right. The broadening of the C–V curve at its midpoint is labeled C. It is a result of the interface states near the conduction band (Deal *et al.*, 1969).

Figure 2.52 illustrates the distortion of a typical high-frequency C–V curve of an MOS capacitor after trapped charge has been created inside the oxide layer and at the oxide–silicon interface (Deal *et al.*, 1967). (The creation of bulk oxide and interface traps by high electric fields will be discussed in Section 2.5.) The oxide trapped charge causes a parallel shift of the C–V curve (dotted line) to the left. The interface-trap capacitance causes the curve to be distorted and shifted to the left by an additional amount.

The C–V distortions depicted in Figs 2.51 and 2.52 are for 200 nm thick oxides having significant oxide charge and surface states. It can be inferred from Eqs. (2.221) and (2.222) that the magnitude of the gate voltage shift caused by a certain areal density of interface states, Q_{it}, is proportional to t_{ox}. The voltage shift caused by a certain uniform volume density of oxide trapped charge, ρ_{net}, is proportional to t_{ox}^2. Therefore, for the same Q_{it} and ρ_{net}, the C–V curves of thinner oxide devices should appear less distorted.

There is a vast amount of published literature on the subject of interface states and the measurement of interface states. Interested readers are referred to the literature for a discussion on the general characteristics of interface states in MOS capacitors (Deal *et al.*, 1969) and on the various techniques for measuring interface states (Schroder, 1990).

2.3.7.4 Surface Generation–Recombination Centers

As discussed in the previous subsection, interface states can serve as generation–recombination centers. In the case of a gated-diode structure, the surface generation–recombination current adds to the diode leakage current. The magnitude of the surface leakage current depends on whether or not the surface states are *exposed*, i.e., whether or not the silicon surface is depleted (Grove and Fitzgerald, 1966). If the surface is inverted, the surface states are all filled with minority carriers and do not function efficiently as generation centers. Similarly, if the surface is in accumulation, the surface states are all filled with majority carriers and do not function efficiently as generation centers either. ***Only when the silicon surface is depleted will the surface states function efficiently as generation centers***. Thus, surface leakage current can be suppressed by biasing the gate to keep the silicon surface either in inversion or in accumulation. The reader is referred to Appendix 5 for a detailed discussion of the physics involved in generation–recombination processes.

As recombination centers, surface states can degrade the minority-carrier lifetime of devices. Consequently, devices where long minority-carrier lifetimes are required are usually designed to confine the minority carriers in them away from the silicon surface. In addition, the device fabrication processes are usually optimized to minimize the density of surface states.

2.3.7.5 Surface-State or Trap-Asssisted Band-to-Band Tunneling

As will be discussed in Subsection 2.5.2, band-to-band tunneling occurs when the electric field across a p–n junction is sufficiently large. For a gated diode, or for a p–n diode with silicon surface components, the presence of surface states or interface traps in the high-field region can enhance the band-to-band tunneling current very significantly. Thus, gate-induced drain leakage currents in MOSFETs, which will be discussed in Section 2.5.5, and emitter–base diode tunneling currents in bipolar transistors, which will be discussed in Section 6.3.4, depend strongly on the density of interface states at the oxide–silicon interface of these devices.

2.4 Metal–Silicon Contacts

The metal–semiconductor contact is a critically important element in all semiconductor devices and technology. As a contact to a silicon device terminal, a metal–silicon contact should be non-rectifying and have a small contact resistance in order to minimize the voltage drop across the contact. Such contacts are usually referred to as *ohmic contacts*. In general, a metal–semiconductor contact has rectifying current–voltage characteristics similar to those of a p–n diode (see Section 2.2). Rectifying metal–semiconductor devices are called *Schottky diodes* or *Schottky barrier diodes*. Here we discuss the basic physics and operation of a metal–silicon contact, focusing on its current–voltage characteristics as a Schottky diode and as an ohmic contact. A brief discussion of Schottky diodes as active devices is also given.

2.4.1 Static Characteristics of a Schottky Barrier Diode

The static characteristics of a Schottky barrier diode can be inferred from those of an MOS capacitor (see Section 2.3) by letting the oxide layer thickness go to zero. Just as the surface potential of the semiconductor in an MOS capacitor is affected by the interface trapped charge, the surface potential of the semiconductor in a Schottky barrier diode is affected by the electron occupation of the surface states on the semiconductor surface. Therefore, the characteristics of a Schottky barrier diode depend on the properties of the metal and the properties of the semiconductor and its surface states. Here we first discuss

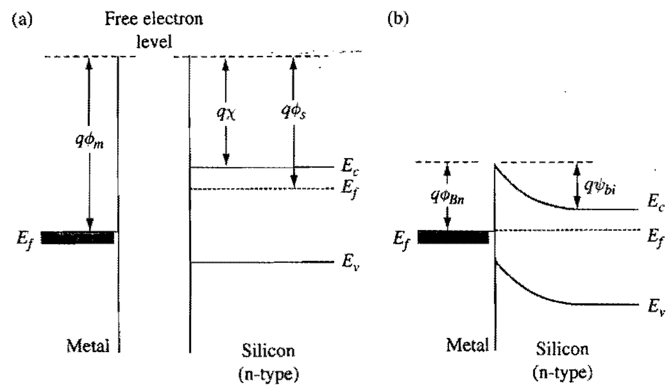

Figure 2.53. Energy-band diagrams of a metal–silicon system where the silicon surface is assumed to be absent of any surface states. (a) When the metal and the silicon are far apart. (b) When the metal is in contact with the silicon, with no externally applied voltage.

the static characteristics of a Schottky barrier diode ignoring all surface states, and then discuss how the surface states can modify the diode characteristics.

2.4.1.1 Schottky Barrier Diodes Without Surface States

When surface states are ignored, the energy-band diagram of an MOS capacitor shown in Fig. 2.29 can be readily adapted to give the energy-band diagram of a metal–silicon system. It was pointed out in Section 2.3.1.1 that when two different materials are brought into contact, they must share the same free electron level at the interface. Also, it was shown in Sections 2.1.1.3 and 2.1.4.5 that at thermal equilibrium or when there is no net electron or hole current through a system, the Fermi level of the system is spatially constant. These two factors together lead to the energy-band diagrams in Fig. 2.53 for a metal–n-silicon contact at thermal equilibrium. From consideration of free electron level at the interface, the Schottky barrier height for electrons, $q\phi_{Bn}$, is

$$q\phi_{Bn} = q(\phi_m - \chi), \tag{2.227}$$

where $q\phi_m$ is the metal work function and $q\chi$ is the electron affinity of silicon. From consideration of Fermi level being spatially constant, the built-in potential ψ_{bi} is

$$\begin{aligned} q\psi_{bi} &= q\phi_{bn} - (E_c - E_f)_{bulk} \\ &= q\left(\phi_{bn} - \frac{E_g}{2q} + \psi_B\right), \end{aligned} \tag{2.228}$$

where $\psi_B = |\psi_f - \psi_i|$ [see Eq. (2.48)].

The built-in potential implies an amount of depletion-layer charge Q_d in the silicon given by Eq. (2.189). To maintain overall charge neutrality of the metal–silicon system, this depletion charge induces a sheet of electronic charge of the same density per unit area in the metal at the metal–silicon interface.

The physical picture for a Schottky diode illustrated in Fig. 2.53 is based on energy-band diagrams for *bulk* silicon and metal. It does not take into consideration any surface properties of the semiconductor. Experimentally measured barrier heights are not consistent with Eq. (2.227). For some semiconductors, the measured barrier heights show little dependence on metal work function. For others, the dependence on work function is weaker than suggested by Eq. (2.227). That the measured barrier height has a weaker dependence on metal work function than suggested by Eq. (2.227) is attributable to the presence of surface states. This is discussed in the next subsection.

2.4.1.2 Schottky Barrier Diodes with Surface States

Many ideas and models for improving the understanding of real Schottky diodes have been proposed. In terms of explaining the relationship between measured barrier heights and metal work functions, the models all include the effects of surface states. Here we discuss qualitatively how surface states can influence the barrier height. A good reference where many ideas and models are discussed at length can be found in Henisch (1984).

The inclusion of surface states makes the physical picture of a metal–semiconductor contact more complex than the bulk model in Fig. 2.53. This is illustrated in Fig. 2.54. Let us first consider the situation where the metal and the silicon are physically and electrically two separate systems, i.e., Fig. 2.54(a). As electrically separate systems, there can be no charge exchange between the metal and the silicon. Occupation of some of the surface states by electrons induces a positive depletion-layer charge of Q_d per unit area in the silicon near the surface, causing the energy bands to bend upward near the surface. The amount of band bending is represented by the surface potential ψ_s. The upward band bending means $\psi_s < 0$. The relationship between the depletion charge density and surface potential is given by Eq. (2.189), namely

$$-q\psi_s(\text{free surface}) = \frac{[Q_d(\text{free surface})]^2}{2\varepsilon_{si}N_d}, \tag{2.229}$$

where N_d is the doping concentration of the n-type silicon. In Eq. (2.229), we have indicated that the surface potential and the corresponding depletion charge are for a free semiconductor surface.

Next, let us consider the metal and the silicon being electrically connected but physically separated, with a physical gap between the metal and silicon surfaces. (In the literature, for purposes of establishing a model for a Schottky diode, this physical gap is often replaced by an oxide layer. In this case, the discussion follows that for an MOS capacitor in Section 2.3.1.1.) The energy bands for this electrically connected system are as illustrated in Fig. 2.54(b). As discussed in Sections 2.1.1.3 and 2.1.4.5, a charge exchange between the metal and the silicon occurs until the Fermi level is spatially constant across the entire system. As discussed in Section 2.3.1.1, the free electron levels in the interfaces are continuous, suggesting the presence of an electric field in the gap between the surfaces of the metal and the silicon. This electric field is supported by a *net* charge Q_M on the metal surface, as required by Gauss's law [Eq. (2.43)]. Q_M is a negative charge for the electric field direction indicated in Fig. 2.54(b). If the amount of net charge

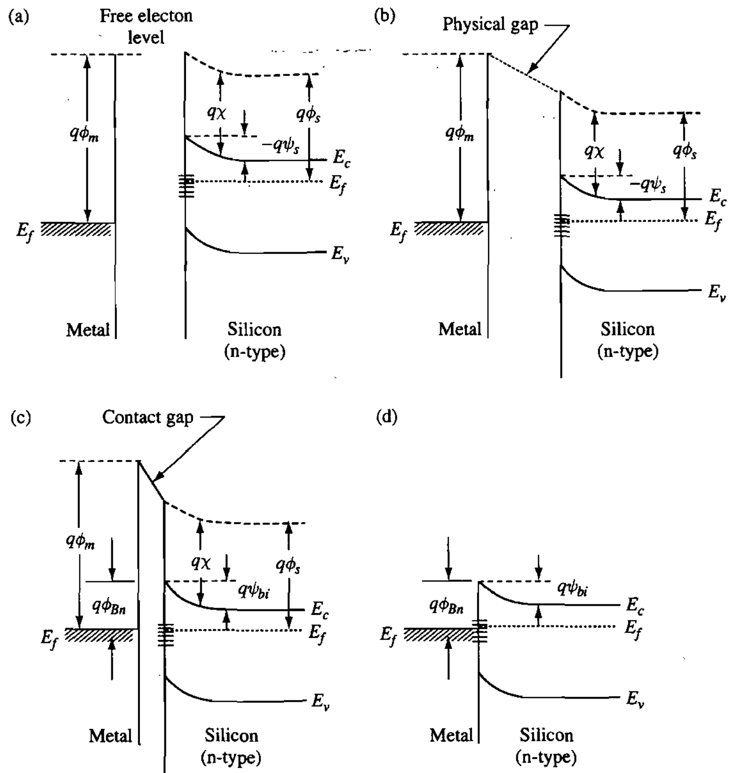

Figure 2.54. Energy-band diagram of a metal–silicon system where the silicon is assumed to have a large density of surface states. (a) When the metal and the silicon are not electrically connected as one system. Some of the interface states are filled with electrons, causing the bands to bend upward. (b) When the metal and the silicon are electrically connected to form one system, but the metal is physically separate from the silicon surface by a gap space. (c) When the metal is in contact with the silicon to form a Schottky diode. A contact gap of atomic dimension is shown. The contact-gap region is discussed in the text. (d) A simplified diagram where the contact gap is omitted.

per unit area in the surface states is Q_{it}, then overall charge neutrality of the metal–silicon system requires that $Q_M = -(Q_{it} + Q_d)$. As the gap between the metal surface and the silicon surface is reduced, the electric field in the gap increases, and hence the magnitude of Q_M increases, requiring the magnitude of $(Q_{it} + Q_d)$ to increase. In other words, a change in Q_M can cause a change in Q_{it} and/or Q_d.

The Bardeen model (Bardeen, 1947) for the roles played by surface states provides a physical picture for explaining how the charges Q_M, Q_{it} and Q_d may change together. According to the Bardeen model, the charges involved in the charge-transfer process in the formation of a metal–semiconductor contact come from four double layers. (A double layer consists of a layer of positive charge and a layer of negative charge of the same magnitude.) These layers are: (i) a double layer of atomic dimensions at the metal surface, (ii) a double layer of atomic dimensions at the semiconductor surface, (iii) a double layer formed from the surface charges on the metal and semiconductor, both of atomic dimensions, and (iv) a double layer formed from a surface charge of atomic dimensions on the semiconductor surface and a depletion charge layer in the semiconductor. According to this model, Q_M has contributions from double layers (i) and (iii); Q_{it} has contributions from double layers (ii), (iii) and (iv); Q_d has contribution from double layer (iv).

Bardeen showed that the strength of double layer (iii) is small for semiconductors where the density of surface states is small (less than about 10^{13} cm^{-2}), and a change in Q_M is balanced by charge transfer primarily among double layers (ii) and (iv). In this case, as the physical gap in Fig. 2.54(b) is reduced, the change in Q_M is balanced primarily by a change in Q_d, with little change in Q_{it}. For semiconductors with sufficiently high (greater than 10^{13} cm^{-2}) density of surface states, double layer (iii) can be the primary source for any change in Q_M. In this case, as the physical gap in Fig. 2.54(b) is reduced, the change in Q_M is balanced primarily by a change in Q_{it}, with little change in Q_d. When there is little change in Q_d, there is little change in the surface potential which in turn implies little change in the energy-band edges at the semiconductor surface relative to the Fermi level. The Fermi level at the semiconductor surface is said to be more or less pinned by the high density of surface states.

When the metal makes contact with the silicon to form a Schottky diode, the energy-band diagram is as illustrated in Fig. 2.54(c). Note that a contact gap is shown to represent the region containing the double layers (i), (ii) and (iii) in the Bardeen model. The width of the contact gap is of atomic dimensions, at least for good contacts where there is no unintended interfacial material. The contact gap is assumed to be sufficiently thin to play no role in the transport of electrons between the metal and the silicon. Notice that, just as in Fig. 2.54(b), the potential difference across this infinitesimal contact gap is equal to $(\phi_m - \chi + \psi_s - E_g/2q + \psi_B)$. In Fig. 2.54(b), before the Schottky diode is formed, the surface potential ψ_s depends on the physical gap space. If there is weak Fermi-level pinning, ψ_s changes with the physical gap space. If there is strong Fermi-level pinning, ψ_s is relatively insensitive to the physical gap space. In Fig. 2.54(c), we have $\psi_s = -\psi_{bi}$, where ψ_{bi} is the built-in potential of the Schottky barrier diode. The electron energy barrier is $q\phi_{Bn}$. (The contact gap is transparent to electron transport between the metal and the silicon.)

Since the contact gap in a good metal-semiconductor contact is assumed to be transparent to electron transport between the metal and the semiconductor, we can omit it from the energy-band diagram completely for purposes of modeling device characteristics of a Schottky diode. (As discussed in the paragraph below, we should do this only with care.) The *simplified* energy-band diagram for a Schottky diode is as illustrated in Fig. 2.54(d).

It should be noted that Fig. 2.54(c) suggests that $\phi_{Bn} \neq \phi_m - \chi$ in cases of high surface-state density. This should be contrasted with the case of no surface states shown in Fig. 2.53, where we have $\phi_{Bn} = \phi_m - \chi$. If we had drawn the contact gap as a layer of zero thickness, the free electron level would have appeared discontinuous at the metal–silicon interface. Therefore, while in common practice Fig. 2.54(d) is shown and used by itself for description of a Schottky diode, the correct and complete physical picture must include a contact gap of finite thickness in order to maintain continuity of free electron level across the metal–semiconductor system. A more detailed discussion on Schottky barriers and surface states with a mathematical model for the barrier height taking into consideration the work function difference, the density of surface states and the contact-gap thickness, can be found in Cowley and Sze (1965).

The energy-band diagrams in Fig. 2.54 are sketched based on the assumption of well defined metal and silicon surfaces and a well defined contact-gap region. In developing the model for Fermi-level pinning by surface states, Bardeen (1947) pointed out explicitly that if the contact between a metal and a semiconductor is very intimate, it may not be possible to distinguish between the double layers (i), (ii) and (iii). For an intimate contact, the metal will tend to broaden the surface levels. Furthermore, Heine (1965) showed that the wavefunction of an electron in a surface state does spread over some finite distance and that the concepts of band bending in a metal–semiconductor contact must not be taken too seriously over distances of the spread of localized electron wavefunctions. Therefore, for an intimate metal–semiconductor contact, the interface boundaries that define the contact-gap region are not as well defined as implied in Fig. 2.54(c). Fortunately, for purposes of describing and establishing the electrical characteristics of a Schottky diode, the details of the contact-gap region are not important because the region is transparent to electron transport.

For a metal–silicon contact represented by Figs 2.54(c) or 2.54(d), the built-in potential is given by

$$q\psi_{bi} = -q\psi_s(\text{contacted surface}) = \frac{[Q_d(\text{contacted surface})]^2}{2\varepsilon_{si}N_d}. \tag{2.230}$$

Once the built-in potential is known, the electron energy barrier can be determined from Eq. (2.228).

2.4.1.3 Measured Barrier Heights

That the Fermi level at the interface of a metal–semiconductor contact tends to be pinned by surface states has been well verified experimentally. The degree of pinning, however, varies with the semiconductor and often depends on the details of the process used for forming the metal–semiconductor contact. In practice, the measured barrier heights vary widely, even when the Schottky diodes are prepared in presumably identical processes. This is likely to be caused by some oxide layer or sublayer at the metal–semiconductor interface which alters the chemical bonding and structure of the interface. For metal contacts to many "clean" group IV and III-V semiconductor surfaces, the Fermi level pinning by surface states appears to be total, with the measured electron barrier heights practically independent of the metal used (Mead and Spitzer, 1964). For silicon and many other semiconductors, the measured barrier heights can be modeled by assuming partial pinning of the Fermi level by surface states (Cowley and Sze, 1965). In the case of silicon, annealing a metal–silicon system to form a metal-silicide–silicon contact can lead to barrier heights that are different from but more reproducible than the corresponding metal–silicon contact (Andrews and Phillips, 1975; Andrews, 1974).

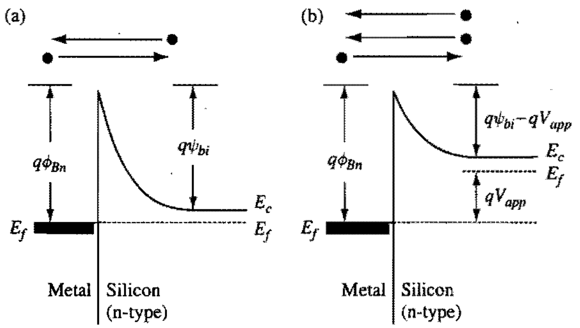

Figure 2.55. Schematic energy-band diagrams illustrating the flow of electrons in an n-type Schottky diode. (a) At thermal equilibrium, there is an equal and opposite flow of electrons. (b) At forward bias, there is a net flow of electrons from the silicon into the metal. For simplicity of illustration, barrier-lowering effect is not shown.

Tung (1992) suggested that there can be lateral inhomogeneity in the distribution of surface states or surface charge as well. Thus, a metal–semiconductor interface can be modeled as consisting of nanometer-sized local patches, with each patch having its own local electron energy barrier (Im *et al.*, 2001). The measured barrier height represents the averaged barrier height of the entire contact. Since there can be contact-to-contact variation in the lateral inhomogeneity, there can be inhomogeneity-induced variation in the measured barrier heights as well.

As can be inferred readily from Figs 2.53(b) and 2.54(d), the hole energy barrier $q\phi_{Bp}$ is related to the electron energy barrier $q\phi_{Bn}$ through

$$q\phi_{Bn} + q\phi_{Bp} = E_g, \tag{2.231}$$

where E_g is the energy gap of the semiconductor. We will focus our discussion on metal contacts to n-type silicon where the barrier height is $q\phi_{Bn}$. Metal contacts to p-type silicon where the barrier height is $q\phi_{Bp}$ will not be discussed explicitly.

2.4.1.4 Effect of Electric Field on Barrier Height

In Appendix 7, it is shown that the image-force effect causes the energy barrier for electron transport across a metal–silicon interface to be lowered by

$$q\Delta\phi \approx \sqrt{\frac{q^3\mathscr{E}_m}{4\pi\varepsilon_{si}}}, \tag{2.232}$$

where $\mathscr{E}_m$ is the maximum electric field in the silicon. The actual energy barrier for electron transport in a Schottky barrier diode is therefore $(q\phi_{Bn} - q\Delta\phi)$. The total band bending in the silicon is $q(\psi_{bi} - V_{app})$, where V_{app} is the forward-bias voltage across the Schottky diode (see Fig. 2.55), and Eq. (2.184) gives $\mathscr{E}_m = \text{sqrt}[2qN_d(\psi_{bi} - V_{app})/\varepsilon_{si}]$ for n-type silicon with a uniform doping concentration of N_d. That is, the effective energy barrier of a Schottky barrier is smaller than that suggested in Figs 2.53 and 2.54. A forward

bias ($V_{app} > 0$) across a diode reduces the electric field and hence increases the effective energy barrier, while a reverse bias ($V_{app} < 0$) increases the electric field and hence reduces the effective energy barrier.

2.4.2 Current Transport in a Schottky Barrier Diode

Consider an n-type silicon Schottky barrier diode. The energy-band diagrams illustrating the flow of electrons across the interface are shown schematically in Fig. 2.55. In modeling the transport of an electron across the interface, we need to consider the kinetic energy of the electron relative to the energy barrier for current flow across the interface, as depicted in the energy-band diagrams. For example, for an electron in the metal having an energy $E = E_f$, it sees an energy barrier of $q\phi_{Bn}$ (barrier-lowering effect is ignored for simplicity of discussion). For an electron having an energy $E = E_f + \Delta E$ it sees an energy barrier of $q\phi_{Bn} - \Delta E$. Similarly, for an electron in the quasineutral silicon region having an energy of $E = E_c$, it sees an energy barrier of $q(\psi_{bi} - V_{app})$. For an electron having an energy ΔE above the conduction-band edge, its barrier for transport across the interface is $q(\psi_{bi} - V_{app}) - \Delta E$. These energy barriers for current flow should not be confused with the energy barrier of the Schottky diode itself, which is $q\phi_{Bn}$.

At thermal equilibrium, there is no net electron flow in either direction in the diode, as indicated in Fig. 2.55(a). If a forward voltage V_{app} is applied to the diode, there will be a net electron flow from the n-silicon to the metal, but there are no holes (minority carriers) flowing into the n-silicon, as indicated in Fig. 2.55(b). Similarly, for a forward-biased p-type silicon Schottky barrier diode, there is a net flow of holes from the p-silicon into the metal, which is equivalent to a net flow of electrons from the metal into the valence band of the p-silicon. There are no excess electrons (minority carriers) injected from the metal into the conduction band of the p-silicon. That is, the current transport in a Schottky barrier diode is mainly due to majority carriers. This should be contrasted with a p–n diode where current transport is mainly due to minority carriers (electrons injected into the conduction band of the p-side and holes injected into the valence band of the n-side). The switching speed of a p–n diode is limited by the time it takes to discharge the minority carriers stored in the diode during forward bias (see Section 2.2.5.3). Therefore, without minority-carrier storage, a Schottky barrier diode is inherently faster than a p–n diode.

2.4.3 Current–Voltage Characteristics of a Schottky Barrier Diode

The processes by which electrons are transported from one side to the other in a Schottky barrier diode are illustrated in Fig. 2.56. Thermionic emission refers to the electrons having sufficient energy to surmount the effective (image-force effect included) energy barrier. Field emission refers to the tunneling of electrons from around the conduction-band edge. Thermionic-field emission describes the tunneling of electrons having energy above the conduction band but not enough energy to surmount the barrier. For a diode designed to function as an active device or a circuit component, the doping concentration is usually sufficiently light, and therefore the depletion layer thickness sufficiently large, so that thermionic emission is the dominant process for electron transport. Field emission and

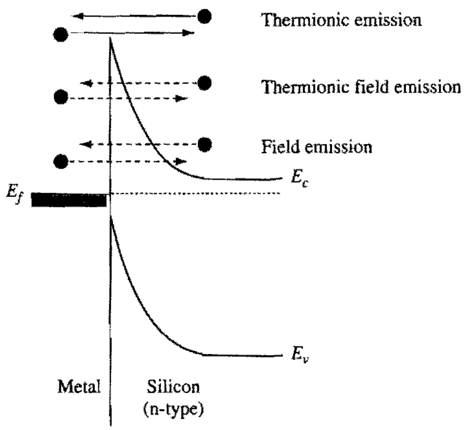

Figure 2.56. Schematic energy-band diagram of a Schottky barrier diode illustrating the principal transport processes.

thermionic-field emission are important transport processes in metal–semiconductor contacts where the doping levels are high (e.g., in ohmic contacts, which will be discussed later).

2.4.3.1 Thermionic Emission

In thermionic emission, the simplest theory is to treat the electrons as an ideal gas that follows Boltzmann statistics in energy distribution. Electron collision within the semiconductor depletion region is ignored, and only those electrons traveling in the direction of emission and having sufficient energy to surmount the barrier are emitted. In the case of a multi-valley semiconductor having anisotropic effective electron masses like silicon, we should consider the emission current from each valley and then sum the currents to obtain the total current.

The conduction band of silicon has six identical valleys located on the k_x-, k_y-, and k_z- axes. Each valley is an ellipsoid, with a longitudinal mass of $m_l = 0.92\, m_0$ and a transverse mass of $m_t = 0.19 m_0$, where m_0 is the free-electron mass. The thermionic electron emission current density from an arbitrarily oriented silicon surface has been derived by Crowell (1965, 1969). The derivation is simplest for <100> silicon. Since <100> is the most commonly used silicon orientation, we shall only consider this orientation here. The reader interested in other orientations is referred to the paper by Crowell (1965).

Since electron collision within the depletion region is ignored, we can consider the thermionic emission current to be due to electrons originating from the quasineutral silicon region having sufficient energy to surmount the energy barrier for emission. As discussed in Section 2.1.1.2, the number of electronic states per unit volume having momenta between p_x and $p_x + dp_x$, between p_y and $p_y + dp_y$, and between p_z and $p_z + dp_z$ in one of the conduction-band valleys is

$$N(p_x, p_y, p_z)dp_x dp_y dp_z = \frac{2}{h^3} dp_x dp_y dp_z. \qquad (2.233)$$

Note that the factor g denoting the number of equivalent minima in the conduction band is unity in Eq. (2.233) because only one valley is being considered. The kinetic energy of an electron in the quasineutral region is $E - E_c$, and the relationship between kinetic energy and momenta is given by Eq. (2.2). In terms of Boltzmann statistics, the probability that an electronic state at energy E is occupied by an electron is $\exp[-(E - E_f)/kT]$ [see Eq. (2.5)]. Therefore, Eq. (2.233) gives the number of electrons per unit volume having momenta between p_x and $p_x + dp_x$, between p_y and $p_y + dp_y$, and between p_z and $p_z + dp_z$ in one of the conduction-band valleys as

$$\begin{aligned} dn(p_x, p_y, p_z) &= \frac{2}{h^3} e^{-(E-E_f)/kT} dp_x dp_y dp_z \\ &= \frac{2}{h^3} e^{-(E_c-E_f)/kT} e^{-(p_x^2/2m_x kT + p_y^2/2m_y kT + p_z^2/2m_z kT)} dp_x dp_y dp_z. \end{aligned} \qquad (2.234.)$$

The number of electrons per unit volume having momenta between p_x and $p_x + dp_x$ is given by integrating Eq. (2.234) over all values of p_y and p_z. That is,

$$\begin{aligned} dn(p_x) &= \frac{2}{h^3} e^{-(E_c-E_f)/kT} e^{-p_x^2/2m_x kT} dp_x \int_{-\infty}^{\infty} e^{-p_y^2/2m_y kT} dp_y \int_{-\infty}^{\infty} e^{-p_z^2/2m_z kT} dp_z \\ &= \frac{4\sqrt{m_y m_z}}{h^3} \pi kT e^{-(E_c-E_f)/kT} e^{-p_x^2/2m_x kT} dp_x. \end{aligned} \qquad (2.235)$$

- *Ignoring Barrier Lowering Effect.* In this case, when a voltage V_{app} is applied to a Schottky diode, the minimum energy an electron traveling perpendicularly to the emission surface must have in order to surmount the emission barrier is $q(\psi_{bi} - V_{app})$. For <100> silicon, the emission surface is perpendicular to the k_x-axis. The current density due to thermionic emission of electrons from a single conduction-band valley into the metal is

$$\begin{aligned} J_{1,s\to m}(V_{app}) &= q \int_{p_x = p_{x0}}^{\infty} v_x dn(p_x) \\ &= \frac{4\sqrt{m_y m_z}}{h^3} q\pi kT e^{-(E_c-E_f)/kT} \int_{p_{x0}}^{\infty} \frac{p_x}{m_x} e^{-p_x^2/2m_x kT} dp_x \\ &= \frac{4\pi q\sqrt{m_y m_z}}{h^3} k^2 T^2 e^{-(E_c-E_f)/kT} e^{-q(\psi_{bi}-V_{app})/kT}, \end{aligned} \qquad (2.236)$$

where $v_x = p_x / m_x$ is the velocity of an electron traveling in the x-direction, and the lower integration limit p_{x0} is given by $p_{x0}^2/2m_x = q(\psi_{bi} - V_{app})$. The subscript 1 indicates that the current density is from only one conduction-band valley. Since $q\phi_{Bn} = q\psi_{bi} + E_c - E_f$, Eq. (2.236) can be rewritten as

$$\begin{aligned} J_{1,s\to m}(V_{app}) &= \frac{4\pi q\sqrt{m_y m_z}}{h^3} k^2 T^2 e^{-q\phi_{Bn}/kT} e^{qV_{app}/kT} \\ &= A \frac{\sqrt{m_y m_z}}{m_0} T^2 e^{-q\phi_{Bn}/kT} e^{qV_{app}/kT}, \end{aligned} \qquad (2.237)$$

where the quantity

$$A \equiv \frac{4\pi q m_0 k^2}{h^3} \tag{2.238}$$

is the Richardson constant for free electrons. A = 120 A/cm^2/K^2.

To obtain the total thermionic electron emission current from the silicon, we need to sum the currents from all six valleys. For <100> silicon, the two valleys on the k_x-axis have $m_y = m_z = m_t$, and the four valleys on the k_y- and k_z-axes have $m_y = m_l$ and $m_z = m_t$. The total thermionic electron emission current density from silicon into metal is

$$\begin{aligned} J_{n-Si<100>,s\to m}(V_{app}) &= \sum_1^6 J_{i,n-Si<100>,s\to m}(V_{app}) \\ &= A\left(\frac{2m_t}{m_0}+\frac{4\sqrt{m_t m_l}}{m_0}\right)T^2 e^{-q\phi_{Bn}/kT} e^{qV_{app}/kT} \\ &= A^*_{n-Si<100>} T^2 e^{-q\phi_{Bn}/kT} e^{qV_{app}/kT}, \end{aligned} \tag{2.239}$$

where

$$A^*_{n-Si<100>} = A\left(\frac{2m_t}{m_0}+\frac{4\sqrt{m_t m_l}}{m_0}\right) = 2.05A \tag{2.240}$$

is the Richardson constant for n-type <100> silicon.

For silicon, the orientation dependence is relatively weak. For n-type <111> silicon, the Richardson's constant is 2.15A (Crowell, 1965). In theory, the measured Richardson's constant contains information about the effective mass tensor of the semiconductor. However, the simple thermionic emission model gives only a qualitative description of experimentally measured currents in a typical Schottky barrier diode. The measured Richardson's constant should not be used to infer information about the effective mass tensor (Crowell, 1969). In practice, the Richardson's constant is often treated as an adjustable parameter for fitting experimental data (Henisch, 1984).

At zero applied bias, the electron emission current from the metal into the silicon is equal in magnitude but opposite in direction to the electron emission current from the silicon into the metal. That is,

$$\begin{aligned} J_{n-Si<100>,m\to s}(V_{app}=0) &= -J_{n-Si<100>,s\to m}(V_{app}=0), \\ &= -A^*_{n-Si<100>} T^2 e^{-q\phi_{Bn}/kT}. \end{aligned} \tag{2.241}$$

When the barrier lower effect is ignored, the energy barrier for electron emission from metal into silicon is independent of V_{app}. Therefore, we expect the electron emission current from metal into silicon to be independent of V_{app} when barrier lowering effect is ignored. The total thermionic emission current density for an n-type <100> silicon Schottky barrier diode, when barrier lower effect is ignored, is therefore

$$\begin{aligned} J_{thermionic,n-Si<100>}(V_{app}) &= J_{n-Si<100>,s\to m}(V_{app}) + J_{n-Si<100>,m\to s}(V_{app}) \\ &= A^*_{n-Si<100>} T^2 e^{-q\phi_{Bn}/kT}\left(e^{qV_{app}/kT}-1\right). \end{aligned} \tag{2.242}$$

For a forward-biased diode ($V_{app} > 0$), the current is dominated by the emission from the semiconductor into the metal. For a reverse-biased diode ($V_{app} < 0$), the current is dominated by the emission from the metal into the semiconductor. Equation (2.242) shows that, when barrier lowering effect is ignored, a Schottky barrier diode has I–V characteristics similar to those of a p–n diode [cf. Eq. (2.102)], with an $\exp(qV_{app}/kT)$ dependence on V_{app} in forward bias, and a saturation current that is independent of V_{app} in reverse bias.

- *Including Barrier Lowering Effect.* There is a subtle difference between a Schottky diode and a p–n diode when barrier lowering effect is included. When barrier lowering effect is included, the Schottky barrier $q\phi_{Bn}$ in Fig. 2.55 and in Eqs. (2.237) to (2.242) should be replaced by an effective Schottky barrier $q(\phi_{Bn} - \Delta\phi)$. The barrier-lowering term $q\,\Delta\phi$ depends on the applied voltage through the electric field $\mathscr{E}_m$ [see Eq. (2.232)]. As discussed in Section 2.4.1.4, a forward bias ($V_{app} > 0$) reduces $q\Delta\phi$ and hence increases the effective Schottky barrier, while a reverse bias ($V_{app} < 0$) increases $q\Delta\phi$ and hence reduces the effective Schottky barrier. Thus, replacing $q\phi_{Bn}$ by $q(\phi_{Bn} - \Delta\phi)$ in Eq. (2.242) suggests that the forward-bias current of a Schottky diode increases with V_{app} at a rate somewhat slower than $\exp(qV_{app}/kT)$. This should be compared with the forward-bias current of a p–n diode which is proportional to $\exp(qV_{app}/kT)$ [see Eq. (2.123)]. See also Fig. 2.23.

In the literature, more complex theories have been proposed for describing the transport process in Schottky diodes. There is a diffusion emission theory which includes the effect of electron collisions within the semiconductor depletion region. There is also a theory which combines the physics involved in the simple thermionic emission process and the diffusion emission process (Crowell and Sze, 1966a). All theories result in an equation similar to Eq. (2.242), with the difference only in the pre-exponential factor. The interested reader is referred to the literature for the details (Sze, 1981; Henisch, 1984). From a device point of view, the important I–V characteristics of a Schottky barrier diode are contained in the exponential factors in Eq. (2.242), namely in the $\exp(-q\phi_{Bn}/kT)$ dependence on $q\phi_{Bn}$ and the $[\exp(qV_{app}/kT) - 1]$ dependence on qV_{app}.

2.4.3.2 Field Emission and Thermionic-Field Emission

If the semiconductor is heavily doped, the depletion region thickness will be thin, and the electron transport can become dominated by a combination of field emission and thermionic-field emission. In this case, large currents can flow even at low applied biases. In general, when field emission and thermionic-field emission dominate the electron transport, a metal–semiconductor contact is no longer useful as a rectifying diode. As a result, we will not consider the general theory of field emission and thermionic-field emission any further. The interested reader is referred to the literature for details (Padovani and Stratton, 1966; Crowell and Rideout, 1969).

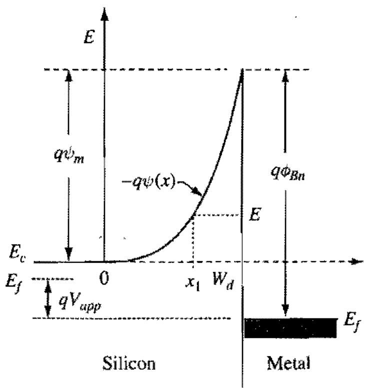

Figure 2.57. Schematic showing the energy bands appropriate for considering field emission in a metal–silicon contact. As illustrated, the Schottky diode is forward biased, as indicated by the Fermi level in the silicon being higher than that in the metal.

2.4.3.3 Schottky Barrier Diode as an Active Device

A rectifying Schottky barrier diode has I–V characteristics similar to those of a p–n diode, but a Schottky barrier diode is a much faster device than a p–n diode because it is a majority-carrier device. As a result, Schottky barrier diodes are often used as microwave diodes and as gates of microwave transistors where speed is important (see e.g., Irvin and Vanderwal, 1969). Also, Schottky barrier diodes are often added to bipolar circuits as voltage clamps to improve circuit speed.

2.4.4 Ohmic Contacts

Ohmic contacts are usually made with metal or metal silicide in contact with heavily doped semiconductor. The electron transport process in this case is dominated by field emission. Let us first consider the tunneling of a conduction-band electron from the quasineutral semiconductor region into the metal. The band bending near the metal–semiconductor contact is illustrated in Fig. 2.57. The total band bending is $q\psi_m = q(\psi_{bi} - V_{app})$ when a forward bias of V_{app} is applied. For a given V_{app}, let us assume the conduction-band starts to bend upward at $x = 0$, and the interface is located at $x = W_d$, where W_d is the depletion-layer thickness. Since we are considering an electron in the conduction band, it is convenient to use the conduction-band edge of the quasineutral silicon region $E_c(x < 0)$ as the energy reference, as indicated in Fig. 2.57. $\psi(x)$ is the electrostatic potential at location x relative to $E_c(x < 0)$, i.e., $-q\psi(0) = E_c(x < 0)$, and $-q\psi(x)$ is the potential energy of an electron at location x. The Poisson equation [Eq. (2.44)] can be integrated twice to give

$$\psi(x) = -\frac{qN_d x^2}{2\varepsilon_{si}} - \frac{E_c(x<0)}{q}, \tag{2.243}$$

where N_d is semiconductor doping concentration. From Eq. (2.188), we have

$$W_d = \sqrt{\frac{2\varepsilon_{si}(\psi_{bi} - V_{app})}{qN_d}}. \tag{2.244}$$

In the WKB approximation for tunneling through an energy barrier, the transmission coefficient through the energy barrier represented by $-q\psi(x)$ for an electron with energy E is

$$\begin{aligned} T(E) &= \exp\left[\frac{-4\pi}{h}\int_{x_1}^{W_d}\sqrt{2m^*}\sqrt{-q\psi(x) - E}dx\right] \\ &= \exp\left[\frac{-4\pi}{h}\int_{x_1}^{W_d}\sqrt{2m^*}\sqrt{\frac{q^2N_d x^2}{2\varepsilon_{si}} + E_c(x<0) - E\,dx}\right], \end{aligned} \tag{2.245}$$

where the lower integration limit x_1 is given by $-q\psi(x_1) = E$. In considering an ohmic contact, we are interested in the current due to electrons tunneling from the quasineutral region of the silicon through the potential barrier into the metal at small applied voltages. These electrons have only thermal energy ($kT \approx 26$ meV at room temperature) which is small compared to the maximum tunneling barrier height $q(\psi_{bi} - V_{app})$, which is approximately equal to $q\psi_{bi}$ at small V_{app}. Therefore, we can assume the tunneling electrons to have an energy $E \approx E_c(x < 0)$. For these electrons, the tunneling process starts at $x_1 = 0$ and the corresponding transmission coefficient is

$$T[E = E_c(x<0)] = \exp\left(\frac{-q(\psi_{bi} - V_{app})}{E_{00}}\right), \tag{2.246}$$

where

$$E_{00} \equiv \frac{qh}{4\pi}\sqrt{\frac{N_d}{m^*\varepsilon_{si}}}. \tag{2.247}$$

For a heavily doped quasineutral silicon region, its Fermi level and conduction-band edge are about equal. That is, referring to Fig. 2.55(a), we have ψ_{bi}(heavily doped) $\approx \phi_{Bn}$. Therefore, for a heavily doped Schottky barrier, we have

$$T[(E = E_c(x<0)] = \exp\left(\frac{-q(\phi_{Bn} - V_{app})}{E_{00}}\right). \tag{2.248}$$

That is, for an ohmic contact, the current density varies as

$$J_{ohmic} \propto \exp\left(\frac{-q(\phi_{Bn} - V_{app})}{E_{00}}\right). \tag{2.249}$$

The specific contact resistance, or contact resistivity, ρ_c, is an important figure of merit for ohmic contacts:

$$\rho_c \equiv \left(\frac{\partial J}{\partial V_{app}}\right)^{-1}_{V_{app}=0}. \tag{2.250}$$

Using Eq. (2.249) for ohmic contacts, we have

$$\rho_c \propto \frac{E_{00}}{q}\exp(q\phi_{Bn}/E_{00}). \tag{2.251}$$

The behavior of ρ_c is dominated by the exponential factor. That is, to ensure a low contact resistance, a low-barrier metal should be used and the silicon should be as heavily doped as possible (to maximize E_{00}). For $N_d = 10^{20}$ cm^{-3} and $T = 300$ K, ρ_c is about 8×10^{-6} Ω-cm^2 for $q\phi_{Bn} = 0.6$ eV, and about 8×10^{-8} Ω-cm^2 for $q\phi_{Bn} = 0.4$ eV (Yu, 1970; Chang, *et al.*, 1971). Experimental determination of the resistance of a contact can be very involved for low-resistance contacts. The reader is referred to the literature for a discussion and comparison of the many contact-resistance measurement methods (Schroder, 1990).

2.5 High-Field Effects

In the presence of an electric field, carriers gain energy from the field as they drift along. These carriers in turn lose energy by emitting phonons. As the field increases, the average energy of the carriers increases. At sufficiently high fields, a number of physical phenomena which have important implications on the design and operation of VLSI devices can occur. In the case of high fields in silicon, these phenomena include impact ionization, or generation of electron–hole pairs, junction breakdown, band-to-band tunneling, and injection of hot carriers from locations near the silicon–oxide interface into the silicon dioxide region. In the case of high fields in silicon dioxide, the important phenomena include tunneling through the oxide layer and dielectric breakdown. The basic physics of these phenomena as they relate to VLSI devices is discussed in this section.

2.5.1 Impact Ionization and Avalanche Breakdown

Consider the depletion region of a p–n diode. At sufficiently high fields, an electron in the conduction band can gain enough energy to "lift" an electron from the valence band into the conduction band, thus generating one free electron in the conduction band and one free hole in the valence band. This process is known as *impact ionization*. Similarly, a hole in the valence band can gain enough energy to cause impact ionization. If the field is high enough, these secondary electrons and holes can themselves cause further impact ionization, thus beginning a process of carrier multiplication in the high-field region. The p–n diode breaks down when the multiplication process runs away or becomes an avalanche. The equations relating the rate of impact ionization to the condition for avalanche breakdown are derived below.

Consider a reverse-biased p–n diode with an electric field in its depletion region high enough to cause impact ionization. Let $x = 0$ and $x = W$ be the locations of the two boundaries of the depletion region. Suppose there is a hole current I_{p0} entering the depletion region at $x = 0$, as illustrated in Fig. 2.58. This hole current will generate electron–hole pairs. The secondary electrons and holes in turn cause further impact ionization as they traverse the depletion region. Thus, the hole current will increase

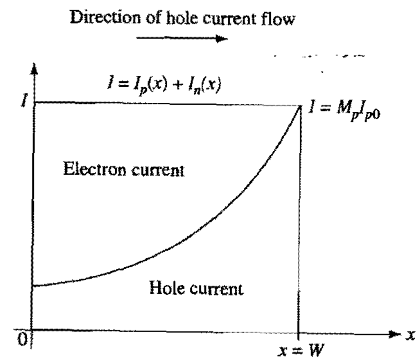

Figure 2.58. Schematic illustration of the steady-state current caused by hole-initiated impact ionization within the depletion region for a p–n diode.

with distance, reaching a value of $M_p I_{p0}$ at $x = W$, where M_p is the multiplication factor for holes. At steady state, the total current I is constant and independent of distance, i.e., $I = M_p I_{p0}$. Within the depletion region, the total current is the sum of the hole and electron currents (Moll, 1964), i.e.,

$$I = I_p(x) + I_n(x). \tag{2.252}$$

These current components are illustrated in Fig. 2.58. The field is such that holes move towards the right ($x = W$), and electrons move towards the left ($x = 0$).

Consider a differential distance between x and $x + dx$. There are $I_p(x)/q$ holes and $I_n(x)/q$ electrons crossing this differential distance per unit time. In crossing this differential distance, the holes cause $\alpha_p(x)I_p(x)dx/q$ electron–hole pairs to be generated, where α_p is the *hole-initiated rate of electron–hole pair generation per unit distance*. Similarly, the number of electron–hole pairs generated by the electrons is $\alpha_n(x)I_n(x)dx/q$, where α_n is the *electron-initiated rate of electron–hole pair generation per unit distance*. Thus the increase in the hole current as the electrons and holes cross the differential distance dx is

$$dI_p = \alpha_p I_p dx + \alpha_n I_n dx. \tag{2.253}$$

Equations (2.252) and (2.253) give

$$\frac{dI_p}{dx} = (\alpha_p - \alpha_n)I_p + \alpha_n I, \tag{2.254}$$

which, subject to the boundary condition $I_p(0) = I/M_p$, has a solution (Sze, 1981)

$$I_p(x) = I\left[\frac{1}{M_p} + \int_0^x \alpha_n \exp\left(-\int_0^{x'}(\alpha_p - \alpha_n)dx''\right)dx'\right]\exp\left(\int_0^x(\alpha_p - \alpha_n)dx'\right). \tag{2.255}$$

Since we are considering hole-initiated impact ionization, and there is no electron current entering the depletion region at $x = W$, the hole current at $x = W$ is simply equal to I. Therefore, Eq. (2.255) gives

$$\frac{1}{M_p} = \exp\left(-\int_0^W (\alpha_p - \alpha_n)dx\right) - \int_0^W \alpha_n \exp\left(-\int_0^x (\alpha_p - \alpha_n)dx'\right)dx. \tag{2.256}$$

Similarly, for impact ionization initiated by electrons, the electron multiplication factor M_n is given by

$$\frac{1}{M_n} = \exp\left(-\int_0^W (\alpha_n - \alpha_p)dx\right) - \int_0^W \alpha_p \exp\left(-\int_x^W (\alpha_n - \alpha_p)dx'\right)dx. \tag{2.257}$$

Avalanche breakdown occurs when carrier multiplication by impact ionization runs away, i.e., when the multiplication factors become infinite. It is shown in Appendix 8 that ***the condition for avalanche breakdown is the same whether the breakdown process is initiated by electrons or by holes***. That is, when a p–n junction breaks down, it does not matter if the avalanche breakdown process is initiated by an electron or by a hole.

It should be noted that the avalanche multiplication of electrons and holes is a positive feedback process where both the electrons and holes generated by impact ionization take part in generating additional electrons and holes. As a result, it is possible to have avalanche breakdown, i.e., M_n and M_p being infinite, for finite values of W, α_n and α_p at some large reverse bias voltage. If the feedback process by either the secondary electrons or the secondary holes were absent, avalanche breakdown would not occur for finite values of α_n and α_p. To demonstrate this point, let us consider the case of impact ionization initiated by holes, where the multiplication factor M_p is given by Eq. (2.256). If there were no positive feedback by the secondary electrons, we would have $\alpha_n = 0$. In this case, Eq. (2.256) shows that M_p is finite (no breakdown) for any finite values of α_p.

If the high-field region where impact ionization occurs is sufficiently wide, the positive feedback process will eventually lead to avalanche breakdown. It is left to the reader to show that, for the special case of both α_n and α_p being constant, independent of distance or electric field, avalanche breakdown occurs when the width of the high-field region approaches a value of $[\ln(\alpha_p/\alpha_n)]/(\alpha_p-\alpha_n)$ (see Exercise 2.20).

In theory, Eqs. (2.256) and (2.257) can be used to calculate the multiplication factors, and hence the breakdown voltage. In practice, however, the ionization rates, as well as the junction doping profiles, are simply not known accurately enough for calculation of breakdown voltages to be made with sufficient accuracy for VLSI device design purposes. Breakdown voltages in modern VLSI devices are usually determined experimentally.

2.5.1.1 Empirical Impact Ionization Rates

The measured ionization rates are often expressed in the ***empirical*** form of

$$\alpha = A \exp(-b/\mathscr{E}) \tag{2.258}$$

where A and b are constants, and $\mathscr{E}$ is the electric field (Chynoweth, 1957). There is quite a bit of spread in the measured impact ionization rates reported in the literature. However, the most recent measurements give similar results (van Overstraeten and de Man, 1970; Grant, 1973). These results are shown in Table 2.2 and plotted in Fig. 2.59.

Two points are clear from Fig. 2.59. First, α_n is much larger than α_p, particularly at low electric fields. This is due to the effective mass of holes being much larger than that of

Table 2.2 Impact-Ionization Rates in Silicon

		α_n (cm^{-1})		α_p (cm^{-1})	
Data	Field Range (V/cm)	A_n (cm^{-1})	b_n (V/cm)	A_p (cm^{-1})	b_p (V/cm)
van Overstraeten and de Man	$1.75 \times 10^5 < \mathscr{E} < 4.0 \times 10^5$	7.03×10^5	1.231×10^6	1.582×10^6	2.036×10^6
	$4.0 \times 10^5 < \mathscr{E} < 6.0 \times 10^5$	7.03×10^5	1.231×10^6	6.71×10^5	1.693×10^6
Grant	$2.0 \times 10^5 < \mathscr{E} < 2.4 \times 10^5$	2.6×10^6	1.43×10^6	2.0×10^6	1.97×10^6
	$2.4 \times 10^5 < \mathscr{E} < 5.3 \times 10^5$	6.2×10^5	1.08×10^6	2.0×10^6	1.97×10^6
	$5.3 \times 10^5 < \mathscr{E}$	5.0×10^5	0.99×10^6	5.6×10^5	1.32×10^6

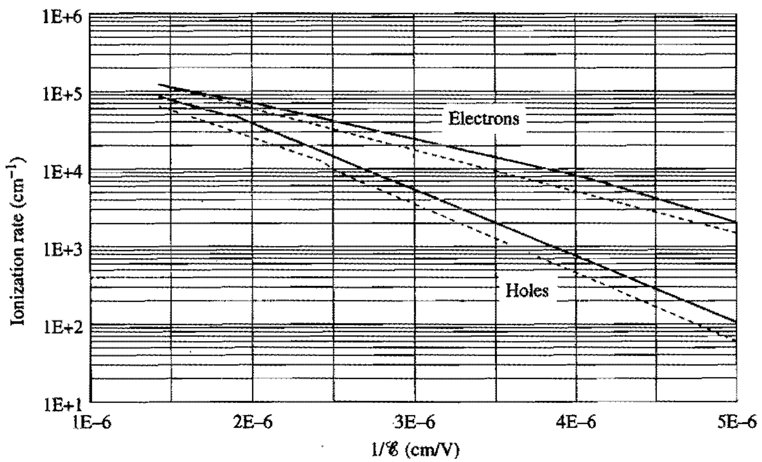

Figure 2.59. Impact-ionization rates in silicon. The solid curves are data of Grant (1973), and the dash curves are data of van Overstraeten and de Man (1970).

electrons. Second, the impact ionization rates increase very rapidly with electric field. For the depletion region of a p–n diode where the electric field is not constant, it is the small region surrounding the maximum-field point that contributes the most to the impact-ionization currents. Thus, to minimize impact ionization in a p–n diode, the maximum electric field should be minimized. As mentioned in Section 2.2.2.4, doping-profile grading, or using lightly doped regions or i-layers, can effectively reduce the peak electric field in a p–n junction.

Impact ionization rates decrease as temperature increases (Grant, 1973). This is due to the increased lattice scattering at higher temperatures. The data in Table 2.2 and Fig. 2.59 are for room temperature.

2.5.2 Band-to-Band Tunneling

When the electric field across a reverse-biased p–n junction approaches 10^6 V/cm, significant current flow can occur due to tunneling of electrons from the valence band

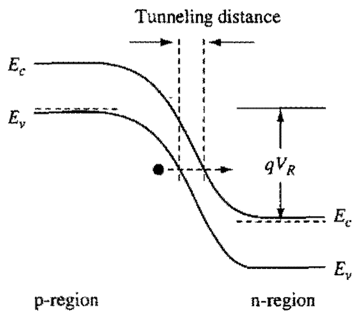

Figure 2.60. Schematic illustrating band-to-band tunneling in a p–n junction.

of the p-region into the conduction band of the n-region. This phenomenon is illustrated schematically in Fig. 2.60. In silicon this tunneling process usually involves the emission or absorption of phonons (Kane, 1961; Chynoweth *et al.*, 1960), and the tunneling current density is given by (Fair and Wivell, 1976)

$$J_{b-b} = \frac{\sqrt{2m^*}q^3\mathscr{E}V_R}{4\pi^3\hbar^2E_g^{1/2}}\exp\left(-\frac{4\sqrt{2m^*}E_g^{3/2}}{3q\mathscr{E}\hbar}\right) \tag{2.259}$$

where $\mathscr{E}$ is the electric field, E_g is the energy bandgap, and V_R is the reverse bias voltage across the junction. An upper-bound estimate of the peak electric field can be made by assuming a one-sided junction. In this case, the analyses in Section 2.2.2 give the upper-bound for the electric field as

$$\mathscr{E}_{\max} = \sqrt{\frac{2qN_a(V_R+\psi_{bi})}{\varepsilon_{si}}}, \tag{2.260}$$

where N_a is the doping concentration of the lightly doped side (assumed p-type) of the diode and ψ_{bi} is the built-in potential of the diode. With these approximations, the band-to-band tunneling current density is about 1 A/cm^2 for $N_a = 5\times10^{18}$ cm^{-3} and $V_R = 1$ V (Taur *et al.*, 1995a). More recently, Solomon *et al.* (2004) showed that band-to-band tunneling current can be modeled using the concept of an effective tunneling distance. In this model, the tunneling current is assumed to be proportional to $\exp(-w_T/\lambda_T)$, where w_T is the tunneling distance, illustrated schematically in Fig. 2.60, and λ_T is an effective tunneling decay length. The reader is referred to Solomon's paper for the details.

As will be discussed in Chapters 4 and 7, in scaling down the dimensions of a transistor, the doping concentrations increase and the junction doping profiles become more abrupt, and hence band-to-band tunneling effect increases. Once the leakage current due to band-to-band tunneling is appreciable, it increases very rapidly with electric field or reduction of the tunneling distance. For modern VLSI devices, band-to-band tunneling is becoming one of the most important leakage-current components, particularly for applications such as DRAM and battery-operated systems where leakage currents must be kept extremely low.

2.5.3 Tunneling into and through Silicon Dioxide

Consider an MOS capacitor discussed in Section 2.3. For simplicity, the gate electrode is assumed to be heavily doped n-type polysilicon. When biased at the flatband condition, the energy-band diagram is as shown in Fig. 2.61(a), where $q\phi_{ox}$ denotes the Si–SiO$_2$ interface energy barrier for electrons which, as indicated in Fig. 2.29, is about 3.1 eV. When a large positive bias is applied to the gate electrode, electrons in the strongly inverted surface can tunnel through the oxide layer and hence give rise to a gate current. Similarly, if a large negative voltage is applied to the gate electrode, electrons from the n$^+$ polysilicon can tunnel through the oxide layer, and again give rise to a gate current.

2.5.3.1 Fowler–Nordheim Tunneling

Fowler–Nordheim tunneling occurs when electrons tunnel into the conduction band of the oxide layer and then drift through the oxide layer. Figure 2.61(b) illustrates Fowler–Nordheim tunneling of electrons from the silicon surface inversion layer. The

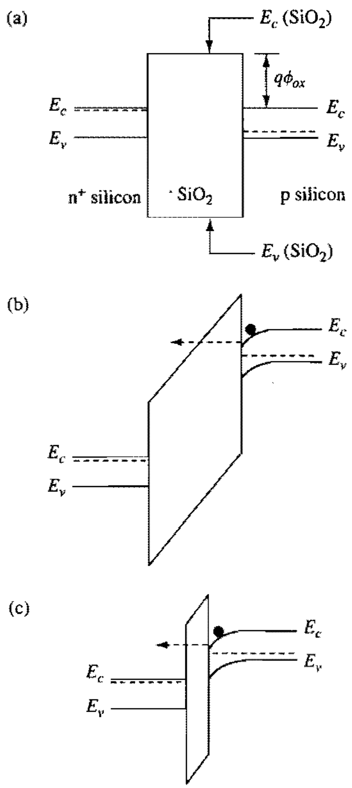

Figure 2.61. Tunneling effects in an MOS capacitor structure: (a) energy-band diagram of an n-type polysilicon-gate MOS structure at flat band; (b) Fowler–Nordheim tunneling; (c) direct tunneling.

complete theory of Fowler–Nordheim tunneling is rather complicated (Good Jr. and Müller, 1956). For the simple case, where the effects of finite temperature and image-force barrier lowering (which is discussed in Appendix 7) are ignored, the tunneling current density is given by (Lenzlinger and Snow, 1969)

$$J_{FN} = \frac{q^2 \mathscr{E}_{ox}^2}{16\pi^2 \hbar \phi_{ox}} \exp\left(-\frac{4(2qm^*)^{1/2} \phi_{ox}^{3/2}}{3\hbar \mathscr{E}_{ox}} \right), \tag{2.261}$$

where $\mathscr{E}_{ox}$ is the electric field in the oxide. Equation (2.261) shows that Fowler–Nordheim tunneling current is characterized by a straight line in a plot of $\log\left(J/\mathscr{E}_{ox}^2\right)$ versus $1/\mathscr{E}_{ox}$.

As discussed later in Section 2.5.3.3, electrons tunneling into an oxide layer can be trapped in an oxide layer. If the tunneling current is measured at a constant voltage, then the trapped electrons in turn can cause the observed tunneling current to decrease with time. Depending on the thickness of the oxide layer and its formation process, this decrease in tunneling current can go on for some time before it reaches a more-or-less steady state. The tunneling currents reported in the classic paper by Lenzlinger and Snow were taken after the samples were first subjected to a current density of about 10^{-10} A/cm^2 for two hours, during which time the tunneling currents decreased by about one order of magnitude from their initial values (Lenzlinger and Snow, 1969). At an oxide field of 8 MV/cm, the measured steady-state Fowler–Nordheim tunneling current density is about 5×10^{-7} A/cm^2. The initial tunneling current is about ten times larger. For normal device operation, Fowler–Nordheim tunneling current is negligible.

The characteristics of the tunneling currents represented by Eqs. (2.259) and (2.261) are determined primarily by their exponential factors. It should be noted that the exponents of the two equations are basically the same. The Fowler–Nordheim tunneling is through a triangular barrier of height $q\phi_{ox}$, slope $q\mathscr{E}_{ox}$ and tunneling distance $\phi_{ox}/\mathscr{E}_{ox}$. It is left as an exercise (Exercise 2.21) for the reader to show that the band-to-band tunneling exponent in Eq. (2.259) can be derived from the WKB approximation, i.e., Eq. (2.245), for tunneling through a triangular barrier of height E_g, slope $q\mathscr{E}$ and tunneling distance $E_g / q\mathscr{E}$.

2.5.3.2 Direct Tunneling

If the oxide layer is very thin, say 4 nm or less, then, instead of tunneling into the conduction band of the SiO_2 layer, electrons from the inverted silicon surface can tunnel directly through the forbidden energy gap of the SiO_2 layer. This is illustrated in Fig. 2.61(c). The theory of *direct tunneling* is even more complicated than that of Fowler–Nordheim tunneling, and there is no simple dependence of the tunneling current density on voltage or electric field (Chang *et al.*, 1967; Scheugraf *et al.*, 1992). Direct-tunneling current can be very large for thin oxide layers. Figure 2.62 is a plot of the measured and simulated thin-oxide tunneling current versus voltage in polysilicon-gate MOSFETs (Lo *et al.*, 1997). For the gate-voltage range shown in Fig. 2.62, the current is primarily a direct-tunneling current. Direct-tunneling current is important in MOSFETs of very small dimensions, where the gate oxide layers can approach 1 nm in thickness.

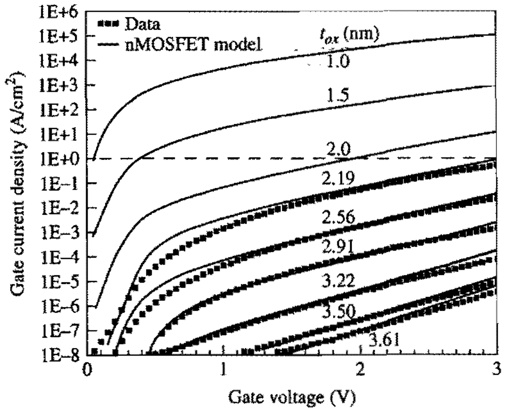

Figure 2.62. Measured (dots) and simulated (solid lines) tunneling currents in thin-oxide polysilicon-gate MOS devices. The dashed line indicates a tunneling-current level of 1 A/cm^2. (After Lo *et al.*, 1997.)

2.5.3.3 Defect Generation Caused By Tunneling Current

The tunneling of electrons into and through a silicon dioxide layer can cause "defects" to be generated within the oxide layer and/or at the oxide–silicon interface. These defects can take the form of electron traps, hole traps, trapped electrons, trapped holes, or interface states (DiStefano and Shatzkes, 1974; Harari, 1978; Chen *et al.*, 1986; DiMaria *et al.*, 1993). These defects govern the time dependent behavior of the tunneling current and play an important role in the wear-out and eventual breakdown of the oxide layer. In this subsection, we briefly discuss how these defects can influence the tunneling process. The reader is referred to the vast literature on the subject for more details (DiMaria and Cartier, 1995, and Suehle, 2002, and the references therein).

- *Tunneling into an electron trap*. As electrons tunnel into an oxide layer, some of the electrons can get trapped. The trapped electrons modify the oxide field such that the field near the cathode (the electrode that acts as an electron source) is decreased, while the field near the anode (the electrode that acts as an electron sink) is increased. This is illustrated in Fig. 2.63. The reduced field near the cathode, in turn, causes the tunneling current to decrease. In a constant-voltage tunneling current measurement, electron trapping is what causes the current to decrease with time. In a ramped-voltage (voltage increasing with time at a constant rate) tunneling current measurement, electron trapping often leads to a hysteresis in the voltage–current plot.
- *Hole generation, injection, and trapping*. As an electron travels in the conduction band of an oxide layer, it gains energy from the oxide field. If the voltage drop across the oxide layer is larger than the bandgap energy of silicon dioxide, which, as indicated in Fig. 2.28, is about 9 eV, the electron can gain sufficient energy to cause impact ionization in the oxide. The holes generated by impact ionization can be trapped in the oxide. Holes can be injected indirectly into the oxide layer during electron tunneling as well. A tunneling electron arriving at the anode can cause impact

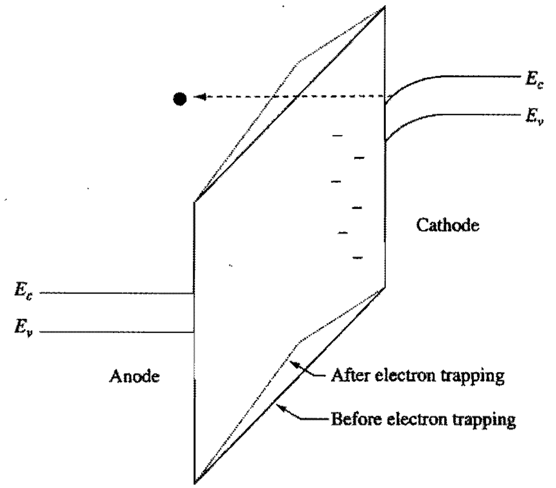

Figure 2.63. Schematic illustrating the trapping of tunneling electrons. As electrons are trapped, the oxide field near the cathode (electron source) is decreased, while the oxide field near the anode (electron sink) is increased.

ionization in the anode near the oxide–anode interface. Depending on the energy of the tunneling electron, the hole thus generated can be from deep down in the valence band of the anode, and thus can be "hot." A hot hole in the anode near the anode–oxide interface can be injected into the oxide layer. This process is illustrated in Fig. 2.64. The injected hole can be trapped in the oxide layer as it travels towards the cathode. The trapped holes in the oxide layer cause the oxide field near the cathode to increase, which in turn causes the tunneling current to increase. This is illustrated in Fig. 2.65. Thus the trapping of holes provides a positive feedback to the electron tunneling process. In a constant-voltage tunneling current measurement, hole trapping is the primary reason the current increases with time.

- *Trap and interface-state generation.* Traps can also be generated in the silicon dioxide layer and at the oxide–silicon interface by the electron current (Harari, 1978; DiMaria, 1987; Hsu and Ning, 1991). In addition to increasing electron and hole trapping, these traps can enhance the tunneling current by assisting in the tunneling process, as discussed in the two subsections below.

2.5.3.4 Bulk-Trap-Assisted Tunneling

Instead of tunneling directly through an oxide layer, an electron can first tunnel from the cathode into an electron trap in the oxide and then tunnel from the trap to the anode. Thus, traps in the oxide layer can act as stepping stones for the tunneling electrons. This is illustrated in Fig. 2.66. The enhanced tunneling current in turn can increase the generation of traps in the oxide. Thus, trap-assisted tunneling plays an important role in the degradation of an oxide under voltage stress because of the positive feedback between trap-generation and trap-assisted tunneling (DiMaria and Cartier, 1995).

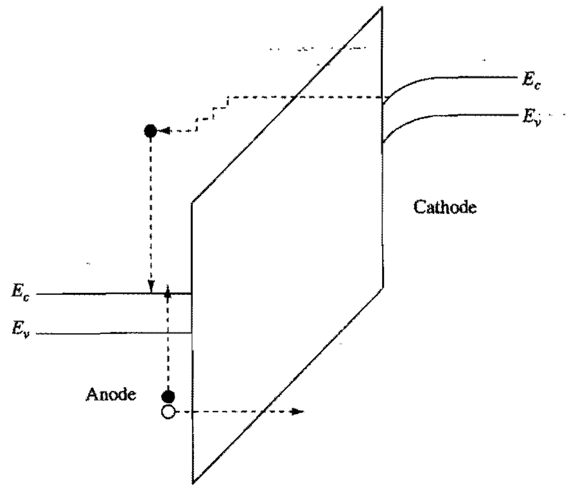

Figure 2.64. Schematic illustrating the generation of an electron–hole pair in the anode by a tunneling electron. The hole thus generated can then be injected (by tunneling in this example) into the oxide layer.

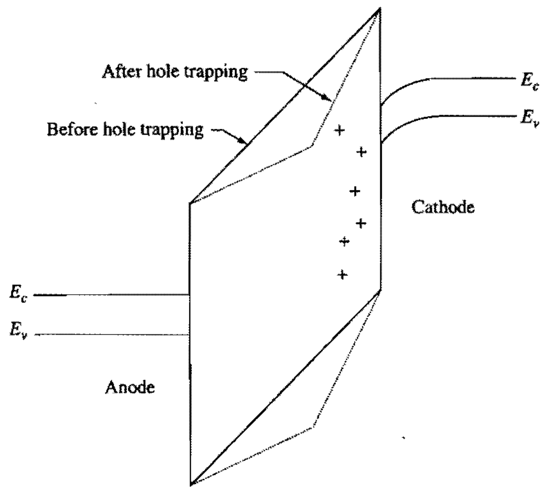

Figure 2.65. Schematic showing the trapping of holes in the oxide layer. The trapped holes enhance the electric field near the cathode, and decrease the electric field near the anode.

2.5.3.5 Interface-Trap-Assisted Tunneling at Low Voltages

Interface traps can also assist in the tunneling process. This is illustrated in Fig. 2.67. Interface states exist at both the substrate silicon–oxide interface and the gate silicon–oxide interface. In general, states above the Fermi level are empty of electrons and states below the Fermi level are filled with electrons. In Fig. 2.67(a), the gate electrode is biased

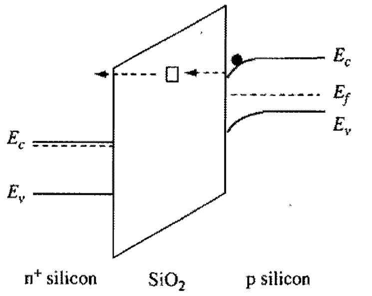

Figure 2.66. Schematic illustrating bulk-trap-assisted tunneling in an MOS capacitor structure.

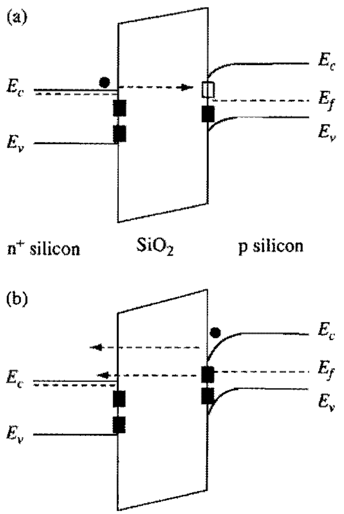

Figure 2.67. Schematics illustrating interface-trap-assisted tunneling of electrons in a silicon-gate MOS structure. (a) The gate electrode has a small negative bias. The Fermi level in the p-silicon lies somewhat below that in the n$^+$ silicon gate. (b) The gate electrode has a small positive bias. The Fermi level in the p silicon lies somewhat above that of the n$^+$ silicon gate.

slightly negatively (towards flatband condition and surface accumulation of the p-silicon). As the surface of the p silicon is driven towards flat band and surface accumulation, more and more states at the interface become empty of electrons. An electron in the conduction band of the gate electrode can tunnel into an empty state at the p-silicon surface. Also, not shown in the figure, an electron in an occupied surface state of the gate electrode can also tunnel into an empty surface state of the p silicon. These interface-trap-assisted tunneling processes are in addition to the normal tunneling of electrons from the conduction band of the gate electrode into the conduction band of the p substrate.

In Fig. 2.67(b), the gate is biased somewhat positively (towards surface inversion of the p silicon). As the surface of the p silicon is driven towards inversion, more and more

states at the surface become filled with electrons. Until the p silicon surface is inverted, there are very few electrons available to tunnel from the conduction band of the p silicon to the conduction band of the gate electrode. However, electrons from the filled surface states of the p silicon can tunnel into the conduction band of the gate electrode. Since interface-trap-assisted tunneling is a direct tunneling process, it is important only for thin oxides. Also, as can be inferred from Fig. 2.67, interface-trap-assisted tunneling is effective only when the p silicon surface potential lies between inversion and weak accumulation. That is, interface-trap-assisted tunneling is important only at low voltages. ***Interface-trap-assisted tunneling can enhance the low-voltage tunneling current in thin oxides whether the gate electrode is biased positively or negatively.*** Interface-trap-assisted tunneling in modern CMOS devices is a widely studied subject. The reader is referred to the literature for more details (see e.g. Crupi *et al.*, 2002, and the references therein).

2.5.4 Injection of Hot Carriers from Silicon into Silicon Dioxide

If a region of sufficiently high electric field is located near the Si–SiO_2 interface, some electrons or holes in the region can gain enough energy from the electric field to surmount the interface barrier and enter the SiO_2 layer. In general, injection from Si into SiO_2 is much more likely for hot electrons than for hot holes because (a) electrons can gain energy from the electric field much more readily than holes due to their smaller effective mass, and (b) the Si–SiO_2 interface energy barrier is larger for holes (≈ 4.6 eV) than for electrons (≈ 3.1 eV), as indicated in Fig. 2.28. The process of hot-electron and hot-hole injection from silicon into silicon dioxide is much too complex to model quantitatively. Thus far, quantitative agreement has been shown only for the special case of hot electrons traveling from the silicon substrate perpendicularly towards the Si–SiO_2 interface, and only with Monte-Carlo models that take into account the correct band structures, all the relevant scattering processes, and nonlocal transport properties (Fischetti *et al.*, 1995). Here we discuss a simple model for the injection of hot electrons and hot holes from Si into SiO_2. The same model can be modified readily to describe the injection of hot holes from Si into SiO_2.

2.5.4.1 Energy Barrier for Hot Electron Injection

The energy barrier $q\phi_{ox}$ shown in Fig. 2.61(a) is the difference in energy between the conduction band of SiO_2 and the conduction band of Si. In Appendix 7, it is shown that the image-force effect causes the barrier for injection of a hot electron from Si into SiO_2 to be lowered by an amount equal to

$$q\Delta\phi = \sqrt{\frac{q^3 \mathscr{E}_{ox}}{4\pi\varepsilon_{ox}}}. \tag{2.262}$$

The actual energy barrier for hot electron emission is therefore $(q\phi_{ox} - q\Delta\phi)$.

For $\mathscr{E}_{ox} = 1 \times 10^6 \text{V/cm}$, $q\Delta\phi = 0.19$ eV. Thus, for practical oxide fields of 10^6 V/cm or larger, image-force barrier lowering is not negligible compared to the interface energy

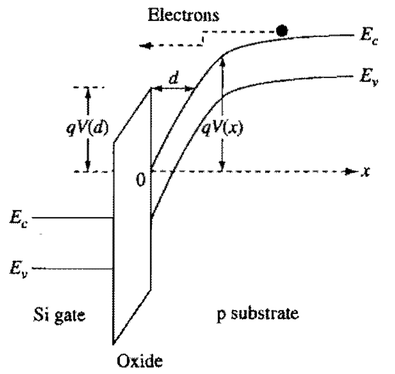

Figure 2.68. Schematic illustrating hot electrons traveling perpendicularly towards the Si–SiO_2 interface and being injected into the SiO_2 layer.

barrier of 3.1 eV. Image-force barrier lowering is included in the more accurate theories of Fowler–Nordheim tunneling (Lenzlinger and Snow, 1969) and direct tunneling (Chang *et al.*, 1967). However, in the literature there are also publications questioning the validity of the concept of image potential at the interface between a semiconductor and an insulator (see e.g. Fischetti *et al.*, 1995, and the references therein).

2.5.4.2 The Lucky Electron Model

The simple one-dimensional injection process is illustrated in Fig. 2.68. The simplest model for describing the injection process is the *lucky electron* model proposed by Shockley (1961). It is an empirical model, but it describes the measured data surprisingly well (Ning *et al.*, 1977a). In this model, the probability that a hot electron at a distance d from the Si–SiO_2 interface will be emitted into the SiO_2 layer is expressed as

$$P(d) = A\exp(-d/\lambda), \tag{2.263}$$

where λ is an effective mean free path for energy loss by hot electrons in silicon, and A is a fitting constant to the experimental data. The relation between the parameter d, the effective hot-electron emission energy barrier, and the electron potential energy, is illustrated in Fig. 2.68. The parameter d can be obtained as follows. Referring to Fig. 2.68, $qV(x)$ is the potential energy of an electron at x. An electron at $x = d$ has just enough potential energy to overcome the effective energy barrier for emission if it can travel from $x = d$ to the interface at $x = 0$ without encountering any energy-losing collision. That is, $qV(d)$ is equal to the effective energy barrier for emission. The injection of hot holes from Si into SiO_2 can be described by a similar *lucky hole* model (Selmi *et al.*, 1993).

It was determined empirically that the effective energy barrier for electron emission can be written as

$$qV(d) = q\phi_{ox} - q\Delta\phi - \alpha\mathscr{E}_{ox}^{2/3}, \tag{2.264}$$

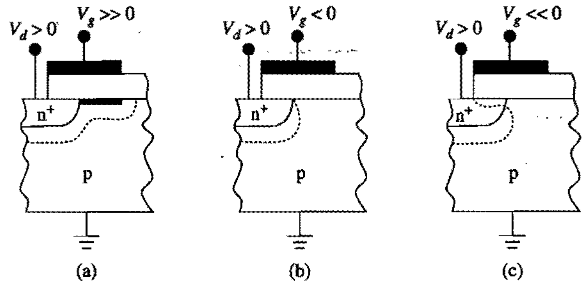

Figure 2.69. Schematics illustrating a gated n^+–p diode when the surface is (a) inverted and (b) accumulated, and (c) when the surface of the n^+ region is depleted or inverted. The dashed lines indicate the boundary of the depletion region.

where the first two terms are the Si–SiO_2 interface energy barrier and the image-force barrier lowering discussed in the previous subsection, and the third term is introduced to account for the fact that hot electrons without enough energy to surmount the image-force-lowered energy barrier can still tunnel into the oxide layer. It was found that setting $\alpha = 1 \times 10^{-5} e(\text{cm}^2\text{-V})^{1/3}$ and $A = 2.9$ fits a wide range of measured emission probabilities (Ning *et al.*, 1977a).

The temperature dependence of the hot-electron injection process is contained in the temperature dependence of the effective mean free path (Crowell and Sze, 1966b)

$$\lambda(T) = \lambda_0 \tanh(E_R/2kT), \tag{2.265}$$

where $E_R = 63$ meV is the optical-phonon energy, and λ_0 is the low-temperature limit of λ. It was found empirically that $\lambda_0 = 10.8$ nm (Ning *et al.*, 1977a). The mean free path associated with hot-hole injection has a comparable value (Selmi *et al.*, 1993).

2.5.5 High-Field Effects In Gated Diodes

Thus far, the effects of high fields have been considered for p–n diodes and MOS capacitors separately. In a gated diode structure, both the location of the peak-field region and the magnitude of the peak field vary with gate voltage. Let us consider a gated n^+–p diode. As discussed in Section 2.3.5, when the gate is biased to invert the silicon surface, the inverted surface region has about the same potential as the n^+ region, and the gated diode behaves like a large-area n^+–p diode. If the p-region is uniformly doped, the depletion-layer width is about the same below the n^+ silicon region as below the surface inversion region, hence the electric field is rather uniformly distributed and is about the same as in a simple p–n diode. This is illustrated schematically in Fig. 2.69(a).

When the gate is biased somewhat negatively to accumulate the silicon surface, the silicon surface under the gate has about the same potential as the p-type substrate. Owing to the presence of the accumulated holes at the surface, the surface behaves like a p-region more heavily doped than the substrate, causing the depletion layer at the surface

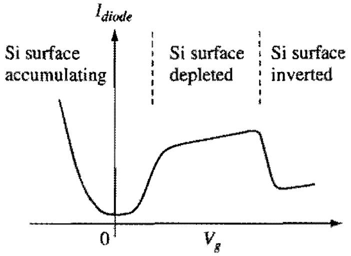

Figure 2.70. Schematic illustrating an n^+–p gated-diode leakage current as a function of gate voltage.

to become narrower than elsewhere. This is illustrated schematically in Fig. 2.69(b). The narrowing of the depletion layer at or near the intersection of the p–n junction and the Si–SiO_2 interface causes *field crowding*, or an increase in the local electric field.

When the negative gate bias is large enough, the n^+ region under the gate can become depleted, and even inverted. This is illustrated in Fig. 2.69(c). In this case, the gate and the n^+ region behave like an MOS capacitor with a heavily doped n-type "substrate." There is more field crowding, and the peak field increases.

As the electric field in and around the gated p–n junction is increased by the gate voltage, all the high-field effects, such as avalanche multiplication and band-to-band tunneling, can increase very dramatically. Thus the leakage current of a reverse-biased gated diode can increase dramatically when the gate voltage begins to cause field crowding in and around the junction region. This is illustrated in Fig. 2.70 which shows the expected gated-diode leakage current as a function of gate voltage. In Fig. 2.70, aside from the increase in current due to field crowding at negative gate voltage, the diode current is simply the sum of the leakage current from the depletion region of a diode and the leakage current from the exposed surface states (Grove and Fitzgerald, 1966). When the Si surface is in accumulation (at $V_g \approx 0$), the leakage current is from the depletion region of the bulk diode alone. When the Si surface is inverted (at large V_g), the leakage current is higher because of the additional leakage current coming from the depletion region of the diode formed by the surface inversion layer and the substrate. When the Si surface is depleted (at intermediate values of V_g), the leakage current is the largest because of the addition leakage current coming from the exposed surface states of the depleted silicon surface. When field crowding occurs in the drain junction of a MOSFET, the increased junction leakage current is called *gate-induced drain leakage*, or GIDL (Chan *et al.*, 1987a; Noble *et al.*, 1989). ***GIDL is an important leakage-current component that must be minimized in modern CMOS devices.***

It should be noted that for the gated n^+–p diode considered here, when the gate is biased to accumulate the silicon surface, the oxide field favors the injection of hot holes from the silicon substrate into the silicon dioxide layer (Verwey, 1972). Similarly, for a gated p^+–n diode, the oxide field favors the injection of hot electrons when the gate is biased to accumulate the silicon surface. Thus injection of majority carriers, instead of minority carriers, from the silicon substrate into the silicon dioxide layer takes place when significant gate-voltage-induced avalanche multiplication occurs in a gated diode.

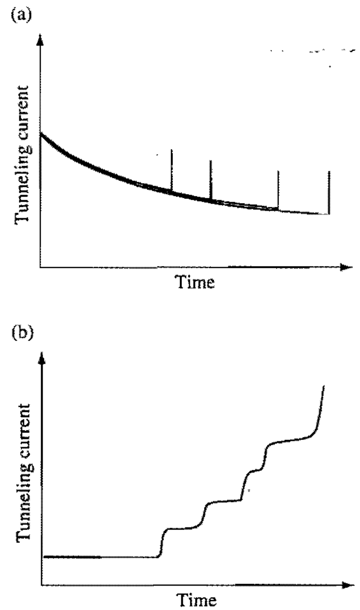

Figure 2.71. Schematics illustrating the typical time dependence of the tunneling current at constant voltage in (a) a thick oxide layer and (b) a thin oxide layer. A sudden jump in tunneling current signals a dielectric breakdown event.

2.5.6 Dielectric Breakdown

As discussed in Section 2.5.3, significant electron tunneling can take place when a large electric field is applied across an oxide layer. Figure 2.71 illustrates schematically the typical time dependence of the tunneling current when a constant voltage is applied across an oxide layer. A sudden jump in the tunneling current indicates that the oxide sample has suffered a *dielectric breakdown* event. For oxides thicker than about 10 nm, the tunneling current typically decreases gradually with time until the oxide breaks down. For these thick oxides, the voltages used to measure tunneling current are usually so large that, unless special care is taken to limit the current, a breakdown event usually leads to the oxide being physically damaged (Shatzkes *et al.*, 1974). There is a distribution in the measured oxide breakdown time (Harari, 1978). This is illustrated in Fig. 2.71(a). For oxides thinner than about 5 nm, the voltages used to measure tunneling current are usually sufficiently small so that not every breakdown event leads to catastrophic breakdown. For most samples, *successive breakdown* events are observed before final or catastrophic breakdown (Suñé *et al.*, 2004). This is illustrated in Fig. 2.71(b). An oxide layer ceases to be a good electrical insulator after it suffers final or catastrophic breakdown.

In thin oxides, the increase in tunneling current from the first breakdown event to final breakdown can occur quite gradually. When the thin gate oxide of a modern MOS transistor in a circuit starts showing signs of breaking down, often the gate tunneling

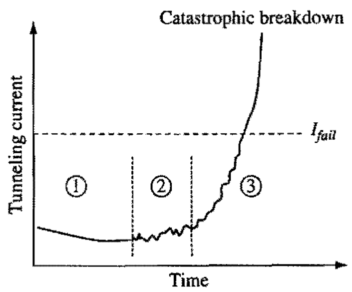

Figure 2.72. Schematic illustrating the evolution of the tunneling current in a thin oxide MOS device at constant applied voltage. I_{fail} indicates the tunneling current at which the device fails to function properly in a circuit. The three stages marked 1, 2, and 3 are discussed in the text.

current can grow to a sufficiently large value to cause the circuit to fail long before the gate oxide layer suffers final breakdown (Kaczer *et al.*, 2000; Linder *et al.*, 2001). Figure 2.72 is a schematic illustrating the time dependence of the tunneling current in a typical thin-oxide MOS device at constant voltage stress. There are roughly three stages in the evolution of the tunneling current. In the initial stage (stage 1 in Fig. 2.72), the current is relatively featureless, typically first decreasing, due to electron trapping, and then rising, due to hole trapping and trap-assisted tunneling, as a function of time. As the current continues to rise, it becomes noisy (stage 2). Finally, the current rises much more rapidly (stage 3) for some time before final breakdown. In Fig. 2.72, I_{fail} denotes the tunneling-current level at which a circuit using the MOS transistor fails to function properly. In the literature, stage 1 is often referred to as the *defect generation* stage or *stress-induced leakage current* stage (DiMaria, 1987; Stathis and DiMaria, 1998); stage 2 as the *soft breakdown* stage (Depas *et al.*, 1996), and stage 3 as the *successive breakdown* or *progressive breakdown* stage (Linder *et al.*, 2002; Okada, 1997; Suñé and Wu, 2002).

2.5.6.1 Breakdown Field

In the literature, the quality of an oxide film is often measured in terms of the electric field at which dielectric breakdown, usually the first breakdown event, occurs. "Good-quality" thick (>100 nm) SiO_2 films typically break down at fields greater than 10 MV/cm, while "good-quality" thin (<10 nm) SiO_2 films usually show larger breakdown fields, often in excess of 15 MV/cm. In bipolar transistors, because there are normally no thin oxide components, the electric fields across the oxide layers are usually so small that dielectric breakdown is not a concern. In CMOS devices, the maximum oxide field varies widely, depending on the application. For devices used in logic and memory circuits, the maximum oxide field is typically in the 3–6 MV/cm range in normal operation, and can reach as high as 5–9 MV/cm in special operations (such as during a device burn-in process). For devices used in electrically programmable nonvolatile memory applications, where normal operation involves tunneling through a thin dielectric layer, the maximum electric field across the thin dielectric layer is in excess of 10 MV/cm. Dielectric breakdown is a real concern in CMOS devices.

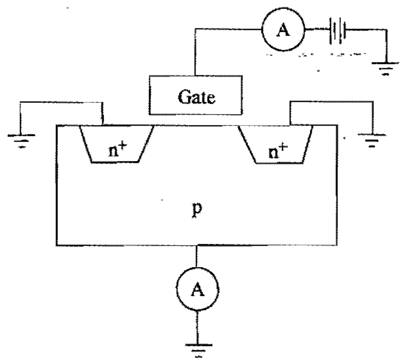

Figure 2.73. Schematic illustrating the bias configuration of an n-channel MOSFET for measuring the charge to breakdown and its hole-charge component.

2.5.6.2 Time to Breakdown and Charge to Breakdown

The breakdown characteristics of an oxide film are often described in terms of its *time to breakdown*, which measures the time needed for the film to reach breakdown, or its *charge to breakdown*, which measures the integrated total tunneling charge leading up to breakdown. In product design, we want to ensure that the gate current of a CMOS device will not grow to the point of causing circuit failure before the product end of life. Therefore, in product design, we want to know the time to breakdown. However, it appears easier to develop physical models relating charge to breakdown to the physical mechanisms involved in the dielectric breakdown process, such as hole current, trapping, trap generation, and interface state generation (Schuegraf and Hu, 1994; DiMaria and Stathis, 1997; Stathis and DiMaria, 1998), than to develop physical models relating time to breakdown to these physical mechanisms. Most publications on the physics of dielectric breakdown discuss the breakdown process in terms of charge to breakdown instead of time to breakdown. Therefore, we will not discuss time to breakdown any further here. The reader is referred to the literature for discussions on time to breakdown and the breakdown statistics in the time domain (see e.g. Suñé *et al.*, 2004, and the references therein).

As discussed in Section 2.5.3.3, a tunneling electron current can generate a hole current. Thus, the charge to breakdown, Q_{BD}, is the sum of the charges due to electrons and holes. If an MOS capacitor structure is used to measure Q_{BD}, then, owing to the two-terminal nature of the device, only the total charge can be measured. However, if an n-channel MOSFET or an n^+–p gated-diode structure is used to measure Q_{BD}, then both the total charge and the hole-charge component can be determined. For the case of an n-channel MOSFET, the bias configuration for such measurements is illustrated in Fig. 2.73. The basic concept of this *charge separation* method is that electron current is measured at the n-type terminal and hole current is measured at the p-type terminal. Integration of the gate current gives the total charge, and integration of the substrate current gives the charge due to the holes. It is shown that, charge for charge, hot holes are much more effective than tunnel electrons in generating defects that lead to oxide breakdown (Li *et al.*, 1999).

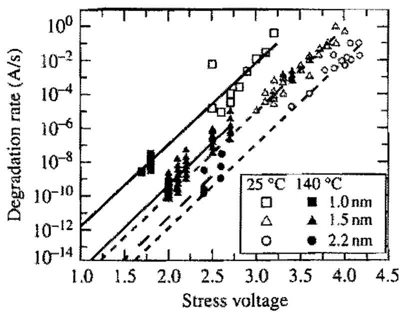

Figure 2.74. Measured rate of increase of breakdown current for typical thin oxides. (After Linder *et al.*, 2002.)

2.5.6.3 Progressive Breakdown and Successive Breakdown

Referring to Fig. 2.72, the evolution of the tunneling current can be described as a process of positive feedback between defect generation and trap-assisted tunneling. When a stress voltage is applied across an oxide layer, at first there are few defects in the oxide and the tunneling current is relatively low, decreasing with time as electrons are trapped. As trapped holes start to accumulate in the oxide, the current will start increasing. The tunneling current generates defects which in turn assist in the tunneling process (DiMaria, 1987; Stathis and DiMaria, 1998). At some point, soft breakdown starts when the defects in the oxide become dense enough such that an electron can tunnel relatively easily from one defect center to another across the oxide layer. This trap-assisted tunneling current tends to be noisy (Depas *et al.*, 1996). As the defect density continues to grow, hard breakdown (a breakdown event or a series of breakdown events) starts when there is a connected path of overlapping defects all the way across the oxide layer (Degraeve *et al.*, 1995; Stathis, 1999). This connected path of defects acts as a low-resistance conduction path for the electrons. Once hard breakdown starts, the electron current is completely dominated by the flow along this low-resistance path and the magnitude of the current is more-or-less independent of the device area. The electron current causes the diameter of the path of connected defects to grow, which in turn causes the current to grow. The current does not grow smoothly, but in a staircase manner. Each time the tunneling current jumps, it represents a breakdown event. Eventually, catastrophic breakdown of the oxide layer occurs. Thus, a thin oxide can go through many successive breakdown events before it breaks down catastrophically (Suñé and Wu, 2002). The oxide degradation rate (the rate at which the average breakdown current increases with time) is a strong function of the stress voltage (Linder *et al.*, 2002; Lombardo *et al.*, 2003). Figure 2.74 is a plot of typical measured degradation rates for thin oxides. It suggests that even after hard breakdown has commenced, the tunneling current in a thin oxide at a low voltage can take a long time to grow to a value sufficiently large to cause circuit failure.

In the literature, most charge to breakdown measurements are made by integrating the tunneling current until the current shows a sudden jump in magnitude, or until the first breakdown event. For a given oxide film, the charge to breakdown Q_{BD} is often plotted

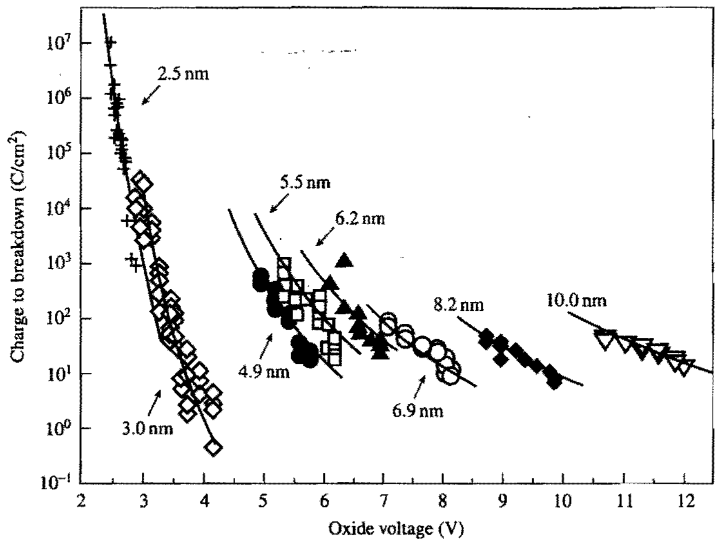

Figure 2.75. Typical plot of charge to breakdown versus oxide voltage for several oxide thickness values. (After Schuegraf and Hu, 1994.)

as a function of oxide voltage. Figure 2.75 is a typical plot for oxide thickness in the 2.5–10 nm range (Schuegraf and Hu, 1994). It shows that, for these relatively thick oxides, Q_{BD} decreases with increasing oxide voltage. It has also been shown that Q_{BD} is about the same for n-channel and p-channel MOSFETs (DiMaria and Stathis, 1997). Since these published Q_{BD} values do not take into account the progressive nature of the breakdown process, they project a lower allowed voltage for an oxide than is justified from a device reliability point of view. The progressive breakdown and the successive breakdown models, which take into account the oxide degradation rate after hard breakdown has commenced (stage 3 in Fig. 2.72), project a larger but more accurate allowed voltage (Linder *et al.*, 2002). The reader is referred to the literature on thin oxide reliability for more details (Degraeve *et al.*, 1998; Suehle, 2002; Suñé *et al.*, 2004).

As illustrated in Fig. 2.72, the gate leakage current of a MOSFET in a circuit has to reach a certain critical level, indicated by I_{fail}, before the circuit ceases to function properly. Thus, in circuit applications, what designers really need to know is the time to critical current instead of time to first breakdown or charge to breakdown.

Exercises

2.1 Show that the values of the Fermi–Dirac distribution function, Eq. (2.4), at a pair of energies symmetric about the Fermi energy E_f, are complementary, i.e., show that $f_D(E_f - \Delta E) + f_D(E_f + \Delta E) = 1$, independent of temperature.

2.2 For a given donor level E_d and concentration N_d of an n-type silicon, solve the Fermi energy E_f from the charge neutrality condition, Eq. (2.19) (neglecting the hole term). Show that $E_c - E_f$ approaches the complete ionization value, Eq. (2.20), under the condition of shallow donor level with low to moderate concentration. What happens if the condition is not satisfied?

2.3 Use the density of states $N(E)$ derived in Section 2.1.1.2 to evaluate the average kinetic energy of electrons in the conduction band:

$$\langle \text{K.E.} \rangle = \frac{\int_{E_c}^{\infty} (E - E_c) N(E) f_D(E) dE}{\int_{E_c}^{\infty} N(E) f_D(E) dE}.$$

(a) For a nondegenerate semiconductor in which can $f_D(E)$ be approximated by the Maxwell–Boltzmann distribution, Eq. (2.5), show that $\langle \text{K.E.} \rangle = \frac{3}{2}kT$.

(b) For a degenerate semiconductor at 0 K, show that $\langle \text{K.E.} \rangle = \frac{3}{5}(E_f - E_c)$.

2.4 The 3-D Gauss's law is obtained after a volume integration of the 3-D Poisson's equation and takes the form

$$\oint\!\!\oint_S \mathscr{E} \cdot dS = \frac{Q}{\varepsilon_{si}},$$

where the left-hand side is an integral of the normal electric field over a closed surface S, and Q is the net charge enclosed within S. Use it to derive the electric field at a distance r from a point charge Q (Coulomb's law). What is the electric potential in this case?

2.5 (a) Use Gauss's law to show that the electric field at a point above a uniformly charged sheet of charge density Q_s per unit area is $Q_s/2\varepsilon$, where ε is the permittivity of the medium.

(b) For two oppositely charged parallel plates with surface charge densities Q_s and $-Q_s$, show that the electric field is uniform and equals Q_s/ε in the region between the two plates and is zero in the regions outside the two plates.

2.6 The total depletion charge and inversion charge densities of a p-type MOS capacitor can be expressed as

$$Q_d = -qN_a \int_0^{W_d} (1 - e^{-q\psi/kT}) dx = -qN_a \int_0^{\psi_s} \frac{1 - e^{-q\psi/kT}}{\mathscr{E}} d\psi,$$

and

$$Q_i = -q\frac{n_i^2}{N_a} \int_0^{W_d} (e^{q\psi/kT} - 1) dx = -q\frac{n_i^2}{N_a} \int_0^{\psi_s} \frac{e^{q\psi/kT} - 1}{\mathscr{E}} d\psi,$$

using Eqs. (2.177) and (2.178). Here $\mathscr{E} = -d\psi/dx$ is given by Eq. (2.181).

(a) Write down the expressions for the small-signal depletion capacitance, $C_d = -dQ_d/d\psi_s$ and the small-signal inversion capacitance (low frequency), $C_i = -dQ_i/d\psi_s$, in silicon as represented in the equivalent circuit in Fig. 2.37.

(b) Show that $C_d + C_i = C_{si}$, where $C_{si} = -dQ_s/d\psi_s$ is evaluated using Eq. (2.182).

(c) Show that $C_d \approx C_i$ at the condition of strong inversion, $\psi_s = 2\psi_B$. (This allows one to use a split C–V measurement to determine the gate voltage where $\psi_s = 2\psi_B$).

(d) From the behavior of C_d beyond strong inversion, explain the "screening" of depletion charge (incremental) by the inversion layer.

2.7 Near the surface of an MOS capacitor biased well into strong inversion, only the $\exp(q\psi/kT)$ term in the square-root expression of Eq. (2.191) needs to be kept (classical model). Solve $\psi(x)$ under the boundary condition $\psi(0) = \psi_s$. Express the inversion electron concentration $n(x)$ in terms of the surface concentration $n(0)$ given by Eq. (2.193).

2.8 Solve the gate voltage equation (2.195) for $\psi_s(V_g)$ under the depletion condition in which $Q_s(\psi_s) = Q_d(\psi_s)$ given by Eq. (2.189). Show that for incremental changes, $\Delta\psi_s = \Delta V_g/(1 + C_d/C_{ox})$, where C_d is the depletion charge capacitance given by Eq. (2.201).

2.9 When the gate voltage greatly exceeds the threshold for strong inversion, a first-order solution of $\psi_s(V_g)$ can be obtained from the coupled equations (2.195) and (2.182), by keeping only the inversion charge term. Show that

$$\psi_s \approx 2\psi_B + \frac{2kT}{q} \ln\left(\frac{C_{ox}(V_g - V_{fb} - 2\psi_B)}{\sqrt{2\varepsilon_{si} kT N_a}}\right)$$

under these circumstances. Estimate how much higher ψ_s can be over $2\psi_B$ by substituting some typical values in the logarithmic expression.

2.10 In the split C–V measurement in Fig. 2.37(b), show that the n$^+$ channel part of the small-signal gate capacitance is

$$\frac{dQ_i}{dV_g} = \frac{C_{ox} C_i}{C_{ox} + C_i + C_d}.$$

Sketch the functional behavior of dQ_i/dV_g versus V_g, and from it describe the behavior of Q_i versus V_g.

2.11 The multiplication factors for holes and for electrons are given by Eqs. (2.256) and (2.257), respectively. For the special case of constant α_p and α_n, show that $M_p \to \infty$ occurs when the depletion-layer width approaches the value of $W = \ln(\alpha_n/\alpha_p)/(\alpha_n - \alpha_p)$. Also show that the condition for $M_n \to \infty$ gives the same result for W.

2.12 Prove the following mathematical identities:

$$\int_0^W f(x) \exp\left(-\int_0^x f(x')\,dx'\right) dx = 1 - \exp\left(-\int_0^W f(x)\,dx\right)$$

and

$$\int_0^W f(x)\exp\left(-\int_x^W f(x')\,dx'\right)dx = 1-\exp\left(-\int_0^W f(x)\,dx\right).$$

These identities are used in Appendix 8 to show that the condition for hole-initiated avalanche breakdown, namely $1/M_p \to 0$, is the same as that for electron-initiated avalanche breakdown, namely $1/M_n \to 0$.

2.13 The depletion-layer capacitance per unit area C_d of a uniformly doped abrupt p–n diode and its dependence on doping concentration and applied voltage are given in Eqs. (2.80), (2.81), and (2.83). Sketch $1/C_d^2$ as a function of the applied reverse-bias voltage V_{app}. Show how this plot can be used to determine N_a and N_d.

2.14 The depletion-layer capacitance of a one-sided p–n diode is often used to determine the doping profile of the lightly doped side. Consider an n$^+$–p diode, with a nonuniform p-side doping concentration of $N_a(x)$. If $Q_d(V)$ is the depletion-layer charge per unit area at bias voltage V, the capacitance per unit area at bias voltage V is $C = dQ_d/dV$. In terms of the depletion-layer width W, we have $C(V) = \varepsilon_{si}/W$, where W is a function of V. (For simplicity, we have dropped the subscripts in C, W, and V here.) Show that the doping concentration at the depletion-layer edge is given by

$$N_a(W) = \frac{2}{q\varepsilon_{si}d(1/C^2)/dV}.$$

2.15 The charge distribution of a p–i–n diode is shown schematically in Fig 2.17. The i-layer thickness is d. The depletion-layer capacitance is given by Eq. (2.96), namely $C_d = \varepsilon_{si}/W_d$, where $W_d = x_n + x_p$ is the total depletion-layer width. Derive this result from $C_d = dQ_d/dV$.

2.16 Consider a p–n diode. Assume the junction is located at $x = 0$, with the n-region to the left (i.e., $x < 0$) and the p-region to the right (i.e., $x > 0$) of the junction. The distribution of the excess electrons is given by Eq. (2.119), and the electron current density entering the p-region is given by Eq. (2.120). Derive the equation for the distribution of the excess holes in the n-region and the equation for the hole current density entering the n-region.

2.17 The minimum leakage current of a reverse-biased diode is determined by its saturation current components. The saturation currents depend on the dopant concentrations of the diode, as well as on the widths of the quasineutral p- and n-regions. They also depend on whether or not heavy-doping effect is included. This exercise is designed to show the magnitude of these effects.

(a) Consider an n$^+$–p diode, with an emitter doping concentration of 10^{20} cm^{-3} and a base doping concentration of 10^{17}cm^{-3}. Assume both the emitter and the base to be wide compared with their corresponding minority-carrier diffusion lengths. Ignore heavy-doping effect and calculate the electron and hole saturation current densities [see Eq. (2.129)].

(b) In most modern MOSFET and bipolar devices, the n$^+$–p diodes have the n$^+$–region width small compared with its hole diffusion length. If we assume the quasineutral n$^+$ region to have a width of 0.1 μm, again ignoring heavy-doping effect, estimate the hole saturation current density [see Eq. (2.133)].

(c) It is discussed in Section 6.1.2 and shown in Fig 6.3 that the effect of heavy doping should be included once the doping concentration is larger than about 10^{17} cm^{-3}. Heavy-doping effect is usually included simply by replacing the intrinsic-carrier concentration n_i by an effective intrinsic-carrier concentration n_{ie}, where n_i and n_{ie} are related by

$$n_{ie}^2 = n_i^2\exp(\Delta E_g/kT).$$

The empirical parameter ΔE_g is called the apparent bandgap narrowing due to heavy-doping effect, and its values are plotted in Fig. 6.3. Repeat (b) including the effects of heavy doping.

2.18 Consider an n$^+$–p diode, with the n$^+$ emitter side being wide compared with its hole diffusion length and the p base side being narrow compared with its electron diffusion length. The diffusion capacitance due to electron storage in the base is C_{Dn}, and that due to hole storage in the emitter is C_{Dp}. Assume the emitter to have a doping concentration of 10^{-20} cm^{-3} and the base to have a width of 100 nm and a doping concentration of 10^{17} cm^{-3}.

(a) If heavy-doping effect is ignored, the capacitance ratio is [see Eq. (2.168)]

$$\frac{C_{Dn}}{C_{Dp}} = \frac{2}{3}\frac{N_E}{N_B}\frac{W_B}{L_{pE}} \quad \text{(heavy-doping effect ignored).}$$

Evaluate this ratio for the n$^+$–p diode.

(b) When heavy-doping effect cannot be ignored, it is usually included simply by replacing the intrinsic-carrier concentration n_i by an effective intrinsic-carrier concentration n_{ie} [see part (c) of Exercise 2.17]. Show that when heavy-doping effect is included, the capacitance ratio becomes

$$\frac{C_{Dn}}{C_{Dp}} = \frac{2}{3}\left(\frac{n_{ieB}^2}{n_{ieE}^2}\right)\left(\frac{N_E}{N_B}\right)\frac{W_B}{L_{pE}} \quad \text{(heavy-doping effect included),}$$

where the subscript B denotes quantities in the base and the subscript E denotes quantities in the emitter. Evaluate this ratio for the n$^+$–p diode.

(This exercise demonstrates that heavy-doping effects cannot be ignored in any quantitative modeling of the switching speed of a diode.)

2.19 As electrons are injected from silicon into silicon dioxide, some of these electrons become trapped in the oxide. Let N_T be the electron trap density, n_T be the density of trapped electrons, and j_G/q be the injected electron particle current density. The rate equation governing $n_T(t)$ is

$$\frac{dn_T}{dt} = \frac{j_G}{q}\sigma(N_T - n_T),$$

where σ is the capture cross section of the traps. If the initial condition for n_T is $n_T(t=0)=0$, show that the time dependence of the trapped electron density is given by

$$n_T(t) = N_T\{1 - \exp\left[-\sigma N_{inj}(t)\right]\},$$

where

$$N_{inj}(t) \equiv \int_0^t \frac{j_G(t')}{q} dt'$$

is the number of injected electrons per unit area. Assume $N_T = 5\times10^{12}$ cm^{-3} and $\sigma = 1\times10^{-13}$ cm^2, sketch a log-log plot of n_T as a function of N_{inj}. (The capture cross section is often measured by fitting to such a plot.)

2.20 The avalanche multiplication factors M_p and M_n are given by Eqs. (2.256) and (2.257). Assume α_p and α_n are constant, independent of distance or electric field. Show that avalanche breakdown occurs when the width W of the high-field region (the region where impact ionization occurs) approaches $\left[\ln(\alpha_p/\alpha_n)\right]/(\alpha_p - \alpha_n)$.

2.21 Show that the band-to-band tunneling exponent in Eq. (2.259) can be derived from the WKB approximation, i.e. Eq. (2.245), for tunneling through a triangular barrier of height E_g, slope $q\mathscr{E}$ and tunneling distance $E_g/q\mathscr{E}$.

2.22 Assume silicon, room temperature, complete ionization. An abrupt p–n junction with $N_a = N_d = 10^{17}$ cm^{-3} is reversed biased at 2.0 V.

(a) Draw the band diagram. Label the Fermi levels and indicate where the voltage appears.

(b) What is the total depletion layer width?

(c) What is the maximum field in the junction?

2.23 For an abrupt n$^+$–p diode in Si, the n$^+$ doping is 10^{20} cm^{-3}, the p-type doping is 3×10^{16} cm^{-3}. Assume room temperature and complete ionization.

(a) Draw the band diagram at zero bias. Indicate $x=0$ as the boundary where the doping changes from n$^+$ to p. Also indicate where the Fermi level is with respect to the midgap.

(b) Write the equation and calculate the built-in potential.

(c) Write the equation and calculate the depletion width.

(d) Will the built-in potential increase or decrease if the temperature goes up and why?

2.24 Sketch the C–V curve (high frequency) of an MOS capacitor consisting of n$^+$ poly gate on n-type Si doped to $N_d = 10^{16}$ cm^{-3}. Calculate and show the flatband voltage on the C–V. Draw the band diagram for $V_g = 0$. Given $t_{ox} = 10$ nm, what is V_{ox} (potential across oxide) at the onset of inversion ($\psi_s = 2\psi_B$)? Ignore quantum and poly depletion effects.

2.25 Consider an MOS device with 20 nm thick gate oxide and uniform p-type substrate doping of 10^{17} cm^{-3}. The gate work function is that of n$^+$ Si.

(a) What is the flatband voltage? What is the threshold voltage for strong inversion?

(b) Sketch the high frequency C–V curve. Label where the flatband voltage and threshold voltage are.

(c) Calculate the maximum and the minimum capacitance (per area) values.

2.26 If the device in Exercise 2.25 is biased at zero gate voltage, determine the surface potential and the electron and hole densities at the surface.